syndrome is caused by vasoactive substances secreted by the tumor that lead to cutaneous flushing, sweating, bronchospasm, colicky abdominal pain, diarrhea, and right-sided cardiac valvular fibrosis. When tumors are confined to the intestine, the vasoactive substances released are metabolized to inactive forms by the liver—a "first-pass" effect similar to that seen with oral drugs. Thus, carcinoid syndrome occurs in less than 10% of patients and is strongly associated with hepatic metastatic disease.

The most important prognostic factor for gastrointestinal neuroendocrine tumors is location:

- *Foregut neuroendocrine (carcinoid) tumors* found within the stomach, duodenum proximal to the ligament of Treitz, and esophagus rarely metastasize and are generally cured by resection. Rare gastrin-producing neuroendocrine tumors, also termed *gastrinomas,* may present with symptoms related to increased acid production, including pain and/or bleeding from gastroduodenal ulcers, refractory gastroesophageal reflux, and diarrhea due to inactivation of pancreatic enzymes by excessive gastric acid. This constellation of findings is referred to as *Zollinger-Ellison syndrome.*
- *Midgut neuroendocrine (carcinoid) tumors* arise in the jejunum and ileum, are often multiple, and tend to be aggressive. In these tumors, depth of local invasion, size, and the presence of necrosis and mitoses are associated with poor outcome.
- *Hindgut neuroendocrine (carcinoid)* tumors arising in the appendix and colorectum are typically discovered incidentally. Those in the appendix occur at any age and almost uniformly pursue a benign course. Rectal tumors tend to produce polypeptide hormones and may manifest with abdominal pain and weight loss. Because they are usually discovered when small, metastasis of rectal neuroendocrine tumors is uncommon.

Gastrointestinal Stromal Tumor

Gastrointestinal stromal tumor (GIST) is the most common mesenchymal tumor of the abdomen and most frequently arises in the stomach. The term stromal is a historical mistake, as GIST is now recognized to arise from the interstitial cells of Cajal, the pacemaker cells of the gastrointestinal muscularis propria. A wide variety of other mesenchymal neoplasms also occur in the stomach. Many are named for the cell type they most closely resemble; for example, smooth muscle tumors are called *leiomyomas* or *leiomyosarcomas,* nerve sheath tumors are termed *schwannomas,* and those resembling glomus bodies in the nail beds and at other sites are termed *glomus tumors.* Only GIST is sufficiently frequent to merit further discussion.

Pathogenesis. **The most common driver mutation in GIST is a gain-of-function mutation in the gene encoding the receptor tyrosine kinase KIT.** These are present in 75% to 85% of all GISTs. An additional 8% of GISTs have mutations that activate a related receptor tyrosine kinase, platelet-derived growth factor receptor A (PDGFRA). For unknown reasons, GISTs bearing *PDGFRA* mutations are overrepresented in the stomach. *KIT* and *PDGFRA* gene mutations are mutually exclusive, reflecting their ability to activate the same downstream signaling pathways, which drive cell growth (Chapter 6). Germline mutations in these genes are present in rare individuals, who have a propensity to develop multiple GISTs as well as diffuse hyperplasia of Cajal cells.

In rare GISTs without *KIT* or *PDGFRA* mutations, genes encoding components of the mitochondrial succinate dehydrogenase (SDH) complex are most commonly affected. These mutations result in loss of SDH function and confer increased risk for both GIST and paraganglioma. One mutant allele is often inherited, with the second copy of the gene being either mutated or otherwise silenced in the tumor. The loss of SDH causes a number of metabolic changes, including increased production of reactive oxygen species, activation of hypoxia inducible factor (HIF), and increased dependency on glycolysis for ATP production; how these alterations lead to transformation is uncertain.

MORPHOLOGY

Primary gastric GISTs usually form a solitary, well-circumscribed, fleshy, submucosal mass. Metastases may form multiple small serosal nodules or fewer large nodules in the liver; spread outside of the abdomen is uncommon. GISTs may be composed of thin, elongated **spindle cells** or plumper **epithelioid cells** (eFig. 13.5). The most useful diagnostic marker is KIT, which is immunohistochemically detectable in 95% of these tumors.

Clinical Features. The peak incidence of gastric GIST is around 60 years of age, with less than 10% occurring in individuals younger than 40 years of age. GIST is slightly more common in males. It may present with symptoms related to mass effects or mucosal ulceration, intestinal obstruction, or gastrointestinal bleeding. Complete surgical resection is the primary treatment for localized gastric GIST.

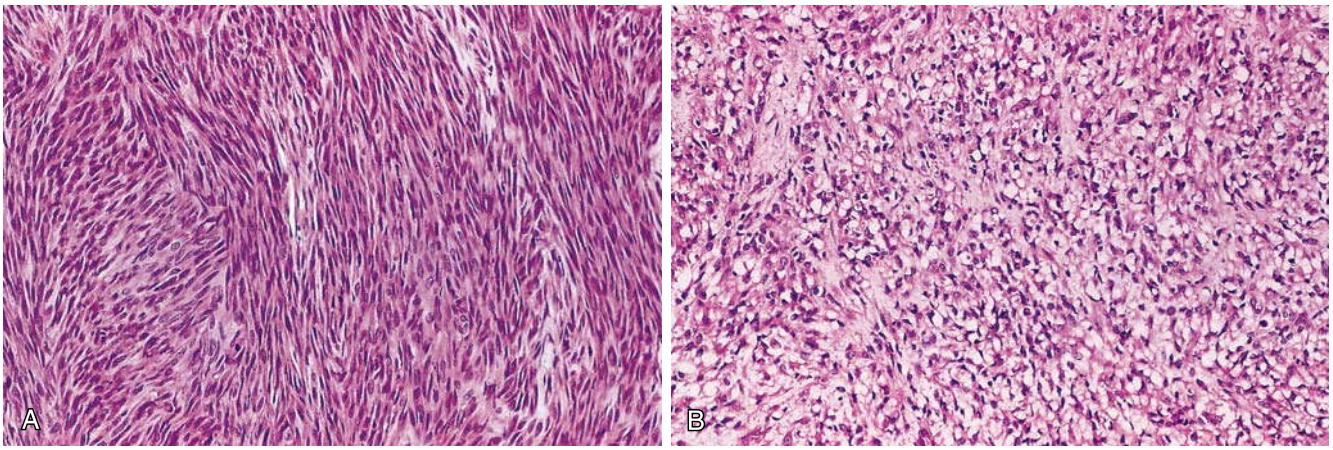
Prognosis correlates with tumor size, mitotic index, and location, with gastric GISTs being somewhat less aggressive than those arising in the small intestine. Recurrence or metastasis is rare for gastric GISTs less than 5 cm in diameter but common for mitotically active tumors larger than 10 cm. Tumors that are unresectable or metastatic often respond, sometimes for years, to tyrosine kinase inhibitors that are active against KIT and PDGFRA, such as imatinib. Unfortunately, as in chronic myeloid leukemia (Chapter 10), resistance to imatinib eventually arises due to outgrowth of subclones with additional mutations in KIT or PDGFRA. In some instances, these tumors respond to different tyrosine kinase inhibitors that bypass resistance to imatinib.

SMALL AND LARGE INTESTINES

The small intestine and colon are the principal sites of nutrient absorption and are exposed to the diverse array of antigens that are present in food and in gut microbes, which create a unique ecosystem referred to as the *microbiome.* The intestines may become involved by a wide variety of vascular, mechanical, infectious, inflammatory, and neoplastic disorders. Many of these diseases affect nutrient and water transport, perturbations that may cause malabsorption and diarrhea. The colon is also the most common site of gastrointestinal neoplasia in Western populations. We will begin our discussion with disorders that lead to mechanical obstruction of the bowel.

INTESTINAL OBSTRUCTION

Obstruction of the gastrointestinal tract may occur at any level, but the small intestine is most often involved because of its relatively narrow lumen. Collectively, *hernias, intestinal adhesions, intussusception,* and *volvulus* account for 80% of mechanical obstructions (Fig. 13.18), while tumors and infarction account for most of the remainder. The clinical manifestations of intestinal obstruction include abdominal pain and distention, vomiting, and constipation. Surgical intervention is usually required in cases involving mechanical obstruction or severe



eFIG. 13.5 Gastrointestinal stromal tumor (GIST). (A) Spindle cell type. (B) Epithelioid cell type with prominent cytoplasmic vacuoles. (From Fletcher CD: *Diagnostic Histopathology of Tumors,* ed 5, Figs. 8.58B, 8.59, Philadelphia, 2021, Elsevier.)

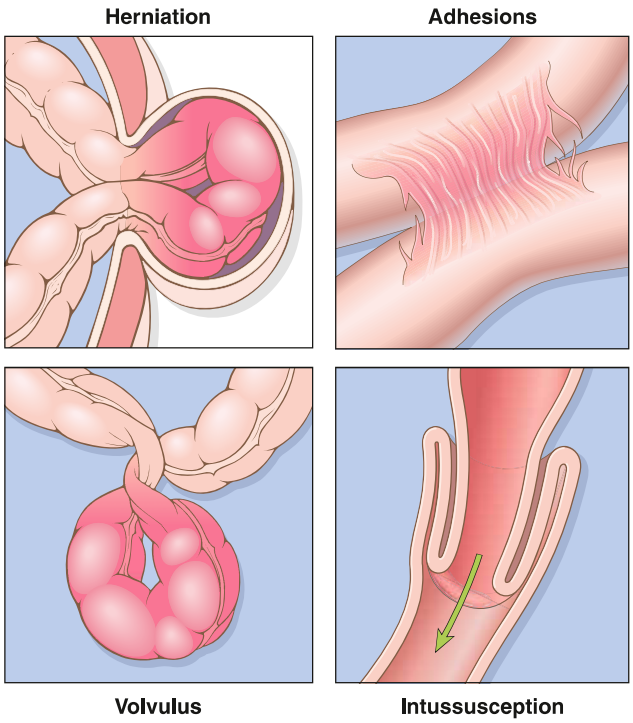


FIG. 13.18 Intestinal obstruction. The four major mechanical causes of intestinal obstruction are (1) herniation of a segment in the umbilical or inguinal regions, (2) adhesion between loops of intestine, (3) volvulus, and (4) intussusception.

infarction. Volvulus and adhesions between bowel loops are illustrated in Fig. 13.18. Brief comments are made next about other causes of obstruction.

Intussusception

Intussusception occurs when a segment of the intestine, constricted by a wave of peristalsis, telescopes into the immediately distal segment. Once trapped, the invaginated segment is propelled by peristalsis and pulls the mesentery along. Intussusception is the most common cause of intestinal obstruction in children younger than 2 years of age. There is usually no underlying anatomic defect and the child is otherwise healthy. Other cases are associated with viral infection and rotavirus vaccines and may be related to reactive hyperplasia of Peyer patches, which can act as the leading edge of the intussusception. If left untreated, intussusception may progress to intestinal obstruction, compression of mesenteric vessels, and infarction. Contrast enemas are diagnostically useful and are also effective in correcting intussusception in infants and young children. However, surgical intervention is necessary when an intraluminal mass or tumor serves as the initiating point of traction, as is typical in older children and in adults.

Hirschsprung Disease

***Hirschsprung*, also known as congenital aganglionic megacolon, *disease* occurs in approximately 1 of 5000 live births and stems from a congenital defect in colonic innervation.** It may be isolated or occur in combination with other developmental abnormalities. It is more common in males but tends to be more severe in females. Siblings of those affected are at increased risk for development of Hirschsprung disease.

Patients typically present as neonates with failure to pass meconium in the immediate postnatal period followed by obstructive constipation. The major threats to life are enterocolitis, fluid and electrolyte disturbances, perforation, and peritonitis. Surgical resection of the aganglionic segment with anastomosis of the normal colon to the rectum is effective, although it may take years for patients to attain normal bowel function and continence.

Pathogenesis. The enteric neuronal plexus develops from neural crest cells that migrate into the bowel wall during embryogenesis. **Hirschsprung disease results when the migration of neural crest cells from cecum to rectum is disrupted.** This produces a distal intestinal segment that lacks both the *Meissner submucosal plexus* and the *Auerbach myenteric plexus* ("aganglionosis") and thus fails to develop coordinated peristaltic contractions. Loss-of-function mutations in the RET receptor tyrosine kinase account for the majority of familial cases and approximately 15% of sporadic cases. Mutations in other genes involved in the development of neural crest cells have been identified in cases without *RET* mutations, and individuals with trisomy 21 are also at increased risk for uncertain reasons.

MORPHOLOGY

Hirschsprung disease always affects the rectum, but the length of the additional involved segments varies. Most cases are limited to the rectum and sigmoid colon, but severe disease can involve the entire colon. The aganglionic region may have a grossly normal or contracted appearance, while the normally innervated proximal colon may undergo progressive dilation as a result of functional distal obstruction (Fig. 13.19). Diagnosis is made by histologically confirming the absence of ganglion cells in the affected segment.

Abdominal Hernia

Any weakness or defect in the wall of the peritoneal cavity may permit protrusion of a serosa-lined pouch of peritoneum called a

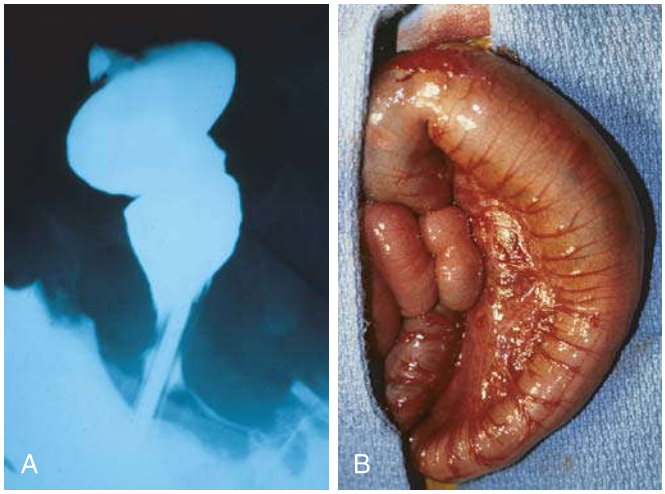


FIG. 13.19 Hirschsprung disease. (A) Preoperative barium enema study showing constricted rectum *(bottom)* and dilated sigmoid colon. Ganglion cells were absent in the rectum, but present in the sigmoid colon. (B) Corresponding intraoperative appearance of the dilated sigmoid colon. (Courtesy of Dr. Aliya Husain, The University of Chicago, Chicago, Illinois.)

hernia sac. Acquired hernias most commonly occur anteriorly, through the inguinal and femoral canals or umbilicus, or at sites of surgical scars. These are of concern because of the risk of visceral protrusion *(external herniation)*. This is most likely to occur with inguinal hernias, which tend to have narrow orifices and large sacs. Small bowel loops herniate most often, but portions of omentum or large bowel may also herniate and become entrapped. Pressure at the neck of the outpouching bowel may impair venous drainage, leading to stasis and edema. These changes increase the bulk of the herniated loop, leading to permanent entrapment *(incarceration)* and, over time, arterial and venous compromise *(strangulation)*, which may result in infarction.

VASCULAR DISORDERS OF BOWEL

Ischemic Bowel Disease

The majority of the GI tract is supplied by the celiac, superior mesenteric, and inferior mesenteric arteries. As they approach the intestinal wall, the superior and inferior mesenteric arteries ramify into the mesenteric arcades. Interconnections between arcades and collateral vessels make it possible for the small intestine and colon to tolerate slowly progressive loss of blood supply from one artery. By contrast, acute compromise of any major vessel may lead to infarction of segments of intestine. In a large majority of cases, acute obstruction is caused by thrombosis or embolism. The most important risk factor for thrombosis is severe atherosclerosis. Obstructive emboli most commonly originate from aortic atheromas or cardiac mural thrombi. Mesenteric venous thrombosis, which can also lead to ischemic disease, is uncommon but can result from inherited or acquired hypercoagulable states, invasive neoplasms, cirrhosis, trauma, or abdominal masses that compress the portal drainage. Intestinal hypoperfusion may also occur in the absence of vascular obstruction in the setting of cardiac failure, shock, dehydration, or use of vasoconstrictive drugs.

Pathogenesis. **The severity of vascular compromise, the time frame of its development, and the vessels that are affected are the major factors that determine the severity of ischemic bowel disease.** At the onset of vascular compromise, some hypoxic injury occurs, but intestinal epithelial cells are relatively resistant to short-term hypoxia. Ironically, the greatest damage appears to be initiated by restoration of the blood supply *(reperfusion injury)*. While the mechanisms of reperfusion injury are incompletely understood, they may involve free radical production, neutrophil infiltration, and locally produced inflammatory mediators (Chapter 1).

Two aspects of intestinal vascular anatomy also contribute to the distribution of ischemic damage:

- *Watershed zone* refers to intestinal segments at the end of their respective arterial supplies that are particularly susceptible to ischemia. These zones include the splenic flexure, where the superior and inferior mesenteric arterial circulations terminate, and to a lesser extent the sigmoid colon and rectum, where the inferior mesenteric, pudendal, and iliac arterial circulations end. Generalized hypotension or hypoxemia may cause localized injury at these vulnerable sites, and ischemic disease should be considered in the differential diagnosis for focal colitis of the splenic flexure or rectosigmoid colon.
- *Patterns of intestinal microvessels.* Intestinal capillaries run alongside the glands from the crypts to the surface before making a hairpin turn and returning to empty into the postcapillary venules. This configuration leaves the surface epithelium particularly vulnerable to ischemic injury.

MORPHOLOGY

While there is increased susceptibility of watershed zones, **mucosal** and **mural infarction** may involve any level of the gut from stomach to anus. Involvement is frequently segmental and patchy, and the affected mucosa is hemorrhagic and often ulcerated. The bowel wall is thickened by edema. With severe disease, pathologic changes include extensive mucosal and submucosal hemorrhage and necrosis. Damage is more pronounced with acute arterial thrombosis, which often leads to **transmural infarction.** Blood-tinged mucus or blood accumulates within the lumen. Coagulative necrosis of the muscularis propria occurs within 1 to 4 days and may be associated with purulent serositis and perforation.

Microscopic examination of ischemic intestine demonstrates **atrophy** or **sloughing of surface epithelium** (Fig. 13.20A). By contrast, crypts may be hyperproliferative. Inflammatory infiltrates are initially absent in acute ischemia, but neutrophils are recruited within hours of reperfusion. Chronic ischemia is accompanied by fibrous scarring of the lamina propria (Fig. 13.20B) and, uncommonly, stricture formation. In acute phases of ischemic damage, bacterial superinfection and enterotoxin release may induce pseudomembrane formation that can resemble *Clostridium difficile*—associated pseudomembranous colitis (discussed later).

Clinical Features. Ischemic bowel disease most commonly occurs in older adults with coexisting cardiac or vascular disease. Acute transmural infarction typically manifests with sudden, severe abdominal pain and tenderness, sometimes accompanied by nausea, vomiting, and bloody diarrhea or grossly melanotic stool. This presentation may progress to shock and vascular collapse within hours due to blood loss from the ischemic bowel. Peristaltic sounds diminish or disappear, and muscular spasm affecting abdominal muscles creates boardlike rigidity of the abdominal wall. Because these physical signs overlap with those of other abdominal emergencies, including acute pancreatitis, acute appendicitis, perforated ulcer, and acute cholecystitis, the diagnosis of intestinal infarction may be delayed or missed, with disastrous consequences. As the mucosal barrier breaks down, bacteria enter the circulation and sepsis may develop; the mortality rate in cases complicated by sepsis exceeds 50%.

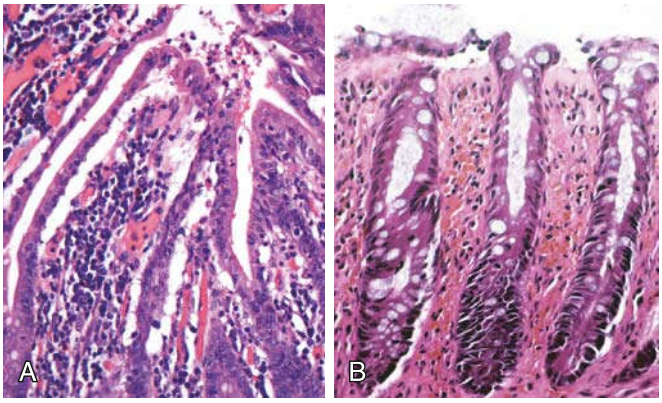


FIG. 13.20 Ischemia. (A) Characteristic attenuated and partially detached villous epithelium in acute jejunal ischemia. Note the hyperchromatic nuclei of proliferating crypt cells. (B) Chronic colonic ischemia with atrophic surface epithelium and fibrotic lamina propria.

The overall progression of ischemic enteritis depends on the underlying cause and severity of injury:

- *Mucosal and mural infarctions* alone may not be fatal. However, these may progress to more extensive, transmural infarction if the vascular supply is not restored by correcting the underlying cause or (in chronic disease) the development of an adequate collateral blood supply.
- *Chronic ischemia* may masquerade as inflammatory bowel disease, with episodes of bloody diarrhea interspersed with periods of healing.
- *Cytomegalovirus (CMV) infection* causes ischemic gastrointestinal disease as a consequence of viral infection of endothelial cells. CMV infection may be a complication of immunosuppressive therapy.
- *Radiation enterocolitis* occurs when the gastrointestinal tract is irradiated. In addition to epithelial damage, radiation-induced vascular injury may produce ischemic disease.
- *Necrotizing enterocolitis* is an idiopathic disorder of the small and large intestines that may result in transmural necrosis (Chapter 4). It is the most common acquired gastrointestinal emergency of neonates, particularly those who are premature or of low birth weight. Its occurrence is often associated with initiation of oral feeding. Although the pathogenesis of necrotizing enterocolitis is not defined, ischemic injury is thought to contribute.

Angiodysplasia

Angiodysplasia is characterized by malformed submucosal and mucosal blood vessels. It occurs most often in the cecum or right colon and usually presents after the sixth decade of life. Although it affects less than 1% of the adult population, angiodysplasia accounts for 20% of major episodes of lower intestinal bleeding.

As with all lower gastrointestinal bleeding, in most instances the blood is maroon or bright red in color. This is in distinction to upper gastrointestinal bleeding, which tends to produce black, tarry stools *(melena)* and so-called coffee ground emesis. These distinctions are not absolute, as a very brisk upper gastrointestinal bleed may produce red blood per rectum and bleeds from the right colon sometimes produce dark stools.

Hemorrhoids

Hemorrhoids are dilated anal and perianal collateral vessels that connect the portal and caval venous systems to relieve elevated venous pressure within the hemorrhoidal plexus. Thus, although hemorrhoids are less serious than esophageal varices, the pathogenesis of these lesions is similar. They are common, affecting about 5% of the population. Frequent predisposing factors include constipation and associated straining (which increase intraabdominal and venous pressures), venous stasis of pregnancy, and portal hypertension.

Collateral vessels within the inferior hemorrhoidal plexus are located below the anorectal line and are termed *external hemorrhoids,* while those that result from dilation of the superior hemorrhoidal plexus within the distal rectum are referred to as *internal hemorrhoids.* On histologic examination, hemorrhoids consist of thin-walled, dilated submucosal vessels beneath the anal or rectal mucosa. These vessels are subject to trauma, which leads to rectal bleeding. In addition, they can become thrombosed and inflamed.

Hemorrhoids often manifest with pain and self-limited rectal bleeding, particularly bright red blood seen on toilet paper. When they develop as a result of portal hypertension, the implications are more ominous. Hemorrhoidal bleeding is generally not a medical emergency; treatment options include sclerotherapy, rubber band ligation, and infrared coagulation. In severe cases, hemorrhoids may be removed surgically.

DIARRHEAL DISEASE

Diarrhea is defined as the passage of loose or watery stools, typically of amounts greater than 200 grams per day. In severe cases, stool volume can exceed 14 L per day and, without fluid resuscitation, result in death. Worldwide, diarrheal diseases are estimated to cause the deaths of 1.5 to 2 million children under 5 years of age annually. Diarrhea is a common symptom of many intestinal diseases, including those due to infection, inflammation, ischemia, malabsorption, and nutritional deficiency. It is subclassified into four major categories:

- *Secretory diarrhea* is characterized by isotonic stool and persists during fasting.
- *Osmotic diarrhea,* such as that occurring with lactase deficiency, is due to osmotic force exerted by unabsorbed luminal solutes. The diarrheal fluid is at least 50 mOsm more concentrated than plasma and the condition abates with fasting.
- *Malabsorptive diarrhea* caused by inadequate nutrient absorption is associated with steatorrhea and is relieved by fasting.
- *Exudative diarrhea* is due to inflammatory disease and characterized by purulent, bloody stools that continue during fasting.

We begin our discussion with malabsorptive diarrhea. Other disorders associated with secretory and exudative types of diarrhea (e.g., cholera and inflammatory bowel disease, respectively) are addressed in separate sections.

Malabsorptive Diarrhea

Malabsorption manifests most commonly as chronic diarrhea and is characterized by defective absorption of fats, fat- and water-soluble vitamins, proteins, carbohydrates, electrolytes, minerals, and water. Chronic malabsorption causes weight loss, anorexia, abdominal distention, borborygmi (a rumbling or gurgling noise of the intestines), and muscle wasting. A hallmark of malabsorption is *steatorrhea,* characterized by excessive fecal fat and bulky, frothy, greasy, yellow or clay-colored stools. The most common chronic malabsorptive disorders in Western countries are pancreatic insufficiency, celiac disease, and Crohn disease. Environmental enteric dysfunction, or environmental enteropathy, which has a malabsorptive component, is another cause that is pervasive in some lower-resource countries.

Malabsorption results from a disturbance in at least one of the four phases of nutrient absorption:

- *Intraluminal digestion,* in which proteins, carbohydrates, and fats are broken down into absorbable forms
- *Terminal digestion,* which involves the hydrolysis of carbohydrates and peptides by disaccharidases and peptidases, respectively, in the brush border of the small-intestinal mucosa
- *Transepithelial transport,* in which nutrients, fluid, and electrolytes are transported across and processed within the small-intestinal epithelium
- *Lymphatic transport* of absorbed lipids

In many malabsorptive disorders, a defect in one of these processes predominates, but more than one usually contributes (Table 13.3). As a result, malabsorption syndromes resemble each other more than they differ. Signs and symptoms include diarrhea (from nutrient

Table 13.3 Defects in Malabsorptive and Diarrheal Disease

Disease	Intraluminal Digestion	Terminal Digestion	Transepithelial Transport	Lymphatic Transport
Celiac disease		+	+	
Tropical sprue		+	+	
Chronic pancreatitis	+			
Cystic fibrosis	+			
Primary bile acid malabsorption	+		+	
Carcinoid syndrome			+	
Autoimmune enteropathy		+	+	
Disaccharidase deficiency		+		
Mycobacterial infection, Whipple disease				+
Abetalipoproteinemia			+	
Viral gastroenteritis		+	+	
Bacterial gastroenteritis		+	+	
Parasitic gastroenteritis		+	+	
Inflammatory bowel disease	+	+	+	

+, Indicates that the process can be abnormal in the disease indicated. Other processes are not typically affected.

malabsorption and excessive intestinal secretion), flatus, abdominal pain, and weight loss. Inadequate absorption of vitamins and minerals may result in anemia and mucositis due to pyridoxine, folate, or vitamin B_{12} deficiency; bleeding due to vitamin K deficiency; osteopenia and tetany due to calcium, magnesium, or vitamin D deficiency; and neuropathy due to vitamin A or B_{12} deficiency. A variety of endocrine and skin disturbances may also occur. Mycobacterial infection, which may lead to lymphatic transport defects, is discussed with infectious causes of diarrhea in the next section.

Cystic Fibrosis

Cystic fibrosis is discussed in greater detail in Chapter 4; only the associated malabsorption is considered here. Due to mutations of the epithelial cystic fibrosis transmembrane conductance regulator (CFTR), individuals with cystic fibrosis have defects in intestinal and pancreatic ductal ion transport. This abnormality interferes with bicarbonate, sodium, and water secretion, ultimately resulting in inadequate luminal hydration. In the pancreas, luminal dehydration results in the production of abnormally viscous mucus, which obstructs the ducts and leads to autodigestion of the pancreas and eventually exocrine pancreatic insufficiency in more than 80% of patients. The failure of nutrient digestion in the small bowel in turn results in osmotic diarrhea, which can be effectively treated in most patients with oral digestive enzyme supplements. Lipase deficiency can lead to steatorrhea.

Celiac Disease

Celiac disease, also known as *celiac sprue* or *gluten-sensitive enteropathy,* is an immune-mediated enteropathy triggered by the ingestion of gluten-containing cereals, such as wheat, rye, or barley, in genetically predisposed individuals. Oats do not contain gluten but are sometimes added to this list as oats processed in factories that also process wheat, rye, or barley may be cross-contaminated. Celiac disease is found globally and has a worldwide estimated prevalence of about 1%. The primary treatment for celiac disease is a *gluten-free diet,* which results in symptomatic improvement for most patients.

Pathogenesis. **Celiac disease is an intestinal immune reaction to gluten, the major storage protein of wheat and similar grains.** Digestion of gluten by luminal and brush border enzymes produces a 33—amino acid peptide known as gliadin that is resistant to further proteolysis (Fig. 13.21). Deamidation of gliadin by tissue transglutaminase in the lamina propria increases its binding to HLA-DQ2 and HLA-DQ8 class II MHC molecules expressed on antigen-presenting cells, allowing gliadin to be presented to CD4+ T cells. These activated CD4+ T cells produce cytokines such as interferon-γ that likely contribute to the tissue damage and characteristic mucosal histopathology. An antibody response follows, leading to the production of antibodies against tissue transglutaminase, deamidated gliadin, and, perhaps due to cross-reactive epitopes, cardiac endomysium. Whether these antibodies contribute to celiac disease pathogenesis or are merely markers of immune activation remains controversial, but their levels are well correlated with the presence of celiac disease and response to a gluten-free diet.

In addition to CD4+ cells, there is an accumulation of CD8+ cells that are not specific for gliadin but that may play an ancillary role in causing tissue damage. It is thought that some gliadin peptides induce epithelial cells to produce cytokines such as IL-15, which in turn trigger activation and proliferation of CD8+ intraepithelial lymphocytes. These T cells are proposed to kill enterocytes that have been induced by various stressors to express molecules such as MIC-A, a surface glycoprotein, that may be recognized by activated CD8+ T cells and NK cells expressing the NKG2D receptor. The resulting epithelial damage is then predicted to increase the movement of gliadin peptides across the epithelium and their deamidation by tissue transglutaminase, perpetuating the cycle of disease. Whatever the precise mechanism, the epithelial injury leads to blunting of villi and altered epithelial cell differentiation, both of which may contribute to malabsorption.

While nearly all individuals eat grain and are exposed to gluten and gliadin, most do not develop celiac disease. Thus, host factors determine whether celiac disease develops. Among these, HLA proteins seem to be critical, since almost all individuals with celiac disease express HLA-DQ2 or HLA-DQ8. There is also an association with

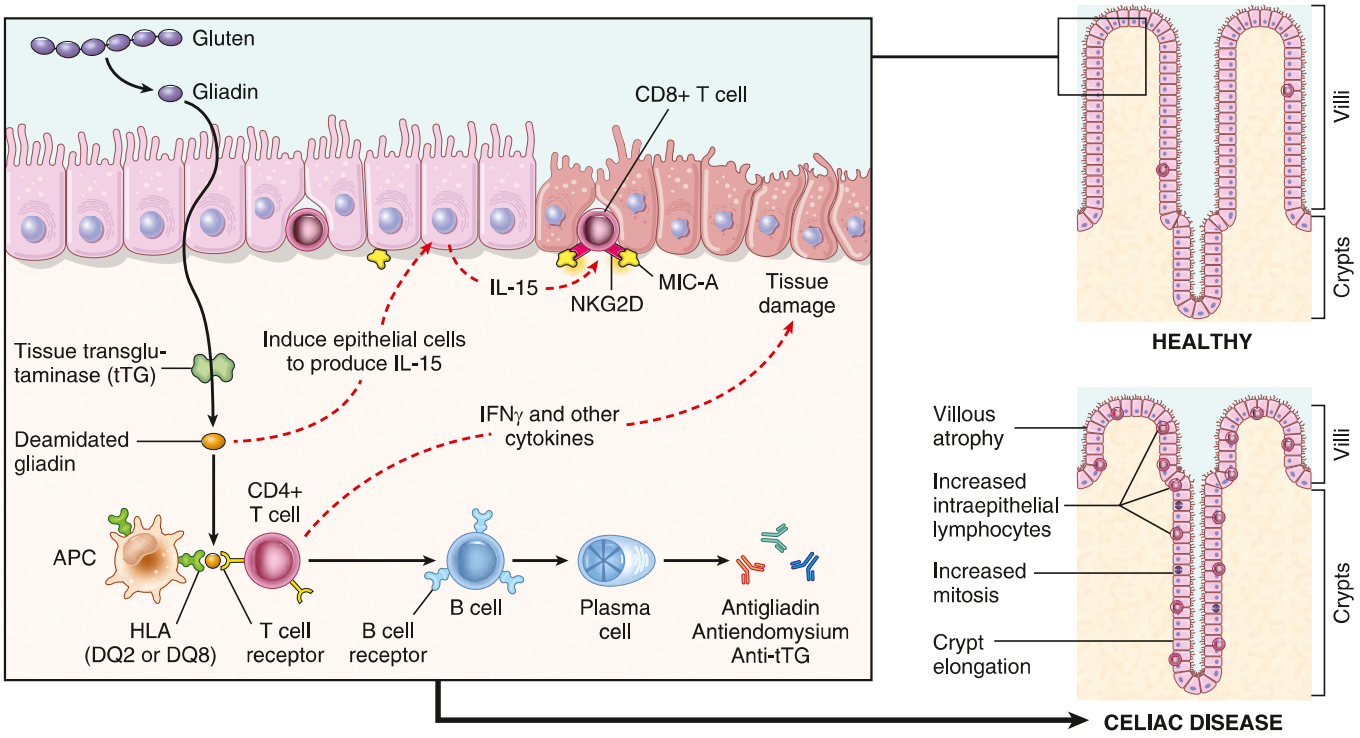


FIG. 13.21 A model for the pathogenesis of celiac disease *(left).* CD4+ T cells are stimulated by antigen-presenting cells displaying deamidated gliadin peptides to produce a variety of cytokines that stimulate antibody production from B cells. The source of the epithelial damage (denoted by *dotted lines*) is less certain. Cytokines such as interferon-γ produced by the CD4+ T cells may directly damage epithelial cells. Alternatively, deamidated gliadin may induce epithelial cells to produce IL-15, stimulating intraepithelial CD8+ T cells expressing the NKG2D receptor to recognize and kill epithelial cells expressing the molecule MIC-A (MHC class I polypeptide-related sequence A). The resulting morphologic alterations *(right)* include varying degrees of villous atrophy, increased numbers of intraepithelial lymphocytes, and epithelial proliferation with crypt elongation.

other immune diseases including type 1 diabetes, thyroiditis, and Sjögren syndrome.

MORPHOLOGY

Biopsy specimens from the second portion of the duodenum or proximal jejunum, which are exposed to the highest concentrations of dietary gluten, are generally diagnostic. The histopathology is characterized by **crypt hyperplasia, villous atrophy** (Fig. 13.22), and increased intraepithelial lymphocytes (eFig. 13.6). The loss of mucosal and brush-border surface area due to villous atrophy probably accounts for the malabsorption. In addition, increased rates of epithelial turnover, reflected in increased crypt mitotic activity, may limit the ability of absorptive enterocytes to fully differentiate and express proteins necessary for terminal digestion and transepithelial transport. Other histologic findings may include increased numbers of plasma cells, mast cells, and eosinophils, especially within the upper part of the lamina propria. It should be noted that intraepithelial lymphocytosis and villous atrophy can be present in other disorders, including viral enteritis. Therefore, serologic findings, combined with histologic examination, are most specific for diagnosis of celiac disease.

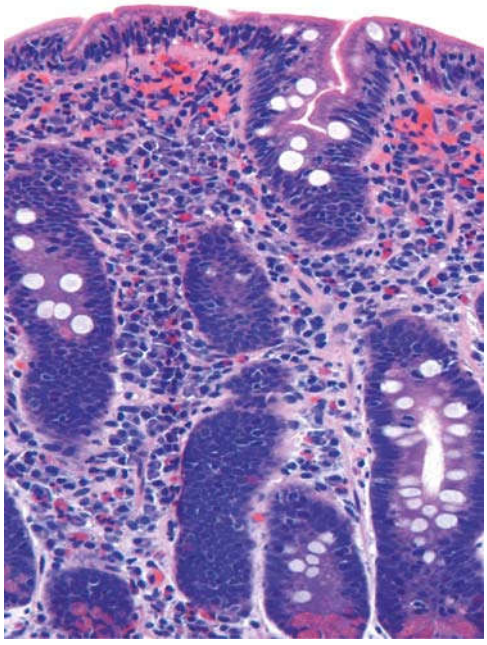
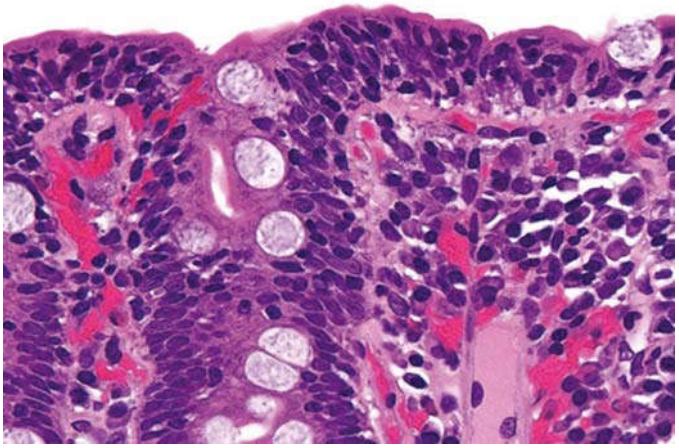


FIG. 13.22 Celiac disease. Advanced cases of celiac disease show complete loss of villi, or total villous atrophy. Note the dense plasma cell infiltrates in the lamina propria and intraepithelial lymphocytes.

Clinical Features. Pediatric celiac disease, which affects male and female children equally, usually manifests between 6 and 24 months of



eFIG. 13.6 Celiac disease. This high power view shows numerous intraepithelial lymphocytes. (© 2022 Elsevier Gattuso's Differential Diagnosis in Surgical Pathology, 4th ed.)

age (after introduction of gluten to the diet) with irritability, abdominal distention, anorexia, diarrhea, failure to thrive, weight loss, or muscle wasting. Older children, by contrast, are more likely to present with abdominal pain, nausea, vomiting, bloating, or constipation. A characteristic pruritic, blistering skin lesion, *dermatitis herpetiformis,* is present in as many as 10% of patients (Chapter 22).

In adults, celiac disease manifests most commonly between 30 and 60 years of age with anemia (due to iron deficiency and, less commonly, B_{12} and folate deficiency), diarrhea, bloating, and fatigue. Many cases, however, escape clinical attention for extended periods because of atypical presentations, and some patients have silent celiac disease, defined as positive serology and villous atrophy without symptoms.

Noninvasive serologic tests are generally performed before biopsy. The most sensitive tests identify the presence of IgA antibodies to tissue transglutaminase or IgA or IgG antibodies to deamidated gliadin. Antiendomysial antibodies are highly specific but less sensitive than other antibodies.

Patients with celiac disease have an elevated risk of small bowel malignancy. The most common celiac disease—associated cancer is *enteropathy-associated T-cell lymphoma,* an aggressive tumor of intraepithelial T lymphocytes. *Small-intestinal adenocarcinoma* is also more frequent in individuals with celiac disease. Thus, when symptoms such as abdominal pain, diarrhea, and weight loss recur despite a strict gluten-free diet, the differential diagnosis includes cancer or *refractory sprue,* in which the response to a gluten-free diet is lost. Gluten exposure is the most common cause of recurrent symptoms; most individuals with celiac disease do well with dietary restrictions and die of unrelated causes.

Environmental Enteric Dysfunction

The term environmental enteric dysfunction, or environmental enteropathy, refers to a syndrome of impaired intestinal function that is prevalent in areas with poor sanitation infrastructure and suboptimal hygiene. It is most common in parts of sub-Saharan Africa, such as Zambia; aboriginal populations within northern Australia; southern India; and some groups living in poverty in South America and Asia. Because of its tendency to occur in a belt close to the equator, it has also been referred to as *tropical enteropathy* or *tropical sprue*. Its etiology is uncertain. Some work has implicated small intestinal colonization by toxigenic coliform bacteria, but oral antibiotic treatment and nutritional supplementation do not usually fully reverse the syndrome. Small intestinal biopsy specimens show blunted villi and chronic inflammation, including intraepithelial lymphocytes, findings similar to those seen in celiac disease, which must be excluded by serologic studies. Although relatively well tolerated in adults, environmental enteropathy may cause sufficient malabsorption to stunt the growth and impair the cognitive development of affected children.

Lactase (Disaccharidase) Deficiency

The term "lactase deficiency" sounds like a descriptor of an abnormality, but in fact expression of lactase wanes following weaning in virtually all mammals. An exception is found in people who descended from certain populations in Africa, Europe, and the Middle East who have acquired mutations that lead to persistent, lifelong lactase expression.

Lactase deficiency gives rise to osmotic diarrhea following the consumption of lactose-rich foods, such as dairy products. The disaccharidases, including lactase, are located in the apical brush border membrane of villous absorptive epithelial cells. Lactase deficiency is of several types:

- *Congenital lactase deficiency* is a rare autosomal recessive disorder caused by mutations in the gene encoding lactase. It manifests as explosive diarrhea with watery, frothy stools and abdominal distention after milk ingestion. Symptoms abate when exposure to milk and milk products is terminated.
- *Acquired lactase deficiency*, as mentioned, is caused by the loss of lactase gene expression early in childhood and is very common, being estimated to affect roughly two-thirds of the world's population and about one-half of adults in the United States. The acquired deficiency typically manifests after the age of weaning from breast milk.
- *Transient lactase deficiency* is caused by small bowel injury, as may occur due to infectious or inflammatory insults, and is reversible if the cause of the injury remits.

Abetalipoproteinemia

Abetalipoproteinemia is a rare autosomal recessive disease marked by an inability of enterocytes to secrete triglyceride-rich chylomicrons. This transepithelial transport defect is caused by mutation of the gene encoding *microsomal triglyceride transfer protein* and renders enterocytes incapable of exporting lipoproteins, free fatty acids, and fat-soluble vitamins. Abetalipoproteinemia manifests in infancy. The clinical picture is dominated by failure to thrive, diarrhea, and steatorrhea. Deficiencies of essential fatty acids and fat-soluble vitamins lead to defects in coagulation (due to vitamin K deficiency) and skeletal, CNS, and retinal abnormalities. A characteristic finding in peripheral blood smears is the presence of acanthocytic red cells *(spur cells)*, owing to an alteration in the lipid content of red cell membranes.

Microscopic Colitis

Microscopic colitis encompasses two idiopathic entities, *collagenous colitis* and *lymphocytic colitis.* Both manifest with chronic, nonbloody, watery diarrhea without weight loss. Findings on radiologic and endoscopic studies are typically normal. Collagenous colitis, which occurs primarily in middle-aged and older women, is characterized by the presence of a dense subepithelial collagen layer, increased numbers of intraepithelial lymphocytes, and a mixed inflammatory infiltrate within the lamina propria. Lymphocytic colitis is histologically similar, but the subepithelial collagen layer is of normal thickness and the increase in intraepithelial lymphocytes tends to be greater. Lymphocytic colitis is associated with celiac disease and, like celiac disease, is more frequent in those with the HLA-DQ2 haplotype. It is also associated with several other autoimmune diseases, including Hashimoto thyroiditis, type I diabetes, and certain forms of arthritis.

Graft-Versus-Host Disease

Graft-versus-host disease occurs after allogeneic hematopoietic stem cell transplantation (Chapter 5). The small bowel and colon are involved in most cases. Graft-versus-host disease is caused by donor T cell—mediated damage to the recipient's epithelial cells. The lymphocytic infiltrate in the lamina propria is typically sparse, suggesting that cytokines secreted by T cells may be the major mediators of tissue injury. Epithelial apoptosis, particularly of crypt cells, is the most common histologic finding. Intestinal graft-versus-host disease often manifests as watery diarrhea.

Infectious Enterocolitis

Globally, infectious enterocolitis is responsible for more than 1 million deaths annually and greater than 10% of deaths in children younger than 5 years of age. Enterocolitis presents with a broad range of signs and symptoms, including diarrhea, abdominal pain, urgency, perianal discomfort, incontinence, and hemorrhage. Bacterial

Table 13.4 Features of Bacterial Enterocolitides

Infection Type	Distribution	Reservoir	Transmission	Epidemiology	Affected GI Sites	Symptoms	Complications
Cholera	India, Africa	Shellfish	Fecal-oral, water	Sporadic, endemic, epidemic	Small intestine	Severe watery diarrhea	Dehydration, electrolyte imbalances
Campylobacter spp.	Higher-resource countries	Chickens, sheep, pigs, cattle	Poultry, milk, other foods	Sporadic; children, travelers	Colon	Watery or bloody diarrhea	Reactive arthritis, Guillain-Barré syndrome
Shigellosis	Worldwide, endemic in lower-resource countries	Humans	Fecal-oral, food, water	Children, migrant workers, travelers, nursing homes	Left colon, ileum	Bloody diarrhea	Reactive arthritis, urethritis, conjunctivitis, hemolytic-uremic syndrome
Salmonellosis	Worldwide	Poultry, farm animals, reptiles	Meat, poultry, eggs, milk	Children, older adults	Colon, small intestine	Watery or bloody diarrhea	Sepsis, abscess
Enteric (typhoid) fever	India, Mexico, Philippines	Humans	Fecal-oral, water	Children. adolescents, travelers	Small intestine	Bloody diarrhea, fever	Chronic infection, carrier state, encephalopathy, myocarditis, intestinal perforation
Yersinia spp.	Northern and central Europe	Pigs, cows, puppies, cats	Pork, milk, water	Clustered cases	Ileum, appendix, right colon	Abdominal pain, fever, diarrhea	Reactive arthritis, erythema nodosum
Enterotoxigenic *E. coli*	Lower-resource countries	Unknown	Food or fecal-oral	Infants, adolescents, travelers	Small intestine	Severe watery diarrhea	Dehydration, electrolyte imbalances
Enteropathogenic *E. coli*	Worldwide	Humans	Fecal-oral	Infants	Small intestine	Watery diarrhea	Dehydration, electrolyte imbalances
Enterohemorrhagic *E. coli*	Worldwide	Widespread, includes cattle	Beef, milk, produce	Sporadic and epidemic	Colon	Bloody diarrhea	Hemolytic-uremic syndrome
Enteroinvasive *E. coli*	Lower-resource countries	Unknown	Cheese, other foods, water	Young children	Colon	Bloody diarrhea	Unknown
Enteroaggregative *E. coli*	Worldwide	Unknown	Unknown	Children, adults, travelers	Colon	Nonbloody diarrhea, afebrile	Poorly defined
Pseudomembranous colitis *(C. difficile)*	Worldwide	Humans, hospitals	Antibiotics allow emergence	Immunocompromised antibiotic-treated	Colon	Watery diarrhea, fever	Relapse, toxic megacolon
Whipple disease	Rural > urban	Unknown	Unknown	Rare	Small intestine	Malabsorption	Arthritis, CNS disease
Mycobacterial infection	Worldwide	Unknown	Unknown	Immunocompromised endemic	Small intestine	Malabsorption	Pneumonia, infection at other sites

CNS, Central nervous system; *GI*, gastrointestinal.

infections, such as enterotoxigenic *Escherichia coli,* are frequently responsible, but the most common pathogens vary with age, nutrition, host immune status, and environment (Table 13.4). For example, epidemics of cholera are common in areas with poor sanitation as a result of inadequate public health measures, natural disasters (e.g., the Haitian earthquake of 2010), or war. Pediatric infectious diarrhea, which may result in severe dehydration and metabolic acidosis, is commonly caused by enteric viruses. A summary of the epidemiology and clinical features of selected causes of bacterial enterocolitis is presented in Table 13.4. Representative bacterial, viral, and parasitic enterocolitides are discussed next.

Cholera

Vibrio cholerae organisms are comma-shaped, gram-negative bacteria that cause cholera, a disease that has been endemic in the Ganges Valley of India and Bangladesh for all recorded history. *V. cholerae* is transmitted primarily by contaminated drinking water. However, person-to-person transmission also occurs and in rare instances it can be transmitted in seafood. There is a marked seasonal variation in most locales due to rapid growth of *Vibrio* bacteria at warm temperatures. The only animal reservoirs are shellfish and plankton.

Pathogenesis. ***Vibrio* organisms cause disease by producing a potent toxin that interferes with the absorptive function of enterocytes.** Flagellar proteins are necessary for efficient colonization of the gut by *Vibrio* organisms, and a secreted metalloproteinase with hemagglutinin activity is important for shedding in the stool. Disease is caused by a preformed enterotoxin known as *cholera toxin* composed of five B subunits that direct endocytosis and an enzymatically active A subunit. After uptake into the endoplasmic reticulum, a fragment of the A subunit is transported into the cytosol, where it interacts with host factors to ribosylate and activate the G protein $G_{s\alpha}$. This stimulates adenylate cyclase, leading to increases in intracellular cyclic adenosine monophosphate (cAMP), which binds and activates the cystic fibrosis transmembrane conductance regulator (CFTR). In enterocytes, activation of CFTR leads to the outflow of chloride ions into the lumen of the gut, creating an osmotic gradient that produces massive *secretory diarrhea.* Mucosal biopsy specimens show only minimal morphologic alterations.

Clinical Features. Most exposed individuals are asymptomatic or suffer only mild diarrhea; unlike many other forms of infectious enterocolitis, fever is uncommon. In those with severe disease, there is abrupt onset of watery diarrhea and vomiting after an incubation period of 1 to 5 days. The volume of diarrhea may reach 1 L per hour, leading to dehydration, hypotension, electrolyte imbalances, muscle cramping, anuria, shock, loss of consciousness, and death. Most deaths occur within the first 24 hours of presentation. The mortality rate for untreated severe cholera is 50% to 70%, but with simple fluid replacement more than 99% of patients survive.

Campylobacter Enterocolitis

***Campylobacter jejuni* is a common cause of acute diarrhea worldwide and is a major foodborne pathogen in the United States.** In resource-limited countries, it is often endemic and frequently afflicts visitors from resource-rich areas (so-called *"traveler's diarrhea"*). Most infections are associated with ingestion of raw or undercooked meat; unpasteurized milk or contaminated water are also culpable in some cases.

Pathogenesis. The number of organisms ingested, the virulence of the infecting strain, and host immunity determine whether *Campylobacter* ingestion results in disease. Virulence factors appear to influence disease development by contributing to four properties: motility, adherence, toxin production, and invasion. Flagella allow *Campylobacter* to be motile and facilitate adherence, colonization, and mucosal invasion. Cytotoxins that cause epithelial damage and a cholera toxin—like enterotoxin are also elaborated by some *C. jejuni* isolates. Dysentery (bloody diarrhea) is generally associated with invasion and only occurs with a small minority of *Campylobacter* strains. *Enteric fever* occurs when bacteria proliferate within the lamina propria and mesenteric lymph nodes.

Campylobacter infection may result in reactive arthritis, primarily in patients with the HLA-B27 allele. Other extraintestinal complications include erythema nodosum and Guillain-Barré syndrome (Chapter 20), a flaccid paralysis caused by autoimmunity-driven inflammation of peripheral nerves that develops in 0.1% or less of individuals infected with *Campylobacter.*

MORPHOLOGY

Campylobacter, Shigella, Salmonella, and many other bacterial infections, including *Yersinia* and *E. coli,* all induce a similar microscopic picture, termed **acute self-limited colitis.** The histology of acute self-limited colitis includes prominent lamina propria and intraepithelial neutrophil infiltrates (Fig. 13.23A); **cryptitis** (neutrophil infiltration of the crypts) and **crypt abscesses** (crypts with accumulations of luminal neutrophils) may also be present. The preservation of crypt architecture in most cases is helpful in distinguishing these infections from inflammatory bowel disease (Fig. 13.23B). Specific diagnosis is primarily by stool culture, but molecular tests are becoming more widely available.

Clinical Features. Ingestion of as few as 500 *C. jejuni* organisms may cause disease after an incubation period of up to 8 days. Watery diarrhea, either acute or with onset after an influenza-like prodrome, is

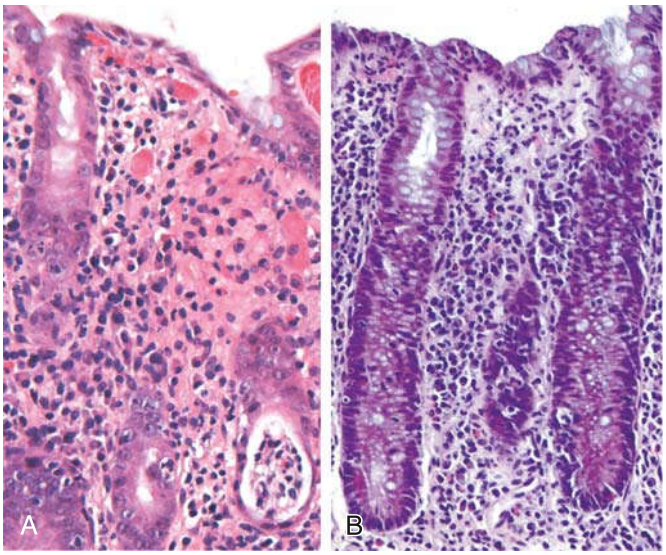


FIG. 13.23 Bacterial enterocolitis. (A) *Campylobacter jejuni* infection produces acute, self-limited colitis. Neutrophils can be seen within surface and crypt epithelium, and a crypt abscess is present *(lower right).* (B) Enteroinvasive *Escherichia coli* infection is similar to other acute, self-limited colitides. Note the maintenance of normal crypt architecture and spacing, despite abundant intraepithelial neutrophils.

the primary manifestation, and dysentery develops in 15% to 50% of patients. Patients may shed bacteria for 1 month or more after clinical resolution. The disease is usually self-limited and antibiotic therapy is generally not required. The fatality rate is low and mainly confined to older adults and those with HIV, in whom *C. jejuni* infection may lead to severe, debilitating illness.

Shigellosis

***Shigella* is one of the most common causes of bloody diarrhea (dysentery).** Shigellae are gram-negative, unencapsulated, nonmotile, facultative anaerobes that are highly transmissible by the fecal-oral route or through ingestion of contaminated water and food. The infective dose is fewer than 100 organisms, and each gram of stool contains as many as 10^9 organisms during acute phases of the disease.

In the United States, the incidence of *Shigella* infection has declined over the past several decades, but it remains an important cause of disease, particularly in underserved populations and among individuals living in poverty. The burden of disease is greater in lower-resource countries in which *Shigella* is endemic, particularly in children under 5 years of age, in whom it is an important cause of morbidity and mortality. Globally, it is estimated that there are approximately 170 million cases of *Shigella* annually, resulting in approximately 160,000 deaths.

Pathogenesis. *Shigella* organisms are resistant to the harsh acidic environment of the stomach, which partially explains the very low infective dose. Once in the intestine, organisms are taken up by M (microfold) epithelial cells, which are specialized for sampling and uptake of luminal antigens. After intracellular proliferation, the bacteria escape into the lamina propria. These bacteria then infect epithelial cells through the basolateral membranes, which express bacterial receptors. Alternatively, luminal organisms may elaborate factors that directly modulate epithelial tight junctions to expose basolateral bacterial receptors and enable binding. This may be mediated by virulence proteins, some of which are directly injected into the host cytoplasm. Certain *Shigella dysenteriae* serotypes also produce the Shiga toxin Stx, which inhibits eukaryotic protein synthesis and causes host cell death.

MORPHOLOGY

Shigella infections are most prominent in the rectosigmoid colon, but the ileum may also be involved, perhaps reflecting the abundance of M cells in the epithelium overlying Peyer patches. The histologic appearance in early cases is similar to other acute self-limited colitides. In more severe cases, the mucosa is hemorrhagic and ulcerated, and pseudomembranes may be present. Perhaps because of the tropism for M cells, which acts to localize the damage, aphthous-appearing ulcers similar to those in Crohn disease may also occur. Distinguishing *Shigella* infection from chronic inflammatory bowel disease is challenging, particularly if there is distortion of crypt architecture.

Clinical Features. ***Shigella* causes self-limited disease characterized by about 6 days of diarrhea, fever, and abdominal pain.** After an incubation period of 1 to 7 days, the initial watery diarrhea progresses to a dysenteric phase in approximately 50% of patients. Nausea and vomiting are notably absent and the diarrhea is commonly described as frequent, small volume, and bloody. Constitutional symptoms can persist for as long as 1 month. A subacute presentation may also occur in a subset of adults. Antibiotic treatment shortens the clinical course and reduces the duration over which organisms are shed in the stool, but antidiarrheal medications are contraindicated because they can prolong symptoms by delaying bacterial clearance.

Complications of *Shigella* infection are uncommon and include *reactive arthritis*, a triad of sterile arthritis, urethritis, and conjunctivitis that preferentially affects HLA-B27–positive men between 20 and 40 years of age. Hemolytic uremic syndrome, which is typically associated with *enterohemorrhagic Escherichia coli (EHEC),* may also occur after infection with shigellae that secrete Shiga toxin.

Escherichia coli

Escherichia coli are gram-negative bacilli that colonize the healthy gastrointestinal tract; most are nonpathogenic, but a subset cause human disease. The latter are classified according to morphology, pathogenesis, and in vitro behavior (see Table 13.4). Here we summarize their pathogenic mechanisms:

- *Enterotoxigenic E. coli (ETEC),* the principal cause of traveler's diarrhea, are spread by the fecal-oral route. They express a heat-labile toxin that is similar to cholera toxin and a heat-stable toxin that increases intracellular cyclic GMP, an alteration that also promotes the development of secretory diarrhea.
- *Enteropathogenic E. coli (EPEC)* are characterized by their ability to attach tightly to the enterocyte apical membranes and cause local loss (effacement) of microvilli. The bacterial proteins necessary for attachment and effacement are encoded by a large genomic pathogenicity island, the locus of enterocyte effacement. This locus also encodes proteins that inject bacterial effector proteins into the epithelial cell cytoplasm. EPEC can cause endemic diarrhea as well as diarrheal outbreaks, particularly in children younger than 2 years of age.
- *Enterohemorrhagic E. coli (EHEC)* are subclassified as O157:H7 and non-O157:H7 serotypes. Outbreaks of *E. coli* O157:H7 in high-resource countries have been associated with the consumption of milk, vegetables, and inadequately cooked ground beef. Both O157:H7 and non-O157:H7 serotypes produce Shiga-like toxins and can cause dysentery. They can also trigger hemolytic-uremic syndrome (Chapter 12).
- *Enteroinvasive E. coli (EIEC)* resemble *Shigella* bacteriologically but do not produce toxins. They invade the gut epithelial cells and cause a bloody diarrhea.
- *Enteroaggregative E. coli (EAEC)* attach to enterocytes by adherence fimbriae but do not invade. Although they cause symptoms by producing heat-labile toxin and Shiga-like toxins, histologic damage is minimal.

Salmonellosis

Salmonella species are divided into *Salmonella typhi,* the causative agent of typhoid fever, and nontyphoid *Salmonella* strains that cause gastroenteritis. Nontyphoid *Salmonella* infection, usually due to *Salmonella enteritidis*, is quite common: more than 1 million cases occur each year in the United States and result in approximately 2000 deaths. It is even more prevalent in lower resource parts of the world; worldwide, *Salmonella* is estimated to cause 550 million episodes of diarrheal illness each year, including 220 million cases in children. Infection is most common in young children and older adults, with a peak incidence in summer and fall. Transmission is usually through contaminated food, particularly raw or undercooked meat, poultry, eggs, and milk. Symptoms usually appear rapidly, within 8 to 72 hours of exposure, and typically consist of diarrhea, nausea, vomiting, fever, and abdominal cramping.

Pathogenesis. **Salmonellae possess virulence genes that, like *Shigella* and enteropathogenic *E. coli,* encode proteins capable of transferring bacterial proteins into M cells and enterocytes.** The

transferred proteins activate host cell Rho GTPases (regulators of cytoskeletal organization), thereby triggering actin rearrangement and bacterial uptake into phagosomes, where the bacteria proliferate. Salmonellae also secrete a molecule that induces epithelial release of a chemoattractant eicosanoid that draws neutrophils into the lumen and potentiates mucosal damage. Very few viable *Salmonella* organisms are necessary to cause infection in healthy individuals; reduced gastric acid, as in individuals with atrophic gastritis or those on acid-suppressive therapy, further reduces the size of the inoculum that is required to produce disease.

Typhoid Fever

Typhoid fever, also referred to as *enteric fever,* affects up to 30 million individuals worldwide each year. It is caused by *Salmonella typhi* and *Salmonella paratyphi*. Infection by *S. typhi* is more common in endemic areas, where children and adolescents are most often affected. By contrast, *S. paratyphi* predominates in lower-resource countries and travelers to those areas. Humans are the sole reservoir for *S. typhi* and *S. paratyphi,* and transmission occurs from person to person or via contaminated food or water. Gallbladder colonization may be associated with gallstones and a chronic carrier state. Acute infection is associated with anorexia, abdominal pain, bloating, nausea, vomiting, and bloody diarrhea followed by a short asymptomatic phase that gives way to bacteremia and fever with flulike symptoms. It is during this phase that detection of organisms by blood culture may prompt antibiotic treatment and prevent further disease progression. Cultures are positive in 90% of cases during the febrile phase. Without such treatment, the febrile phase is followed by up to 2 weeks of sustained high fevers with abdominal tenderness that may mimic appendicitis. *Rose spots,* small erythematous maculopapular lesions, are seen on the chest and abdomen. Systemic dissemination may cause *extraintestinal complications* including encephalopathy, meningitis, seizures, endocarditis, myocarditis, pneumonia, and cholecystitis.

Like *S. enteritidis, S. typhi* and *S. paratyphi* are taken up by M cells and then engulfed by mononuclear cells in the underlying lymphoid tissue. This causes Peyer patches in the terminal ileum to enlarge into plateaulike elevations up to 8 cm in diameter. Mucosal shedding creates oval ulcers oriented along the long axis of the ileum. However, unlike *S. enteritidis, S. typhi* and *S. paratyphi* can disseminate via lymphatic and blood vessels. This causes reactive hyperplasia of draining lymph nodes, in which bacteria-containing phagocytes accumulate. In addition, the red pulp of the spleen expands due to prominent phagocyte hyperplasia. Randomly scattered small foci of parenchymal necrosis with macrophage aggregates, termed *typhoid nodules,* may be found in the liver, bone marrow, and lymph nodes.

Pseudomembranous Colitis

Pseudomembranous colitis caused by *Clostridioides difficile* is an important cause of morbidity and mortality in the hospital setting. It can be placed in the categories of *antibiotic-associated colitis* or *antibiotic-associated diarrhea.* While antibiotic-associated diarrhea may also be caused by other organisms such as *Salmonella, C. perfringens* type A, or *S. aureus,* of these only *C. difficile* causes pseudomembrane formation. *C. difficile* is an anaerobic, gram-positive, spore-forming bacillus. It is likely that disruption of the normal colonic microbiota by antibiotics allows *C. difficile* overgrowth. Almost any antibiotic may be responsible; the most important factors for disease development are the frequency of antibiotic use and its effect on colonic microbiota. Toxins released by *C. difficile* cause the ribosylation of small GTPases and lead to disruption of the epithelial cytoskeleton, tight junction barrier loss, cytokine release, and apoptosis.

MORPHOLOGY

Fully developed *C. difficile*–associated colitis is accompanied by formation of **pseudomembranes** (Fig. 13.24A), made up of an adherent layer of inflammatory cells and debris at sites of colonic mucosal injury. The surface epithelium is denuded, and the superficial lamina propria contains a dense infiltrate of neutrophils and occasional fibrin thrombi within capillaries. Damaged crypts are distended by a mucopurulent exudate that "erupts" to the surface in a fashion reminiscent of a volcano (Fig. 13.24B).

Clinical Features. In addition to antibiotic exposure, risk factors for *C. difficile*–associated colitis include age greater than 65 years, use of proton pump inhibitors, hospitalization, and immunosuppression. The organism is particularly prevalent in hospitals; as many as 20% of hospitalized adults are colonized with *C. difficile* (a rate 10 times higher than in the general population), but most colonized patients are free of disease. Individuals with *C. difficile*–associated colitis usually present with watery diarrhea and abdominal cramping; dehydration, fever, and leukocytosis are seen in more severe cases. Fecal leukocytes and occult blood may be present, but grossly bloody diarrhea is rare. Diagnosis of *C. difficile*–associated colitis in symptomatic patients is made with a positive nucleic acid amplification test and a positive stool test for *C. difficile* toxin; patients with only a positive nucleic amplification test are considered to be colonized. Regimens of selected antibiotics are generally effective treatments, but antibiotic-resistant and hypervirulent *C. difficile* strains are increasingly common. In patients with severe or recurrent disease, fecal microbial transplantation may clear the infection, speaking to the role of a disturbed microbiome in the pathogenesis of *C. difficile* colitis.

Mycobacterial Infection

Mycobacterial species, including *M. tuberculosis, M. bovis,* and *M. avium* species, may infect the gastrointestinal tract primarily or

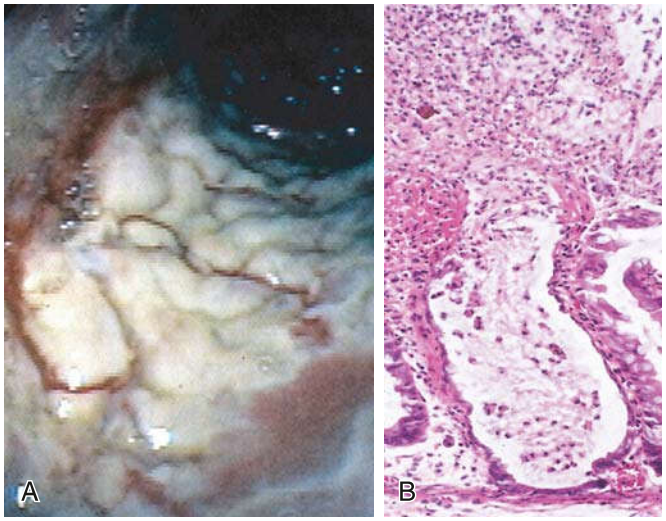


FIG. 13.24 *Clostridioides difficile* colitis. (A) The colon is coated by tan pseudomembranes composed of neutrophils, dead epithelial cells, and inflammatory debris (endoscopic view). (B) Typical pattern of neutrophils emanating from a crypt is reminiscent of a volcanic eruption.

as part of a disseminated infection. Gastrointestinal tuberculosis is rare in Western countries, except in individuals who are immunocompromised, in whom *M. avium* infection is most common. In areas in which intestinal disease is endemic, including India and Pakistan, *M. bovis* is the usual cause. Risk factors for *M. bovis* infection include consumption of unpasteurized dairy products. Clinical features of *M. bovis* infection are similar to those caused by *M. tuberculosis*. Any part of the gut may be affected, but infection is most common in the ileocecal region, which is involved in 75% of cases.

Intestinal mycobacterial infection usually induces a granulomatous host response with caseous necrosis, often associated with systemic symptoms such as fever and night sweats. With chronicity, fibrosis, mucosal ulceration, and bleeding may develop. The granulomas may extend through the full thickness of the bowel, potentially leading to perforation, stenoses, or strictures. Patients may also develop peritoneal dissemination with ascites. By contrast, because *M. avium* infection typically affects individuals who are immunocompromised, granulomas may be absent. Instead, *M. avium* tends to accumulate in macrophages within the lamina propria. In some cases, the sheetlike expansion of macrophages may compress lymphatic vessels in the small bowel, leading to malabsorption. Such cases may mimic Whipple disease, a rare infection caused by *Tropheryma whipplei*, clinically and morphologically but can be easily distinguished by acid-fast stains, as *M. avium* is acid-fast and *T. whipplei* is not.

Norovirus

Norovirus (also known as Norwalk virus) causes approximately half of all gastroenteritis outbreaks worldwide and is a common cause of sporadic gastroenteritis in high-resource countries. It is very infectious; ingestion or inhalation of as few as 10 viral particles can transmit the disease. Local outbreaks are usually related to contaminated food or water, whereas person-to-person transmission underlies most sporadic cases. Infections spread easily within schools, hospitals, and nursing homes as well as other settings where people are packed closely together, such as cruise ships. After a short incubation period, affected individuals develop nausea, vomiting (often severe), explosive watery diarrhea, and abdominal pain. Biopsy morphology is nonspecific. The disease is self-limited.

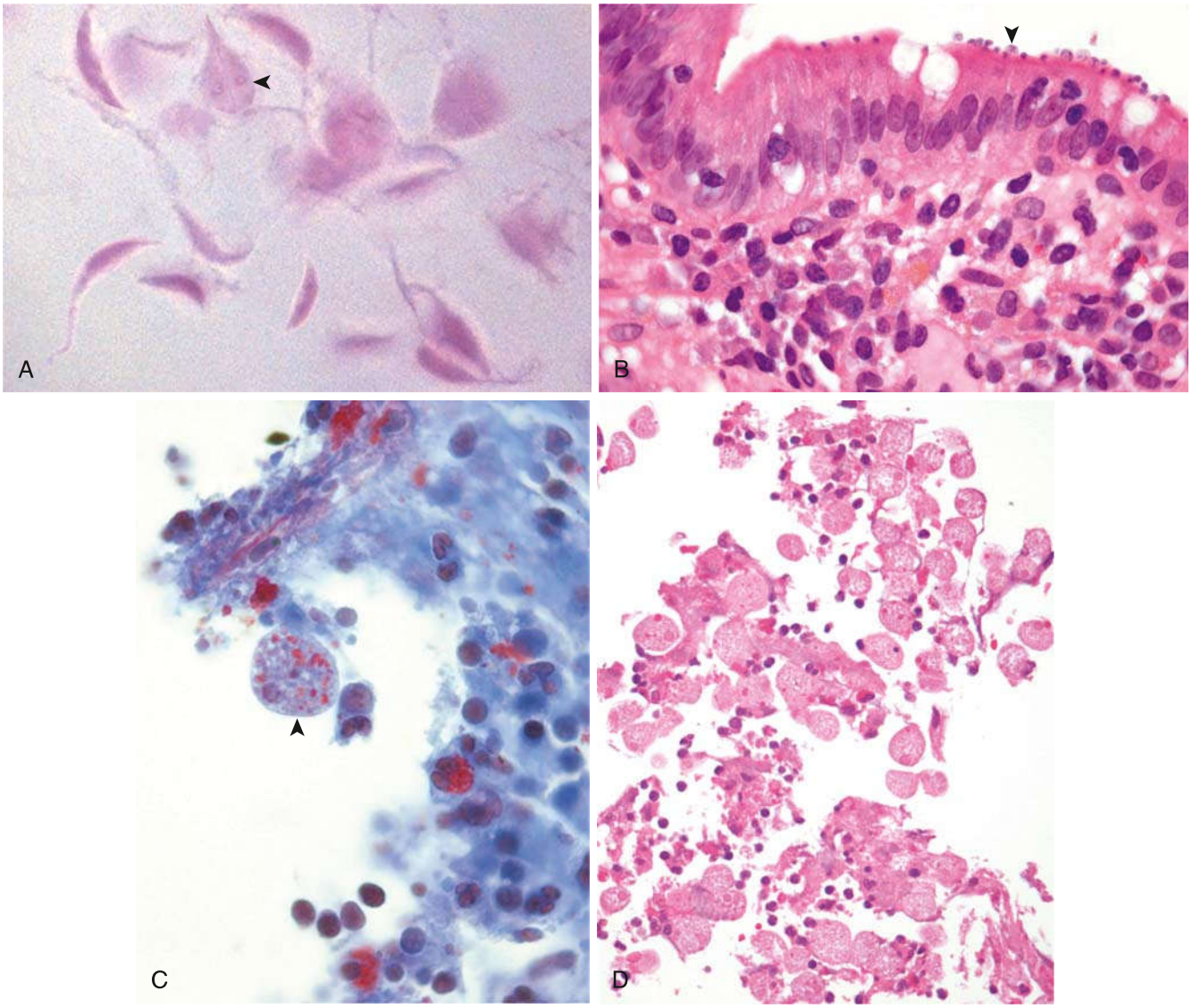
Rotavirus

Rotavirus is most common cause of severe childhood diarrhea and diarrhea-related deaths worldwide. Children between 6 and 24 months of age are most vulnerable. Protection in the first 6 months of life is due to the presence of antibodies to rotavirus in breast milk, while protection beyond 2 years of age is due to immunity that develops after the first infection. Like norovirus, it is highly infectious; the estimated minimal infective inoculum is only 10 viral particles. Rotavirus selectively infects and destroys mature (absorptive) enterocytes in the small intestine, and the villus surface is then first repopulated by immature secretory cells. This change in functional capacity results in loss of absorptive function and net secretion of water and electrolytes, compounded by an osmotic diarrhea from incompletely absorbed nutrients. Rotavirus infection becomes symptomatic after a short incubation period and manifests as several days of vomiting and watery diarrhea. Fortunately, the introduction of effective vaccines in 2006 has substantially reduced the burden of rotavirus disease in higher-resource countries and is beginning to have a positive impact in lower-resource countries as well.

Parasitic Disease

Parasitic disease and protozoal infections affect over half of the world's population on a chronic or recurrent basis. Intestinal parasites infesting humans include nematodes, such as the roundworms *Ascaris* and *Strongyloides*, hookworms, pinworms, cestodes (e.g., flatworms and tapeworms), trematodes (flukes), and protozoa. Some of the more common parasites are discussed briefly.

- *Ascaris lumbricoides.* This nematode is spread by human fecal-oral contamination and infects more than 1 billion individuals worldwide. Ingested eggs hatch in the intestine and the larvae penetrate the intestinal mucosa. From there, the larvae migrate via the splanchnic circulation to the liver, creating hepatic abscesses, and then through the systemic circulation to the lung, where they may cause pneumonitis. From the lung, larvae migrate up the trachea to the oropharynx and are swallowed, allowing them to again reach the intestine and mature into adult worms. Diagnosis is made by detection of eggs in the stool.
- *Strongyloides. Strongyloides* is widely found in tropical and subtropical regions worldwide. The larvae of *Strongyloides* live in ground soil contaminated by feces. The larvae can penetrate unbroken skin of a person walking barefoot. From here the blood stream carries them to the lungs, from which they travel to the trachea and the oropharynx, allowing them to be swallowed and to reach the intestines, where they mature into adult worms. Unlike other intestinal worms, whose life cycle requires an ovum or larval stage outside the human, the eggs of *Strongyloides* can hatch within the intestine and release larvae that penetrate the mucosa, creating a cycle of *autoinfection.* Hence, *Strongyloides* infection may persist for life and immunocompromised individuals may develop overwhelming infections. The disease is asymptomatic in immunocompetent individuals.
- *Necator americanus* and *Ancylostoma duodenale.* These hookworms infect more than 500 million individuals worldwide and cause significant morbidity. Infection is initiated by larval penetration of the skin. After further development in the lungs, the larvae migrate up the trachea and are swallowed. Once in the duodenum, the larvae mature and the adult worms attach to the mucosa, suck blood, and reproduce. Hookworms are a leading cause of iron deficiency anemia in low-resource parts of the world. The diagnosis is made by detecting the eggs in stool.
- *Giardia lamblia.* This flagellated protozoan, also referred to as *Giardia duodenalis* or *Giardia intestinalis,* is the most common pathogenic parasite in humans. It is spread by fecally contaminated water or food. Infection may occur after ingestion of as few as 10 cysts and is characterized by acute or chronic malabsorptive diarrhea. Because cysts are resistant to chlorine, *Giardia* organisms are endemic in unfiltered public and rural water supplies. In the acidic environment of the stomach, excystation occurs and trophozoites are released. Secretory IgA and mucosal IL-6 responses are important for clearance of *Giardia* infections. Individuals who are immunocompromised, particular those who fail to mount antibody responses, are often severely affected. *Giardia* evades immune clearance through continuous modification of its major surface antigen, variant surface protein, and can persist for months or years while causing intermittent symptoms. *Giardia* infection reduces expression of brush border enzymes, including lactase, and causes microvillous damage and apoptosis of small-intestinal epithelial cells. *Giardia* trophozoites are noninvasive and can be identified in duodenal biopsy specimens or stool preps by their characteristic pear shape (eFig. 13.7A). The diagnosis is made with tests on stool for *Giardia*-specific antigens or nucleic acids.



eFIG. 13.7 Gastrointestinal parasites. (A) *Giardia lamblia* organisms, seen in a stool prep. A characteristic trophozoite is marked by the *arrowhead*. (B) *Cryptosporidium* organisms *(arrowhead)* are seen coating the surface of epithelial cells in the small bowel. (C) *Entamoeba histolytica.* The *left panel* shows an amoeba that has phagocytosed red cells (*arrowhead*, trichrome stain). (D) shows a large cluster of amoebae within an area of colonic ulceration. (From Klatt EC: *Robbins and Cotran Atlas of Pathology*, ed 4, eFigs. 7.8, 7.11, and 7.10, Philadelphia, 2021, Elsevier.)

- *Cryptosporidium.* Along with *Giardia*, *Cryptosporidium* is one of the most common enteric parasites of humans. Infections in higher-resource countries have been linked to contaminated public water supplies, whereas in lower-resource countries infection is associated with poor sanitary conditions and overcrowding; in these areas, *Cryptosporidium* is an important cause of childhood diarrhea and is responsible for as many as 200,000 deaths per year, a toll that may be second only to rotavirus. The organisms are typically found on the surface of small bowel epithelial cells (see eFig. 13.7B) and characteristically lead to secretory diarrhea and malabsorption. The diagnosis is made by identifying the organisms in stool preps by morphology or with tests for *Cryptosporidium*-specific nucleic acids.
- *Entamoeba histolytica.* This protozoan causes amebiasis and is spread by fecal-oral transmission, primarily in low-resource parts of the world with poor sanitation. Areas with high rates of infection include Mexico, parts of Central and South America, Africa, and India. In some areas the prevalence of infection may be as high as 50%.

  Amebiasis most often affects the cecum and ascending colon (see eFig. 13.7C, D). Dysentery develops when the amebae attach to the colonic epithelium, induce apoptosis, invade crypts, and burrow laterally into the lamina propria. The resulting damage leads to recruitment of neutrophils and creates a flask-shaped ulcer with a narrow neck and broad base. Parasites penetrate splanchnic vessels and embolize to the liver to produce abscesses in about 40% of patients with amebic dysentery. *Amebic liver abscesses,* which may exceed 10 cm in diameter, have a scant inflammatory reaction at their margins and a shaggy fibrinous lining. The abscesses persist after the acute intestinal illness has passed, and on rare occasions the parasites may subsequently spread to the lung, heart, kidneys, or brain. Individuals with amebiasis may present with abdominal pain, bloody diarrhea, or weight loss. Occasionally, acute necrotizing colitis and megacolon occur, both of which are associated with significant mortality.

INFLAMMATORY INTESTINAL DISEASE

Sigmoid Diverticulitis

In general, the term diverticular disease is a misnomer, as it usually refers to acquired pseudodiverticula rather than true diverticula. Colonic diverticula are rare in individuals younger than 30 years of age but rise in incidence sharply thereafter; their prevalence approaches 50% in Western adult populations older than 60 years of age. Diverticula are generally multiple, a condition referred to as *diverticulosis.* This disease is much less common in lower-resource countries, probably because of dietary differences.

Pathogenesis. **Colonic diverticula develop due to elevated intraluminal pressure.** The location of diverticula is explained by the unique structure of the colonic muscularis propria, where nerves, arterial vasa recta, and their connective tissue sheaths penetrate the inner circular muscle coat to create discontinuities in the muscle wall. In other parts of the intestine, these gaps are reinforced by the external longitudinal layer of the muscularis propria, but in the colon, this muscle layer is discontinuous, being gathered into three bands termed *taeniae coli.* This creates a point of structural weakness through which the mucosa and submucosa may herniate outward in response to high intracolonic luminal pressures. Most diverticula occur in the narrowest part of the colon, the sigmoid, which experiences the highest pressures during peristaltic contractions. The pathogenesis also appears to involve exaggerated peristaltic contractions and spasmodic sequestration of bowel segments. The etiology of these abnormal contractions is unknown, but they are hypothesized to be associated with consumption of Western diets high in red meat and low in fiber. Other risk factors include a sedentary lifestyle, obesity, smoking, and use of certain medications, including steroids and opiates.

MORPHOLOGY

Colonic diverticula are small, flasklike outpouchings, usually 0.5 to 1 cm in diameter, that occur in a regular distribution between the taeniae coli (Fig. 13.25A and B). They are most common in the sigmoid colon, but other regions of the colon may also be affected. The diverticula have a thin wall composed of a flattened or atrophic mucosa, compressed submucosa, and attenuated muscularis propria—often, this last component is totally absent (Fig. 13.25C and D). Obstruction of diverticula and stasis of contents leads to inflammation, producing **diverticulitis** and peridiverticulitis. Because the wall of the diverticulum is supported only by the muscularis mucosa and a thin layer of subserosal adipose tissue, inflammation, increased pressure, and mucosal ulceration within an obstructed diverticulum may result in **perforation.** With or without perforation, recurrent diverticulitis may cause segmental colitis, fibrotic thickening in and around the colonic wall, or stricture formation.

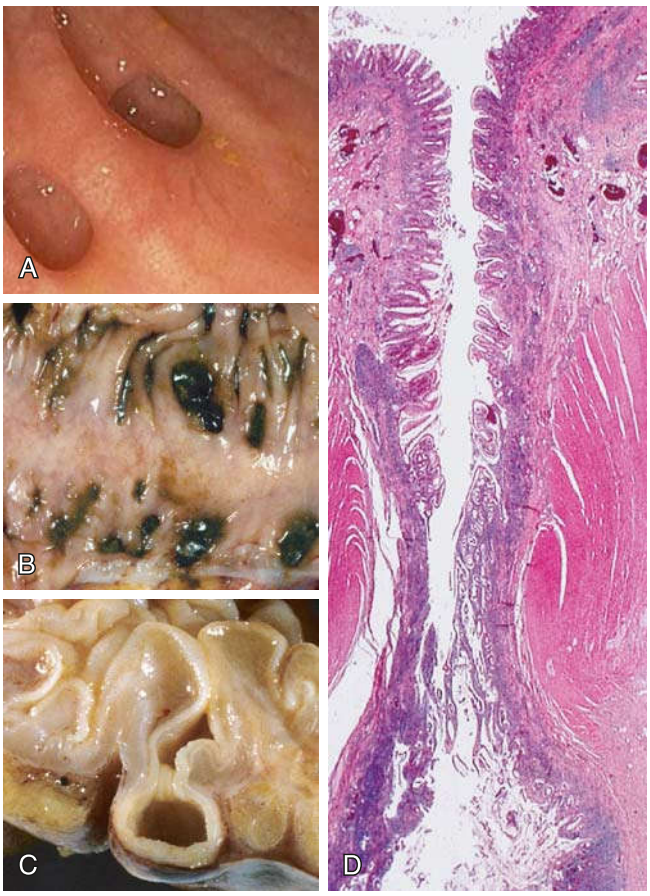


FIG. 13.25 Sigmoid diverticular disease. (A) Endoscopic view of two sigmoid diverticula. Compare to (B). (B) Gross examination of a resected sigmoid colon shows regularly spaced stool-filled diverticula. (C) Cross-section showing the outpouching of mucosa beneath the muscularis propria. (D) Low-power photomicrograph of a sigmoid diverticulum showing protrusion of the mucosa and submucosa through the muscularis propria. (Endoscopic image courtesy of Dr. Ira Hanan, The University of Chicago, Chicago, Illinois.)

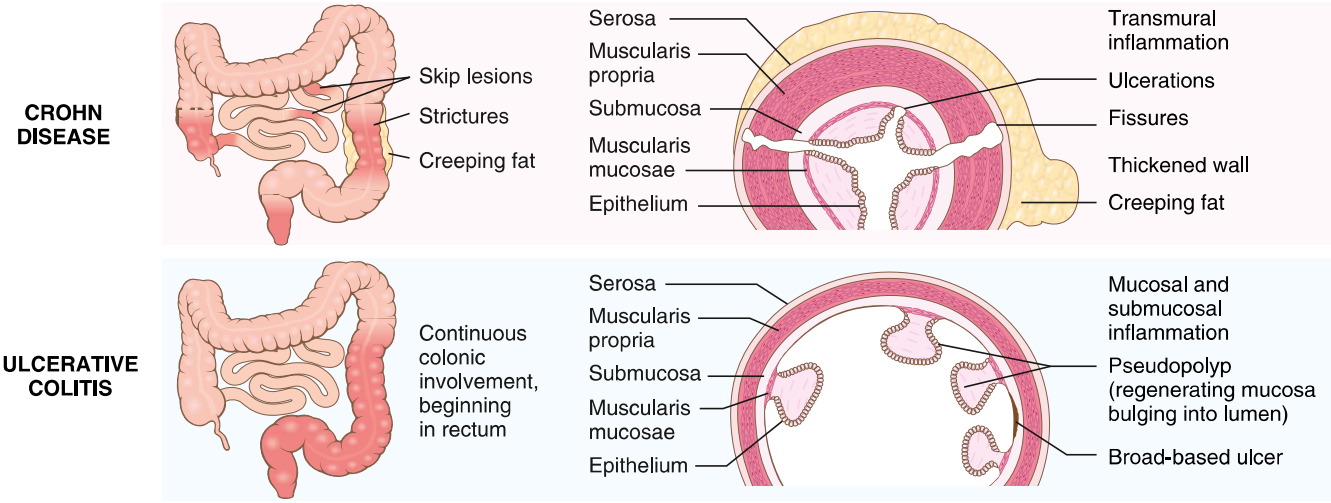


FIG. 13.26 Distribution and types of lesions in inflammatory bowel disease. The distinction between Crohn disease and ulcerative colitis is based primarily on morphology.

Clinical Features. Most individuals with diverticular disease remain asymptomatic throughout their lives. About 20% of those affected develop symptoms, including intermittent cramping, continuous lower abdominal discomfort, constipation, and diarrhea. Longitudinal studies have shown that while diverticula may regress early in their development, they often become larger and more numerous over time. Whether a high-fiber diet prevents such progression or protects against diverticulitis is unclear. Even when diverticulitis occurs, it most often resolves spontaneously or after antibiotic treatment, and relatively few patients require surgical intervention for perforation.

Inflammatory Bowel Disease

Inflammatory bowel disease (IBD) is a chronic inflammatory condition triggered by the host immune response to intestinal microbes in genetically predisposed individuals. IBD encompasses two entities, *Crohn disease* and *ulcerative colitis*. The distinction between Crohn disease and ulcerative colitis is based, in large part, on the distribution of affected sites and the morphologic expression of disease at those sites (Fig. 13.26; Table 13.5). Ulcerative colitis is limited to the colon and rectum and involves only the mucosa and submucosa. By contrast, Crohn disease, also referred to as *regional enteritis* (because of frequent ileal involvement), may involve any area of the gastrointestinal tract and often produces transmural inflammation.

Epidemiology. Both ulcerative colitis and Crohn disease frequently present in adolescents or in young adults. Worldwide, incidence varies according to a number of factors including population density (rural versus city), the extent of industrialization, and geography, as IBD is more prevalent in higher latitude locations. IBD appears to be increasing in incidence, including in regions in which the prevalence was historically low. One proposed explanation is the *hygiene hypothesis,* first applied to asthma, which suggests that childhood and even prenatal exposure to environmental microbes stimulates the immune system in a way that prevents excessive reactions. Applied to IBD, it suggests that a reduced frequency of enteric infections early in life due to improved hygiene results in inadequate development of regulatory processes that limit mucosal immune responses. While attractive and commonly stated, firm evidence is lacking and hence the increasing incidence of IBD remains mysterious.

Pathogenesis. **IBD appears to result from the combined effects of alterations in host interactions with intestinal microbiota, intestinal epithelial dysfunction, and aberrant mucosal immune responses.**

Table 13.5 Features of Crohn Disease and Ulcerative Colitis

Feature	Crohn Disease	Ulcerative Colitis
Morphology		
Bowel region affected	Ileum ± colon	Colon only
Rectal involvement	Sometimes	Always
Distribution	Skip lesions	Diffuse
Stricture	Yes	Rare
Bowel wall appearance	Thick	Thin
Inflammation	Transmural	Limited to mucosa and submucosa
Pseudopolyps	Moderate	Marked
Ulcers	Deep, knifelike	Superficial, broad-based
Lymphoid reaction	Marked	Moderate
Fibrosis	Marked	Mild to none
Serositis	Yes	No
Granulomas	Yes (~35%)	No
Fistulas/sinuses	Yes	No
Clinical		
Perianal fistula	Yes (in colonic disease)	No
Fat/vitamin malabsorption	Yes	No
Malignant potential	Yes	Yes
Recurrence after surgery	Common	No
Toxic megacolon	No	Yes

NOTE: Not all features may be present in a single case.

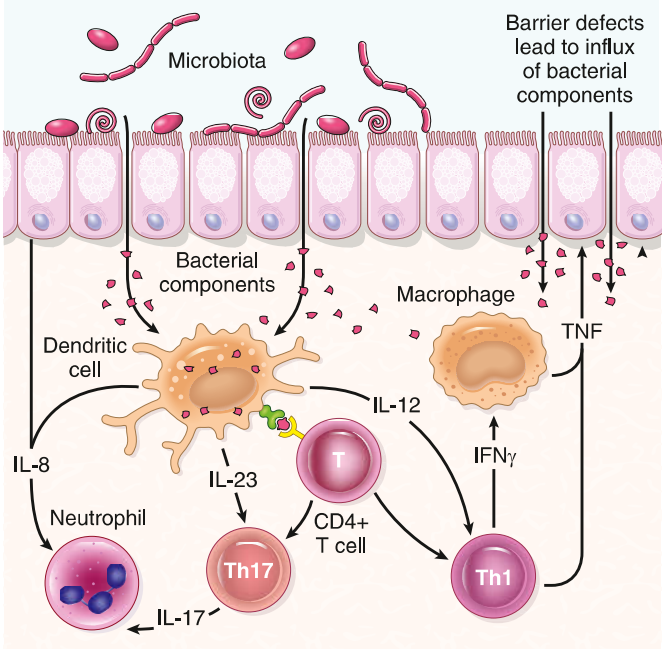


FIG. 13.27 A model of pathogenesis of inflammatory bowel disease (IBD). *IFN-γ*, Interferon gamma; *TNF*, tumor necrosis factor. See text for details.

This view is supported by epidemiologic, genetic, and clinical studies as well as data from laboratory models of IBD (Fig. 13.27). The associated factors can be broken down into four categories, as follows:

- *Genetic factors.* Risk for disease is increased when there is an affected family member. In Crohn disease, the concordance rate for monozygotic twins is approximately 50%. By contrast, concordance of monozygotic twins for ulcerative colitis is approximately 20%, suggesting that genetic factors are less dominant in this form of IBD.
  - Molecular linkage analyses of affected families have identified *NOD2* (nucleotide oligomerization binding domain 2) as a susceptibility gene in Crohn disease. *NOD2* encodes a protein that binds to intracellular bacterial peptidoglycans and subsequently activates NF-κB. Some studies suggest that the disease-associated form of NOD-2 is ineffective at defending against intestinal bacteria. This defect allows increased numbers of bacteria to penetrate past the epithelium into the intestinal wall, where they trigger inflammatory reactions. However, disease develops in less than 10% of individuals carrying specific *NOD2* polymorphisms, and these polymorphisms are uncommon in patients of African and Asian descent with Crohn disease; therefore, other factors are also important.
  - The search for IBD-associated genes using genome-wide association studies (GWAS) and other approaches have identified over 200 genes with nucleotide variants that are associated with IBD. In addition to *NOD2* (discussed previously), multiple variants associated with risk lie in autophagy-related genes. The latter encode components of the autophagosome pathway and, like NOD-2, are involved in host cell responses to intracellular bacteria, supporting the hypothesis that inadequate defense against luminal bacteria is important in the pathogenesis of IBD. However, none of these genes is associated with ulcerative colitis, and the associations hold mainly for people of European descent.
- *Abnormal mucosal immune responses.* How altered mucosal immunity contributes to the pathogenesis of ulcerative colitis and Crohn disease is still being deciphered, but it is notable that immunosuppressive and immunomodulatory agents are mainstays of IBD therapy, supporting a role for inappropriately sustained immune responses in these diseases. It is proposed that the initial trigger is the presentation of microbial antigens to CD4+ helper T cells, which are induced to differentiate into Th1 and Th17 cells by cytokines such as IL-12 and IL-23 (see Fig. 13.27). These then activate macrophages, recruit neutrophils, and release proinflammatory cytokines such as TNF. The presence of microbial antigens in the lamina propria may stem from defects in epithelial barrier function that are exacerbated by inflammation (see below). Whether the disease takes the form of ulcerative colitis or Crohn disease likely depends upon differences in the cytokine milieu that is induced in response to microbial products. Defects in regulatory T cells, especially the IL-10-producing subset that dampens the immune response, may also contribute to the inflammation; consistent with this idea, rare individuals with mutations in the IL-10 or IL-10 receptor genes are susceptible to severe, early-onset colitis. Other data suggest that mucosal production of the Th2-derived cytokine IL-13 is increased in ulcerative colitis, and, to a lesser degree, in Crohn disease. Thus, defective immune regulation likely contributes to the chronic inflammation in IBD. Notably, patients with IBD benefit from treatment with antibodies that inhibit TNF or IL-12/IL-23, supporting a role for these cytokines in its pathogenesis. The contribution of true autoimmunity is unclear, and no initiating or target self antigens have been definitively identified.
- *Epithelial defects.* A variety of epithelial defects have been described in Crohn disease, ulcerative colitis, or both. For example, defects in intestinal epithelial tight junction barrier function occur in patients with Crohn disease and a subset of their healthy first-degree relatives. This barrier dysfunction cosegregates with disease-associated *NOD2* polymorphisms, and experimental models demonstrate that barrier dysfunction can activate innate and adaptive mucosal immunity and sensitize subjects to disease. Interestingly, Paneth cell granules, which contain antimicrobial peptides that affect the composition of luminal microbiota, are abnormal in some patients with Crohn disease, suggesting a mechanism in which a defective "crosstalk" between the epithelium and the microbiota contributes to disease pathogenesis.
- *Changes in the microbiota.* The quantity of microbial organisms in the gastrointestinal lumen is enormous, amounting to as many as 10^{12} organisms/mL of fecal material in the colon (50% of fecal mass). There is significant interindividual variation in the composition of the microbiota, which is modified by diet and disease. Microbial transfer may either promote or reduce disease in animal models of IBD, and clinical trials suggest that probiotic (beneficial) bacteria or even fecal microbial transplants from healthy individuals may benefit patients with IBD.

We will discuss next the morphologic and clinical features of each of the two forms of IBD.

Crohn Disease

MORPHOLOGY

The most common sites of Crohn disease at presentation are the **terminal ileum, ileocecal valve,** and **cecum.** Disease is limited to the small intestine in about 40% of cases; the small intestine and the colon are both involved in 30% of patients; and the remainder of cases are characterized by colonic involvement only. Infrequently, Crohn disease may involve the

esophagus or stomach. The presence of **skip lesions** (multiple, separate, sharply delineated areas of disease interspersed with areas of normal mucosa) is characteristic of Crohn disease and distinguishes Crohn disease from ulcerative colitis. Strictures are common (Fig. 13.28A).

The earliest lesion is the **aphthous ulcer;** these may progress, become multiple, and coalesce into elongated, serpentine ulcers oriented along the long axis of the bowel. Edema and loss of normal mucosal folds are common. Sparing of interspersed mucosa results in a coarsely textured, **cobblestone** appearance, in which diseased tissue is depressed below the level of normal mucosa (Fig. 13.28B). Those with colon involvement are susceptible to developing **toxic megacolon** before fibrosis develops, which prevents dilatation of the colon. **Fissures** frequently develop between mucosal folds and may extend deeply to become sites of perforation or fistula tracts. The intestinal wall is thickened as a consequence of transmural edema, inflammation, submucosal fibrosis, and hypertrophy of the muscularis propria, all of which contribute to formation of **strictures.** In cases with extensive transmural disease, mesenteric fat frequently becomes adherent to the serosal surface **(creeping fat)** (Fig. 13.28C).

The microscopic features of active Crohn disease include abundant neutrophils that infiltrate and damage crypt epithelium. Clusters of neutrophils within a crypt are referred to as a **crypt abscess** and are often associated with crypt destruction. Ulceration is common in Crohn disease, and there may be an abrupt transition between ulcerated and normal mucosa. Repeated cycles of crypt destruction and regeneration lead to **distortion of mucosal architecture;** the normally straight and parallel crypts take on branching shapes and unusual orientations to one another (Fig. 13.29A). Epithelial metaplasia, another consequence of chronic injury, often takes the form of gastric antral-appearing glands (pseudopyloric metaplasia). **Paneth cell metaplasia** may occur in the left colon, where Paneth cells are normally absent. These architectural and metaplastic changes may persist, even when active inflammation has resolved. Mucosal atrophy, with loss of crypts, may follow years of disease. **Noncaseating granulomas** (Fig. 13.29B), a hallmark of Crohn disease, are found in approximately 35% of cases and may arise in areas of active disease or uninvolved regions in any layer of the intestinal wall (Fig. 13.29C). Granulomas also may be found in mesenteric lymph nodes and sometimes even the skin, representing extraintestinal manifestations of Crohn disease. Importantly, the absence of granulomas does not preclude the diagnosis.

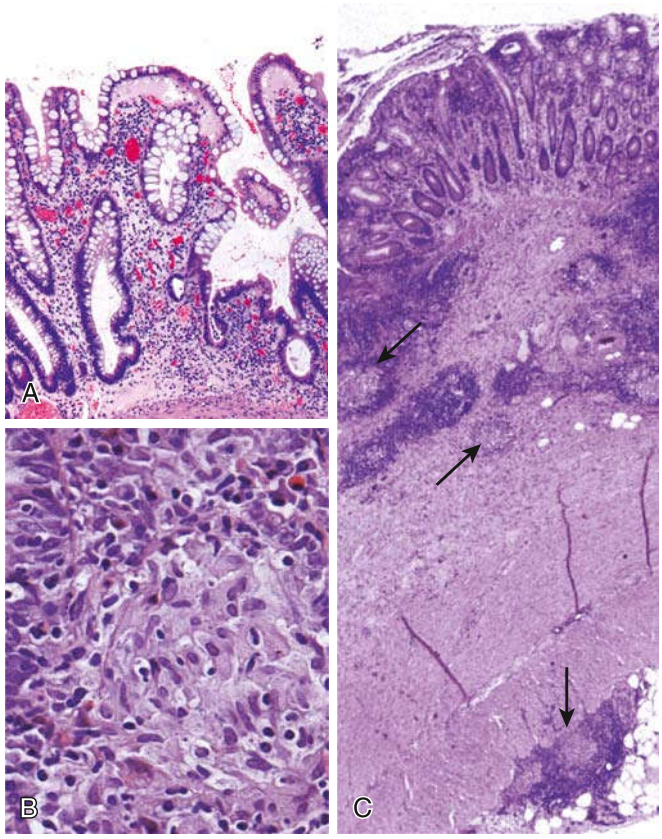


FIG. 13.29 Microscopic pathology of Crohn disease. (A) Haphazard crypt organization results from repeated injury and regeneration. (B) Noncaseating granuloma. (C) Transmural Crohn disease with markedly thickened wall and submucosal and serosal granulomas *(arrows).*

Clinical Features. The clinical manifestations of Crohn disease are extremely variable. In most patients, disease begins with intermittent attacks of relatively mild diarrhea, fever, and abdominal pain. Approximately 20% of patients present acutely with right lower quadrant pain and fever, features that mimic acute appendicitis or bowel perforation. Patients with colonic involvement may present with bloody diarrhea and abdominal pain, suggesting a colonic infection. Periods of disease activity are typically interrupted by asymptomatic intervals that last for weeks to many months. Disease reactivation may be associated with a variety of external triggers, including physical or emotional stress, specific dietary items, NSAID use, and cigarette smoking.

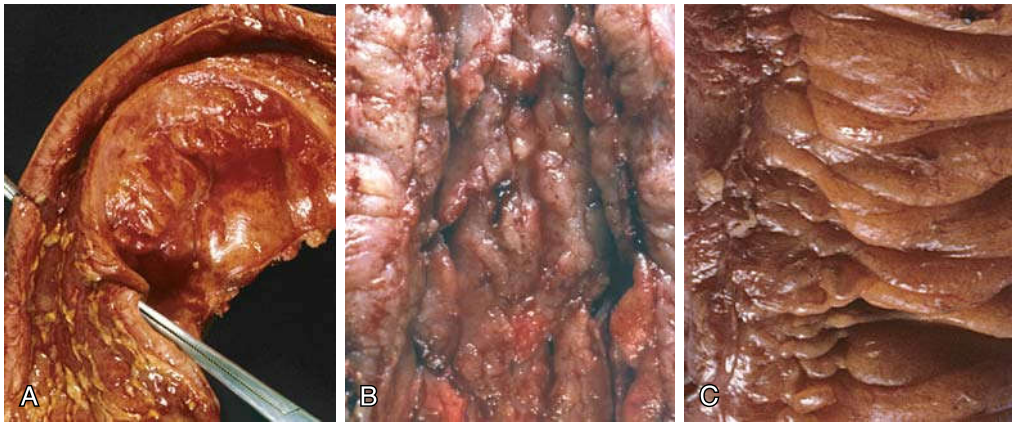


FIG. 13.28 Gross pathology of Crohn disease. (A) Small-intestinal stricture. (B) Linear mucosal ulcers and thickened intestinal wall. (C) Creeping fat.

Iron deficiency anemia due to blood loss may develop in individuals with colonic disease, while extensive small bowel disease and associated malabsorption may result in hypoalbuminemia, generalized nutrient malabsorption, or vitamin B_{12} deficiency. Fibrosing strictures, particularly of the terminal ileum, are common and require surgical resection. Disease often recurs at the site of anastomosis, and as many as 40% of patients require additional resections within 10 years. Fistulas can develop between loops of bowel and may also involve the urinary bladder, vagina, and abdominal or perianal skin. Perforations and peritoneal abscesses may also occur.

Extraintestinal manifestations of Crohn disease include uveitis, migratory polyarthritis, sacroiliitis, ankylosing spondylitis, erythema nodosum, and clubbing of the fingertips, any of which may develop before intestinal disease is recognized. Pericholangitis and primary sclerosing cholangitis may occur in those with Crohn disease but are more common in patients with ulcerative colitis. As discussed later, the risk for development of colonic and small intestinal adenocarcinoma is increased in patients with long-standing colonic Crohn disease.

Ulcerative Colitis

MORPHOLOGY

Ulcerative colitis always involves the rectum and extends proximally in a continuous fashion to involve part or all of the colon (Fig. 13.30A). Skip lesions are not seen. Disease of the entire colon is termed **pancolitis** (Fig. 13.30B). Disease limited to the rectum or rectosigmoid may be referred to descriptively as **ulcerative proctitis** or **ulcerative proctosigmoiditis.** The small intestine is normal, although mild mucosal inflammation of the distal ileum **(backwash ileitis)** may be present in severe cases.

On gross evaluation, involved colonic mucosa may be slightly red and granular appearing or exhibit extensive **broad-based ulcers.** The transition between diseased and uninvolved colon is abrupt (Fig. 13.30C). Ulcers are aligned along the long axis of the colon but typically do not replicate the serpentine ulcers of Crohn disease. Isolated islands of regenerating mucosa often bulge into the lumen to create small elevations, termed **pseudopolyps.** Chronic disease may lead to **mucosal atrophy** and a flat, smooth mucosal surface lacking normal folds. Unlike in Crohn disease, mural thickening is absent, the serosal surface is normal, and strictures do not occur. However, inflammation and inflammatory mediators may damage the muscularis propria and disturb neuromuscular function, leading to colonic dilation and **toxic megacolon,** which carries a significant risk for perforation. Patients with toxic megacolon are quite sick with fever, tachycardia, hypotension.

Histologic features of mucosal disease in ulcerative colitis are similar to colonic Crohn disease and include inflammatory infiltrates, crypt abscesses, crypt distortion, and epithelial metaplasia. However, skip lesions are absent, and inflammation is generally limited to the mucosa and superficial submucosa (Fig. 13.30D). In severe cases, mucosal damage may be accompanied by ulcers that extend more deeply into the submucosa, but the muscularis propria is rarely involved. Submucosal fibrosis, mucosal atrophy, and distorted mucosal architecture remain as residua of healed disease, but the histologic pattern may also revert to near normal after prolonged remission. Granulomas are not present.

Some extraintestinal manifestations of ulcerative colitis overlap with those of Crohn disease, including migratory polyarthritis, sacroiliitis, ankylosing spondylitis, uveitis, skin lesions, pericholangitis, and primary sclerosing cholangitis.

Clinical Features. Ulcerative colitis is a relapsing disorder characterized by attacks of bloody, stringy, often mucoid diarrhea associated with lower abdominal pain and cramps that are temporarily relieved by defecation. These symptoms may persist for days, weeks, or months before they subside, and occasionally the initial attack may be sufficiently severe to constitute a medical or surgical emergency. More than half of patients have mild disease, but almost all experience at least one relapse during a 10-year period. Colectomy cures intestinal disease, but extraintestinal manifestations may persist.

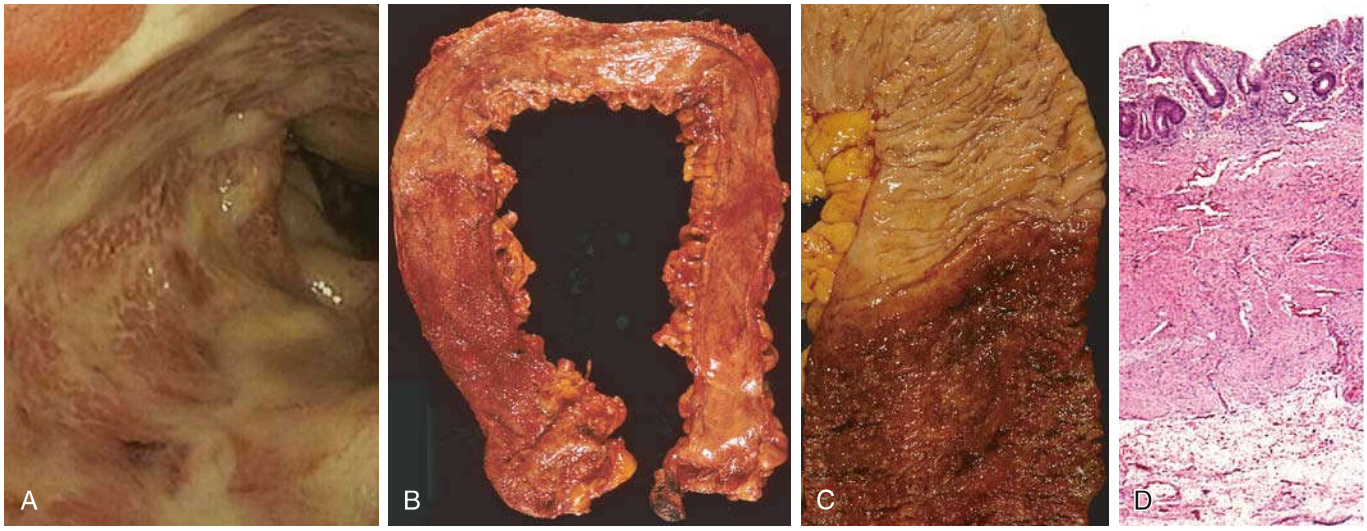


FIG. 13.30 Pathology of ulcerative colitis. (A) Endoscopic view of severe ulcerative colitis with ulceration and adherent mucopurulent material. (B) Total colectomy with pancolitis showing active disease, with red, granular mucosa in the cecum *(left)* and smooth, atrophic mucosa distally *(right).* (C) Sharp demarcation between active ulcerative colitis *(bottom)* and normal *(top).* (D) This full-thickness histologic section shows that disease is limited to the mucosa. Compare with Fig. 13.29C. (Endoscopic image courtesy of Dr. Ira Hanan, The University of Chicago, Chicago, Illinois.)

The factors that trigger ulcerative colitis are not known; infectious enteritis precedes disease onset in some cases. The onset of symptoms may occur shortly after smoking cessation in some patients, and smoking may partially relieve symptoms. However, studies of nicotine as a therapeutic agent have been disappointing.

Therapy is focused on antiinflammatory agents. Individuals with mild disease are often treated with glucocorticoids or aminosalicylates. Patients with more severe disease often benefit from treatment with biologic agents, such as antibodies that bind and neutralize TNF or IL-12/IL-23.

Colitis-Associated Neoplasia

One of the most serious long-term complications of ulcerative colitis and colonic Crohn disease is the development of adenocarcinoma. This risk is confined to the colon in patients with ulcerative colitis, but those with Crohn disease are also at increased risk of small intestinal adenocarcinoma. This process begins as dysplasia, which, just as in Barrett esophagus and chronic gastritis, is part of the progression to carcinoma. The risk for development of dysplasia is related to several factors:

- *Duration of disease.* Risk increases beginning 8 to 10 years after disease initiation.
- *Extent of involvement.* Patients with involvement of the entire colon are at greater risk than those with only partial involvement.
- *Inflammation.* Greater frequency and severity of active inflammation (characterized by the presence of neutrophils) may increase risk. This is another example of the enabling effect of inflammation on carcinogenesis (Chapter 6).

To facilitate early detection of neoplasia, patients are typically enrolled in colonoscopic surveillance programs approximately 8 years after diagnosis of IBD. An important exception to this approach is in patients with primary sclerosing cholangitis, who are at markedly greater risk for development of dysplasia; in this instance, surveillance is generally initiated at the time of diagnosis. Surveillance requires regular and extensive mucosal biopsy, which is costly and invasive. In many cases, dysplasia occurs in flat areas of mucosa that do not appear abnormal by eye. Thus, advanced endoscopic imaging techniques are being developed to try to improve the detection of early dysplastic changes.

COLONIC POLYPS AND NEOPLASTIC DISEASE

Polyps are most common in the colon and may also occur in the esophagus, stomach, or small intestine. Those without stalks are referred to as *sessile.* As sessile polyps enlarge, traction on the luminal protrusion may create a stalk. Polyps with stalks are termed *pedunculated.* In general, intestinal polyps can be classified as *nonneoplastic* or *neoplastic.* The most common neoplastic polyp is the adenoma, which has the potential to progress to cancer. Nonneoplastic colonic polyps can be further classified as inflammatory, hamartomatous, or hyperplastic.

Inflammatory Polyps

Inflammatory polyps are associated with the *solitary rectal ulcer syndrome.* Patients present with the triad of rectal bleeding, mucus discharge, and an inflammatory polyp on the anterior rectal wall. The underlying cause is impaired relaxation of the anorectal sphincter, creating a sharp angle at the anterior rectal shelf. This leads to recurrent abrasion and ulceration of the overlying rectal mucosa. With repeated cycles of injury and healing, a polypoid mass forms that is composed of inflamed and reactive mucosal tissue.

Hamartomatous Polyps

Hamartomatous polyps occur sporadically and as components of various genetically determined or acquired syndromes (Table 13.6). As described in Chapter 6, hamartomatous polyps are disorganized, tumorlike growths composed of mature cell types normally present at

Table 13.6 Gastrointestinal (GI) Polyposis Syndromes

Syndrome	Mean Age at Presentation (Years)	Mutated Gene(s)	GI Lesions	Selected Extragastrointestinal Manifestations
Peutz-Jeghers syndrome	10–15	*LKB1/STK11*	Arborizing polyps—small intestine > colon > stomach; colonic adenocarcinoma	Mucocutaneous pigmentation; increased risk for thyroid, breast, lung, pancreas, gonadal, and bladder cancers
Juvenile polyposis	<5	*SMAD4, BMPR1A*	Juvenile polyps; increased risk for gastric, small-intestinal, colonic, and pancreatic adenocarcinoma	Pulmonary arteriovenous malformations, digital clubbing
Cowden syndrome	<15	*PTEN*	Hamartomatous polyps, lipomas, ganglioneuromas, inflammatory polyps; increased risk for colon cancer	Benign skin tumors, benign and malignant thyroid and breast lesions
Tuberous sclerosis	Infancy to adulthood	*TSC1, TSC2*	Hamartomatous polyps (rectal)	Facial angiofibroma, cortical tubers, renal angiomyolipoma
Familial adenomatous polyposis (FAP)				
Classic FAP	10–15	*APC*	Multiple adenomas	Congenital RPE hypertrophy
Attenuated FAP	40–50	*APC*	Multiple adenomas	
Gardner syndrome	10–15	*APC*	Multiple adenomas	Osteomas, desmoids, skin cysts
Turcot syndrome	10–15	*APC*	Multiple adenomas	CNS tumors, medulloblastoma

CNS, Central nervous system; *RPE,* retinal pigment epithelium.

the site at which the polyp develops. Hamartomatous polyposis syndromes are rare but must be recognized because of associated intestinal and extraintestinal manifestations, the need to screen family members, and, in some instances, an increased risk of cancer.

Juvenile Polyps

Juvenile polyps are the most common type of hamartomatous polyp. They may be sporadic or syndromic. Sporadic juvenile polyps are usually solitary, whereas *juvenile polyposis*, an autosomal dominant syndrome, may be associated with dozens of polyps. Most juvenile polyps occur in children younger than 5 years. They are characteristically located in the rectum and usually manifest with rectal bleeding. In some cases, prolapse occurs and the polyp protrudes through the anal sphincter. Dysplasia occurs in a small proportion of juvenile polyps, and juvenile polyposis is associated with an increased risk for adenocarcinoma of the colon and other sites (see Table 13.6). This is not surprising, as juvenile polyposis is associated with germline mutations in genes that encode components of the TGFβ/BMP signaling pathway (e.g., *SMAD4*), which is frequently mutated in certain cancers. Colectomy may be required to limit the hemorrhage associated with polyp ulceration in juvenile polyposis.

MORPHOLOGY

Sporadic and syndromic juvenile polyps are similar in appearance. They are typically pedunculated, smooth-surfaced, reddish lesions less than 3 cm in diameter that display characteristic cystic spaces on cut sections. These spaces are dilated glands filled with mucin and inflammatory debris (Fig. 13.31A).

Peutz-Jeghers Syndrome

Peutz-Jeghers syndrome is a rare autosomal dominant disorder defined by the presence of multiple gastrointestinal hamartomatous polyps, mucocutaneous hyperpigmentation, and an increased risk for development of numerous malignancies. Included among the malignancies are cancers of the colon, pancreas, breast, lung, ovaries, uterus, and testes, as well as other unusual neoplasms. Germline loss-of-function mutations in the *LKB1/STK11* gene are present in approximately half of the patients with the familial form of Peutz-Jeghers syndrome. *LKB1/STK11* encodes a tumor suppressive protein kinase that regulates cellular metabolism, another example of the link between altered metabolism, abnormal cell growth, and cancer risk.

MORPHOLOGY

Hamartomatous polyps are most common in the small intestine, although they may also occur in the stomach, colon, and, rarely, in the bladder and lungs. On gross evaluation, the polyps are large and pedunculated with a lobulated contour. Histologic examination demonstrates a characteristic arborizing network of connective tissue, smooth muscle, lamina propria, and glands lined by normal-appearing intestinal epithelium (Fig. 13.31B).

Hyperplastic Polyps

Colonic hyperplastic polyps are common epithelial proliferations that typically occur in the sixth and seventh decades of life. Their pathogenesis is incompletely understood, but their formation is thought to result from decreased epithelial cell turnover and delayed shedding of surface epithelial cells, leading to a "pileup" of goblet cells. Although these lesions have no malignant potential, they are difficult to distinguish endoscopically from sessile serrated adenomas, histologically similar lesions that do have malignant potential, and so are frequently biopsied.

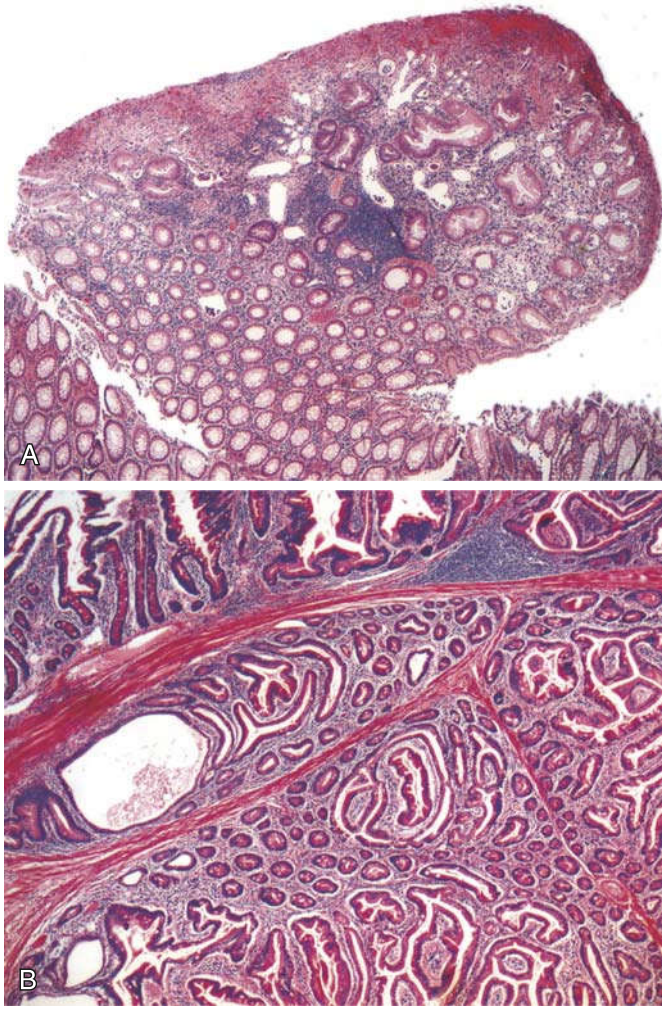


FIG. 13.31 Hamartomatous polyps. (A) Juvenile polyp. Note the surface erosion and cystically dilated crypts filled with mucus, neutrophils, and debris. (B) Peutz-Jeghers polyp. Complex glandular architecture and bundles of smooth muscle help to distinguish Peutz-Jeghers polyps from juvenile polyps.

MORPHOLOGY

Hyperplastic polyps are most commonly found in the left colon and are typically less than 5 mm in diameter. They are smooth, nodular protrusions of the mucosa, often on the crests of mucosal folds. They may occur singly but more frequently are multiple, particularly in the sigmoid colon and rectum. Histologically, they are composed of an expanded population of mature goblet and absorptive cells, crowding of which creates a serrated surface, which is the morphologic hallmark of these lesions (Fig. 13.32).

Adenomas

The most common and clinically important neoplastic polyps are colonic adenomas, which are precursor lesions of colorectal

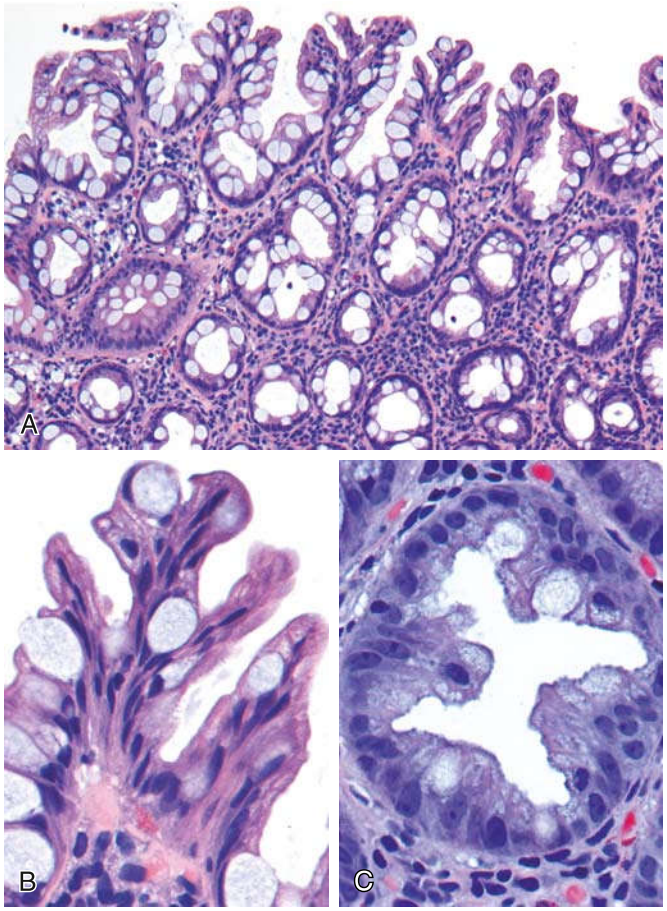


FIG. 13.32 Hyperplastic polyp. (A) Polyp surface with irregular tufting of epithelial cells. (B) Tufting results from epithelial overcrowding. (C) Epithelial crowding produces a serrated architecture when glands are cut in cross-section.

adenocarcinomas. There is no sex predilection, and they are present in nearly 50% of adults living in the Western world over 50 years of age. Because these polyps are precursors of colorectal cancer, current recommendations are that all adults in the United States undergo screening colonoscopy starting at 45 years of age. Since individuals with a family history are at risk for developing colon cancer earlier in life, they are typically screened at least 10 years before the youngest age at which a relative was diagnosed. While adenomas are less common in Asia, their frequency has risen (in parallel with an increasing incidence of colorectal adenocarcinoma) as Western diets and lifestyles become more common.

MORPHOLOGY

Typical adenomas range from 0.3 to 10 cm in diameter and can be **pedunculated** (Fig. 13.33A) or **sessile,** with both types having a velvety texture (Fig. 13.33B) and bumpy surface due to the abnormal epithelial growth pattern. Histologically, the cytologic hallmark is **epithelial dysplasia** (Fig. 13.34C), marked by nuclear hyperchromasia, elongation, and stratification. These changes are most easily appreciated at the surface of the adenoma because the epithelium fails to mature as cells migrate from the crypt. Pedunculated adenomas have slender fibromuscular stalks (Fig. 13.33C) containing prominent blood vessels derived from the submucosa. The stalk is usually covered by nonneoplastic epithelium, but dysplastic epithelium is sometimes present.

Adenomas may be classified as **tubular, tubulovillous,** or **villous** based on their architecture. Tubular adenomas tend to be small, pedunculated polyps composed of small, rounded, or tubular glands (Fig. 13.34A). By contrast, villous adenomas, which are often larger and sessile, are covered by slender villi (Fig. 13.34B). Tubulovillous adenomas have a mixture of tubular and villous elements. Although foci of invasion are more frequent in villous adenomas than in tubular adenomas, villous architecture alone does not increase cancer risk which is related to size, as discussed below.

The histologic features of **sessile serrated adenomas** overlap with those of hyperplastic polyps and these lesions typically lack dysplasia (Fig. 13.34D). Nevertheless, sessile serrated adenomas have a malignant potential similar to conventional adenomas. The most useful feature distinguishing sessile serrated adenoma from hyperplastic polyp is the presence of serrated architecture throughout the full length of the glands, including the crypt base, in association with crypt dilation and lateral growth (Fig. 13.34D). By contrast, serrated architecture is typically confined to the surface of hyperplastic polyps.

Although most colorectal adenomas behave in a benign fashion, a small proportion harbor invasive cancer at the time of detection. **Size is the most important characteristic that correlates with risk for**

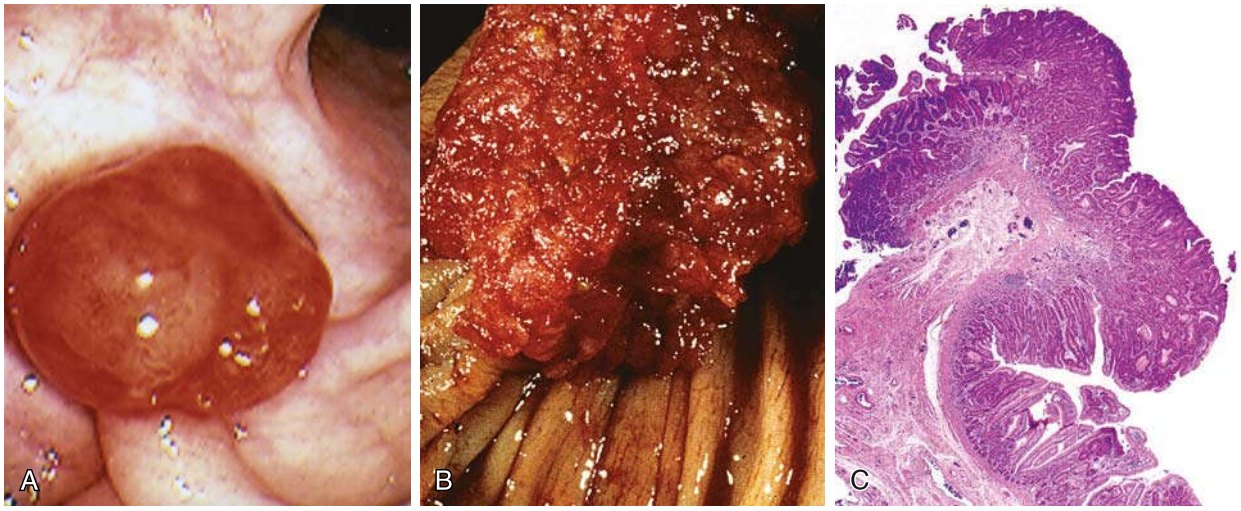


FIG. 13.33 Colonic adenomas. (A) Pedunculated adenoma (endoscopic view). (B) Adenoma with a velvety surface. (C) Low-magnification photomicrograph of a pedunculated tubular adenoma.

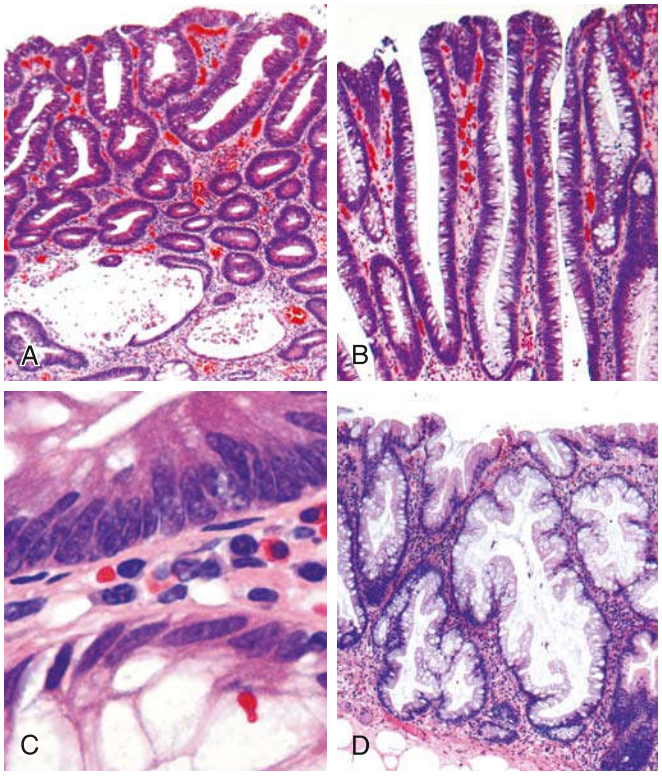


FIG. 13.34 Histologic appearance of colonic adenomas. (A) Tubular adenoma with a smooth surface and rounded glands. In this case, crypt dilation and rupture, with associated reactive inflammation, can be seen at the bottom of the field. (B) Villous adenoma with long, slender projections that are reminiscent of small-intestinal villi. (C) Dysplastic epithelial cells *(top)* with an increased nuclear-to-cytoplasmic ratio, hyperchromatic and elongated nuclei, and nuclear pseudostratification. Compare with the nondysplastic epithelium *(bottom)*. (D) Sessile serrated adenoma lined by goblet cells without typical cytologic features of dysplasia. This lesion is distinguished from a hyperplastic polyp by involvement of the crypts. Compare with the hyperplastic polyp in Fig. 13.32.

malignancy. For example, while cancer is extremely rare in adenomas less than 1 cm in diameter, some studies suggest that nearly 40% of lesions larger than 4 cm in diameter contain foci of invasive cancer. High-grade dysplasia is a second, less important risk factor for the presence of cancer in a polyp.

Familial Colonic Neoplasia Syndromes

Several syndromes associated with colonic polyps and increased rates of colon cancer have been described. The genetic bases of these disorders have been established and have provided insights that have greatly enhanced the understanding of sporadic colon cancer (Table 13.7).

Familial Adenomatous Polyposis

Familial adenomatous polyposis (FAP) is an autosomal dominant disorder marked by the appearance of numerous colorectal adenomas by the teenage years that is caused by mutation of the adenomatous polyposis coli *(APC)* gene. A count of at least 100 polyps is necessary for a diagnosis of classic FAP, and as many as several thousand may be present (Fig. 13.35). Except for their remarkable numbers, these growths are morphologically indistinguishable from sporadic adenomas. Colorectal adenocarcinoma develops in 100% of patients with untreated FAP, often before 30 years of age. As a result, prophylactic colectomy is standard therapy for individuals carrying *APC* mutations. However, patients remain at risk for extraintestinal manifestations, including neoplasia at other sites. Specific *APC* mutations are also associated with the development of other manifestations of FAP such as *Gardner syndrome* and *Turcot syndrome* (see Table 13.6). In addition to intestinal polyps, clinical features of Gardner syndrome include osteomas of the mandible, skull, and long bones; epidermal cysts; desmoid and thyroid tumors; and dental abnormalities, including unerupted and supernumerary teeth. Turcot syndrome is rarer and is characterized by intestinal adenomas and tumors of the central nervous system. Two-thirds of patients with Turcot syndrome have *APC* gene mutations and develop medulloblastomas. The remaining one-third have mutations in one of several genes involved in DNA repair and develop glioblastomas. The role of these base excision repair genes in carcinogenesis is discussed next.

Hereditary Nonpolyposis Colorectal Cancer

Hereditary nonpolyposis colorectal cancer (HNPCC), also known as *Lynch syndrome,* is an autosomal dominant condition marked by an increased risk for cancers of the colorectum, endometrium, stomach, ovary, ureters, brain, small bowel, hepatobiliary tract, and skin. Colon cancers in patients with HNPCC tend to occur at younger ages than do sporadic colon cancers and are often located in the right colon (see Table 13.7). In contrast to familial adenomatous polyposis, only a few adenomatous precursors are found in the colon of affected individuals. These precursors tend to be sessile serrated adenomas, which often give rise to adenocarcinomas with abundant mucin production.

Table 13.7 Common Patterns of Sporadic and Familial Colorectal Neoplasia

Etiology	Molecular Defect	Target Gene(s)	Transmission	Predominant Site(s)	Histology
Familial adenomatous polyposis	APC/WNT pathway	*APC*	Autosomal dominant	None	Tubular, villous; typical adenocarcinoma
Hereditary nonpolyposis colorectal cancer	DNA mismatch repair	*MSH2, MLH1*	Autosomal dominant	Right side	Sessile serrated adenoma; mucinous adenocarcinoma
Sporadic colon cancer (80%)	APC/WNT pathway	*APC*	None	Left side	Tubular, villous; typical adenocarcinoma
Sporadic colon cancer (10%–15%)	DNA mismatch repair	*MSH2, MLH1*	None	Right side	Sessile serrated adenoma; mucinous adenocarcinoma

FAP, Familial adenomatous polyposis.

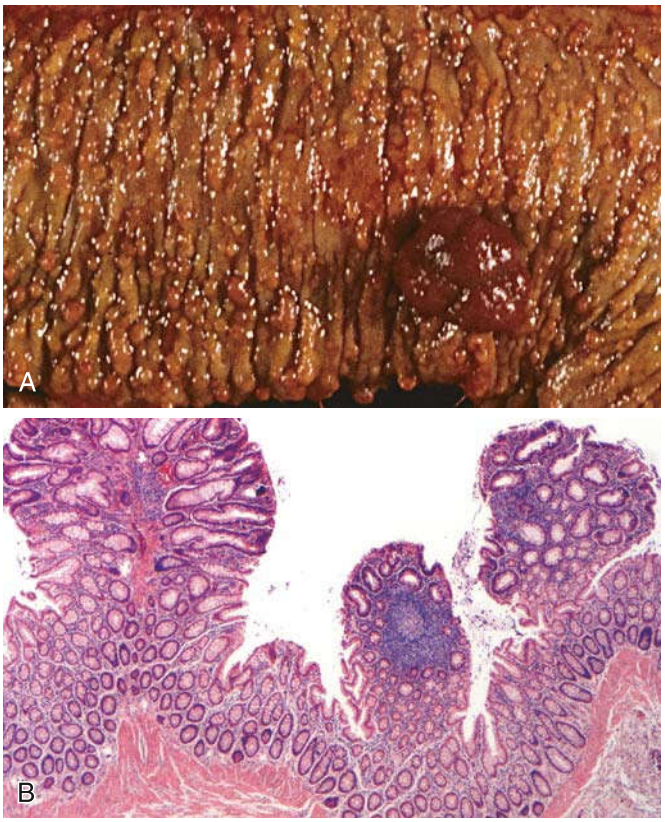


FIG. 13.35 Familial adenomatous polyposis. (A) Hundreds of small colonic polyps are present along with a dominant polyp *(right).* (B) Three tubular adenomas are present in this single microscopic field.

HNPCC is caused by inherited germline mutations in genes that encode proteins responsible for the detection, excision, and repair of errors that occur during DNA replication. Five mismatch repair genes have been identified, but a majority of HNPCC cases are caused by mutations in *MSH2* or *MLH1*. Patients with HNPCC inherit one mutated DNA repair gene and one normal allele. When the second copy is lost through mutation or epigenetic silencing, the resulting defect in mismatch repair leads to the accumulation of mutations at rates up to 1000 times higher than normal, mostly in regions containing short repeating DNA sequences referred to as *microsatellite DNA*. The human genome contains approximately 50,000 to 100,000 microsatellites, which are prone to undergo expansion during DNA replication and are the most frequent sites of acquired mutations in HNPCC. Understanding of the genetic underpinnings of HNPCC has shed light on the mechanisms responsible for a subset of sporadic colon cancers, which also carry mutations in mismatch repair genes. The consequences of mismatch repair defects and the resulting *microsatellite instability* are discussed next in the context of colonic adenocarcinoma.

Adenocarcinoma

Adenocarcinoma of the colon is the most common malignancy of the gastrointestinal tract and is a major contributor to morbidity and mortality worldwide. By contrast, the small intestine, which accounts for 75% of the overall length of the gastrointestinal tract, is an uncommon site for benign and malignant tumors. Among malignant small-intestinal tumors, adenocarcinoma and carcinoid (neuroendocrine) tumors have roughly equal rates of occurrence, followed by lymphoma and sarcoma.

Epidemiology. In the United States in 2022, it is estimated that approximately 151,000 individuals will be newly diagnosed with colorectal adenocarcinoma and that approximately 53,000 people will die of the disease. This represents nearly 15% of all cancer-related deaths, second only to lung cancer. Colorectal cancer incidence peaks at 60 to 70 years of age. Less than 20% of cases occur before 50 years of age, but for unclear reasons deaths from colorectal carcinoma have increased in individuals younger than age 50 in recent years, and as a result it is now recommended that screening for colorectal cancer begins at the age of 45 years. Males are affected slightly more often than females. Colorectal carcinoma is most prevalent in the United States, Canada, Australia, New Zealand, Denmark, Sweden, and other higher income countries that share lifestyles and diet. The incidence of this cancer is as much as 30-fold lower in India, South America, and Africa. In Japan, where the incidence was previously very low, the incidence has risen to an intermediate level (similar to those in the United Kingdom), presumably as a result of changes in lifestyle and diet.

The factors most closely associated with a high incidence of colorectal cancer are diets with a low intake of unabsorbable vegetable fiber and high intake of refined carbohydrates and fat. In addition to dietary modification, pharmacologic chemoprevention has become an area of interest. Colorectal cancer is also associated with obesity, tobacco smoking, and alcohol consumption. By contrast, aspirin and other NSAIDs appear to have a protective effect. This is consistent with studies showing that some NSAIDs cause polyp regression in patients with familial adenomatous polyposis in whom the rectum was left in place after colectomy. It is suspected that this effect is mediated by inhibition of the enzyme cyclooxygenase-2 (COX-2), which is highly expressed in 90% of colorectal carcinomas and 40% to 90% of adenomas and is known to promote epithelial proliferation, particularly in response to injury.

Pathogenesis. **Studies of colorectal carcinogenesis have provided fundamental insights into the general mechanisms of cancer evolution.** The combination of molecular events that lead to colonic adenocarcinoma is heterogeneous and includes genetic and epigenetic abnormalities. At least two distinct genetic pathways, the APC/β-catenin pathway and the microsatellite instability pathway, have been described. In simplest terms, mutations involving the APC/β-catenin pathway lead to increased WNT signaling, whereas those involving the microsatellite instability pathway are associated with defects in DNA mismatch repair (see Table 13.7). Both pathways involve the stepwise accumulation of multiple mutations, but the genes involved and the mechanisms by which the mutations accumulate differ. Epigenetic events, the most common of which is methylation-induced gene silencing, may enhance progression along both pathways.

- *APC/β-catenin pathway.* The classic *adenoma-carcinoma sequence,* which accounts for as many as 80% of sporadic colon tumors, typically involves mutation of the *APC* tumor suppressor early in the neoplastic process (Fig. 13.36). For adenomas to develop, both copies of the *APC* gene must be functionally inactivated, either by mutation or epigenetic silencing. APC is a key negative regulator of β-catenin, a component of the WNT signaling pathway (Chapter 6). The APC protein normally binds to and promotes the degradation of β-catenin. With loss of APC function, β-catenin accumulates and translocates to the nucleus, where it activates the

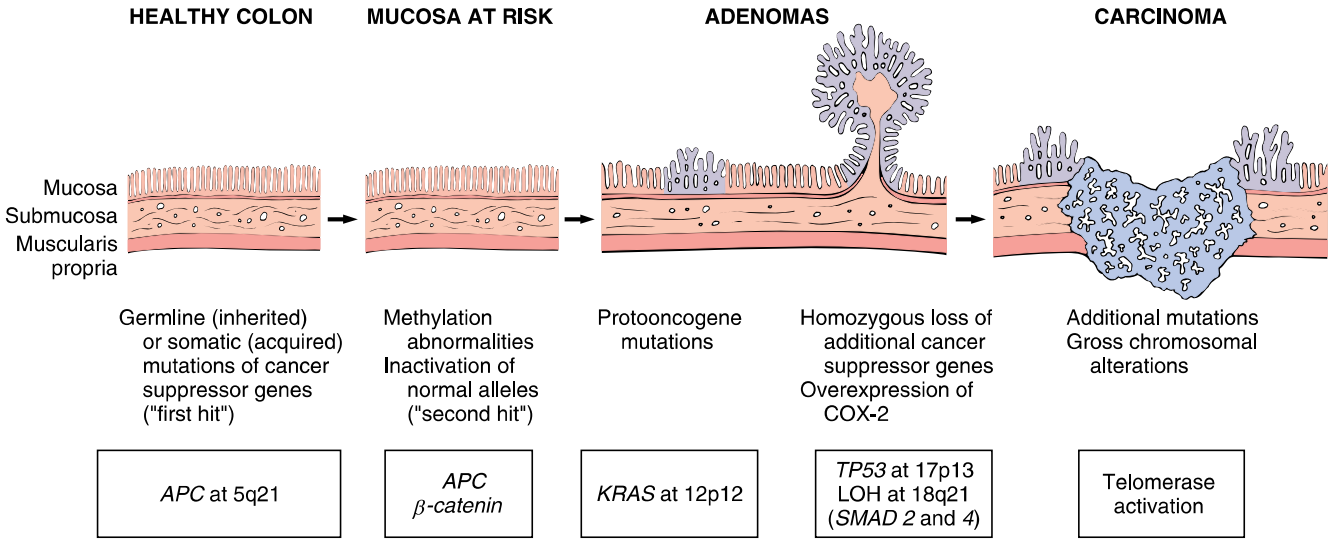


FIG. 13.36 Morphologic and molecular changes in the adenoma-carcinoma sequence. It is postulated that loss of one normal copy of the tumor suppressor gene *APC* occurs early. Individuals with familial adenomatous polyposis are born with one mutant allele, making them extremely prone to the development of colon cancer. Other mutations involving *KRAS, SMAD2,* and *SMAD4,* and the tumor suppressor gene *TP53,* as well as activation of telomerase, lead to the emergence of a full-blown cancer. Although there may be a preferred temporal sequence for these changes, it is the aggregate effect of the mutations, rather than their order of occurrence, that appears most critical. *APC,* Adenomatous polyposis coli; *COX-2,* cyclooxygenase-2; *LOH,* loss of heterozygosity.

transcription of genes, such as those encoding MYC and cyclin D1, that promote proliferation. This is followed by additional mutations, including activating mutations in *KRAS,* which also promote growth and prevent apoptosis. The conclusion that mutation of *KRAS* is a late event is supported by the observations that *KRAS* mutations are present in fewer than 10% of adenomas less than 1 cm in diameter, 50% of adenomas greater than 1 cm in diameter, and 50% of invasive adenocarcinomas. Neoplastic progression is also associated with mutations in other tumor suppressor genes such as *SMAD2* and *SMAD4,* which encode effectors of TGF-β signaling. Because TGF-β signaling normally inhibits the cell cycle, loss of these genes may allow unrestrained cell growth. The tumor suppressor gene *TP53* is mutated in 70% to 80% of colon cancers but is uncommonly affected in adenomas, suggesting that *TP53* mutations also occur at late stages of tumor progression. Loss of function of *TP53* and other tumor suppressor genes is often caused by chromosomal deletions, highlighting chromosomal instability as a hallmark of the APC/β-catenin pathway. Telomerase activation also likely plays a role in tumor progression.

- *Microsatellite instability pathway.* In patients with DNA mismatch repair deficiency (due to loss of mismatch repair genes, as discussed earlier), mutations accumulate in microsatellite repeats, a condition referred to as *microsatellite instability.* These mutations are generally silent because microsatellites are typically located in noncoding regions, but some microsatellite sequences are located in the coding or promoter regions of genes involved in regulation of cell growth, such as those encoding the type II TGF-β receptor and the proapoptotic protein BAX (Fig. 13.37). Because TGF-β inhibits colonic epithelial cell proliferation, type II TGF-β receptor mutations can contribute to uncontrolled cell growth, while loss of BAX may enhance the survival of genetically abnormal clones. In a subset of colon cancers with microsatellite instability, mutations in DNA mismatch repair genes are absent. Instead, in these tumors the *MLH1* promoter region is hypermethylated, thereby reducing MLH1 expression and repair function. These features define the *CpG island hypermethylation phenotype* (CIMP). Activating mutations in the *BRAF* oncogene are common in these cancers, whereas *KRAS* and *TP53* are not typically mutated.

MORPHOLOGY

Overall, adenocarcinomas are distributed approximately equally over the entire length of the colon. **Tumors in the proximal colon often grow as polypoid, exophytic masses** that extend along one wall of the large-caliber cecum and ascending colon; these tumors rarely cause obstruction (Fig. 13.38A). By contrast, **carcinomas in the distal colon tend to be annular lesions that produce "napkin ring"** constrictions and luminal narrowing (Fig. 13.38B), sometimes to the point of obstruction. Both forms grow into the bowel wall over time and may be palpable as firm masses (Fig. 13.38C). The general microscopic characteristics of right- and left-sided colonic adenocarcinomas are similar. Most tumors are composed of tall columnar cells that resemble dysplastic epithelium found in adenomas (Fig. 13.39A). The invasive component of these tumors elicits a strong stromal desmoplastic response, which is responsible for their characteristic firm consistency. Some poorly differentiated tumors form few glands (Fig. 13.39B). Others produce abundant mucin that accumulates within the intestinal wall; these so-called mucinous carcinomas are associated with a poor prognosis. Tumors may also be composed of signet ring cells similar to those in gastric cancer (Fig. 13.39C).

Clinical Features. The availability of endoscopic screening combined with the recognition that most carcinomas arise within adenomas presents a unique opportunity for cancer prevention. Unfortunately, colorectal cancers develop insidiously and may therefore go

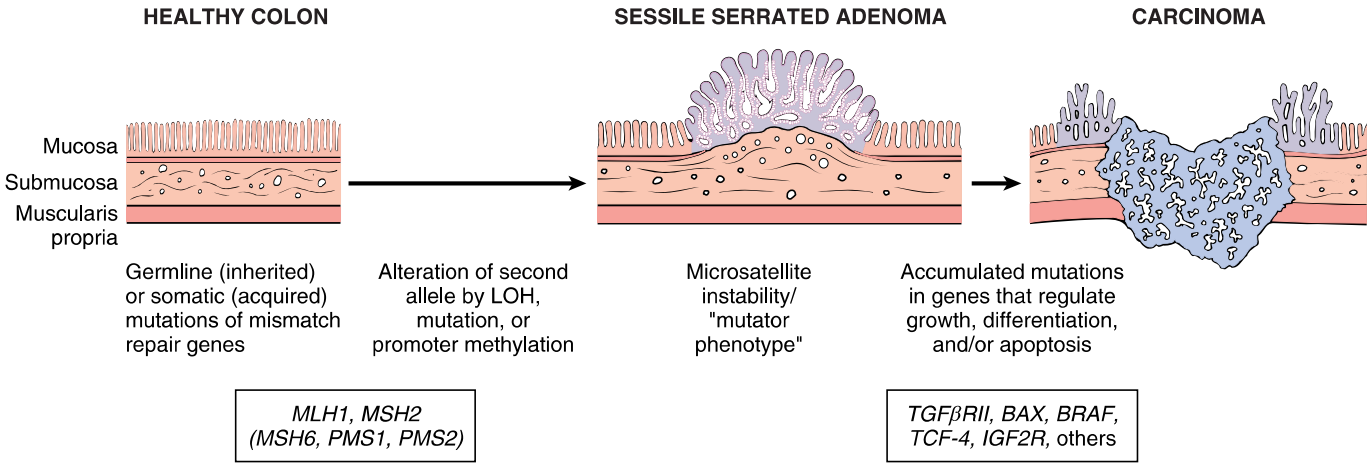


FIG. 13.37 Morphologic and molecular changes in the mismatch repair pathway of colon carcinogenesis. Defects in mismatch repair genes (most commonly *MLH1* or *MSH2*) result in microsatellite instability and permit accumulation of mutations in numerous genes. If these mutations affect genes involved in cell survival and proliferation, cancer may develop. *IGF2R*, Insulin-like growth factor 2 receptor; *LOH*, loss of heterozygosity; *TCF-4*, transcription factor-4; *TGFβRII*, transforming growth factor beta receptor II.

undetected for long periods. Cecal and other right-sided colon cancers are most often called to clinical attention by fatigue and weakness due to iron deficiency anemia. Thus, it is a clinical maxim that the underlying cause of iron deficiency anemia in an older male or postmenopausal female is gastrointestinal cancer until proven otherwise. Left-sided colorectal adenocarcinomas may produce occult bleeding, changes in bowel habits, or cramping, left lower-quadrant discomfort.

Although poorly differentiated and mucinous histologic patterns are associated with poor prognosis, the two most important prognostic factors are depth of invasion and the presence or absence of lymph node metastases. These factors form the core of the TNM (tumor-node-metastasis) classification and staging system from the American Joint Committee on Cancer (Chapter 6). Staging systems have become more complex with time, reflecting more nuanced treatment approaches and personalized therapeutic strategies. The most important considerations are:

- *Depth of invasion.* Tumors limited to the mucosa (i.e., those that do not cross the muscularis mucosae) have 5-year survival rates approaching 100%, while invasion into the submucosa or muscularis propria reduces 5-year survival to 95% and 70% to 90%, respectively (for tumors limited to the primary site). Invasion through the visceral serosal surface or into adjacent organs and tissues further reduces survival.
- *The presence of lymph node metastases* (Fig. 13.40A) also reduces survival. As a result, most cases with lymph node metastases receive radiation or chemotherapy. In some cases, these treatments may be administered prior to primary tumor resection, a process termed neoadjuvant therapy. Molecular characterization of the tumor can help guide the specific therapeutic approach.

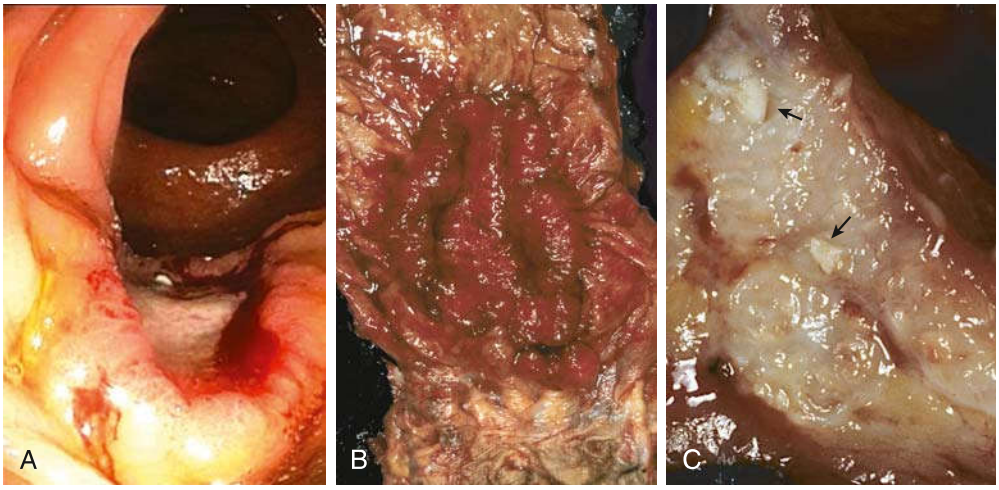


FIG. 13.38 Colorectal carcinoma. (A) Endoscopic view of ulcerated ascending colon adenocarcinoma. (B) Resected rectum showing a circumferential adenocarcinoma. Note the anal mucosa at the bottom of the image. (C) Cancer of the sigmoid colon that has invaded through the muscularis propria and is present within subserosal adipose tissue *(left)*. Areas of chalky necrosis are present within the colon wall *(arrows)*. (Endoscopic image courtesy of Dr. Ira Hanan, The University of Chicago, Chicago, Illinois.)

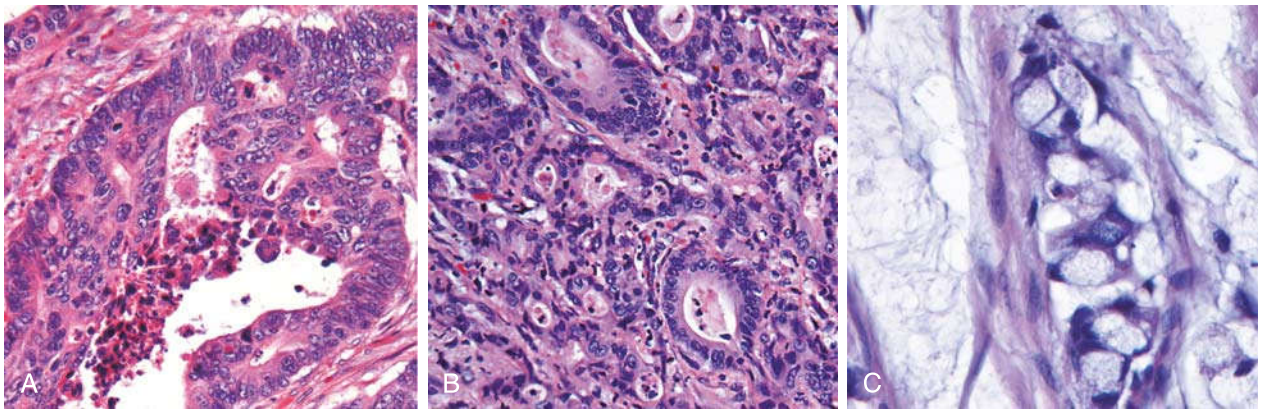


FIG. 13.39 Histologic appearance of colorectal carcinoma. (A) Well-differentiated adenocarcinoma. Note the elongated, hyperchromatic nuclei. Necrotic debris, present in the gland lumen, is typical. (B) Poorly differentiated adenocarcinoma forms a few glands but is largely composed of infiltrating nests of tumor cells. (C) Mucinous adenocarcinoma with signet ring cells and extracellular mucin pools.

- *Distant metastasis* to the lung (Fig. 13.40B), liver (Fig. 13.40C), or other sites also limits survival, and only 15% or fewer of patients with tumors at this stage are alive 5 years after diagnosis. Because of portal drainage of the colon, the liver is the most common site of metastatic lesions. However, the rectum does not drain by way of the portal circulation, and metastases from carcinomas of the anorectum region may circumvent the liver and lodge in the lung or other sites.

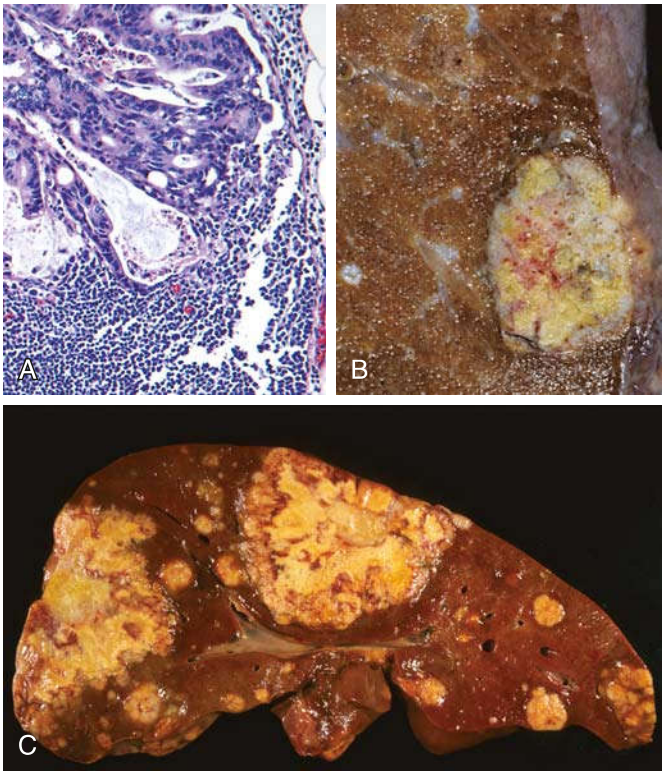


FIG. 13.40 Metastatic colorectal carcinoma. (A) Lymph node metastasis. Note the glandular structures within the subcapsular sinus. (B) Solitary subpleural nodule of colorectal carcinoma metastatic to the lung. (C) Liver containing two large and many smaller metastases. Note the central necrosis within metastases.

Regardless of stage, some patients with small numbers of metastases do well for years after resection of distant tumor nodules. This is particularly true of solitary metastases to the liver or lung and emphasizes the clinical and molecular heterogeneity of colorectal carcinomas. Tumors that exhibit microsatellite instability are often responsive to immune checkpoint inhibitor therapies, presumably because these tumors express particularly large numbers of tumor-specific neoantigens.

APPENDIX

The appendix is a true diverticulum of the cecum. Like any diverticulum, it is prone to acute and chronic inflammation, and acute appendicitis is a relatively common entity. Other lesions, including tumors, can also occur in the appendix but are far less common.

ACUTE APPENDICITIS

Acute appendicitis is most common in adolescents and young adults but may occur in any age group. The lifetime risk for appendicitis is 7%; males are affected slightly more often than females. Despite the prevalence of acute appendicitis, the diagnosis can be difficult to confirm preoperatively, and the condition may be confused with mesenteric lymphadenitis, acute salpingitis, ectopic pregnancy, mittelschmerz (pain associated with ovulation), and Meckel diverticulitis.

Pathogenesis. Acute appendicitis is thought to be initiated by a progressive increase in intraluminal pressure that compromises venous outflow. In 50% to 80% of cases, acute appendicitis is associated with luminal obstruction, usually by a small, stonelike mass of stool, or *fecalith,* or, less commonly, a gallstone, tumor, or mass of worms. Ischemic injury and stasis of luminal contents, which favor bacterial proliferation, trigger inflammatory responses including tissue edema and neutrophilic infiltration of the lumen, muscular wall, and periappendiceal soft tissues.

MORPHOLOGY

In early acute appendicitis, subserosal vessels are congested and a modest perivascular neutrophilic infiltrate is present within all layers of the wall. The inflammatory reaction transforms the normal glistening serosa into a dull, granular, erythematous surface. Although mucosal neutrophils and focal

superficial ulceration are often present, these findings are not specific, and diagnosis of acute appendicitis requires neutrophilic infiltration of the muscularis propria. In more severe cases, focal abscesses may form within the wall **(acute suppurative appendicitis),** and these may even progress to large areas of hemorrhagic ulceration and gangrenous necrosis that extend to the serosa, creating **acute gangrenous appendicitis,** which is often followed by rupture and suppurative peritonitis.

Clinical Features. Typically, early acute appendicitis produces periumbilical pain that then moves to the right lower quadrant, followed by nausea, vomiting, low-grade fever, and a mildly elevated peripheral white blood cell count. A classic physical finding is *McBurney sign,* deep tenderness noted at a location two-thirds of the distance from the umbilicus to the right anterior superior iliac spine *(McBurney point).* These signs and symptoms, however, are often absent, creating difficulty in clinical diagnosis. Imaging studies such as CT scanning are helpful in narrowing the differential diagnosis.

TUMORS OF THE APPENDIX

Several tumors occur in the appendix, the most important of which are the following:

- *The most common tumor of the appendix is carcinoid,* a well-differentiated neuroendocrine tumor (discussed earlier). It is usually discovered incidentally at the time of surgery or on examination of a resected appendix. This neoplasm most frequently involves the distal tip of the appendix, where it produces a solid bulbous swelling up to 2 to 3 cm in diameter. Although intramural and transmural extension may be evident, nodal metastases are very infrequent and distant spread is exceptionally rare.
- *Adenomas and non—mucin-producing adenocarcinomas* occur in the appendix and may cause obstruction and enlargement that mimics the changes of acute appendicitis.
- *Mucinous cystadenoma* and *mucinous cystadenocarcinoma* also occur in the appendix. When they cause obstruction of the lumen, the appendix may be dilated by inspissated mucin, an appearance referred to as a *mucocele.* Rupture of the swollen appendix may lead to intraperitoneal seeding and spread. In women, the resulting peritoneal implants may be mistaken for mucinous ovarian tumors. In the most advanced cases, the abdomen fills with tenacious, semisolid mucin, a condition called *pseudomyxoma peritonei* (eFig. 13.8). This disseminated intraperitoneal disease may be held in check for years by repeated debulking but is ultimately fatal in most instances.

RAPID REVIEW

Diseases of Teeth and Supporting Structures

- *Caries:* Most common cause of tooth loss in individuals younger than 35 years of age, results from loss of tooth structure due to acids produced by fermentation of sugar by bacteria
- *Gingivitis:* Common and reversible inflammation of the mucosa surrounding the teeth associated with buildup of dental plaque and calculus
- *Periodontitis:* Chronic inflammatory condition associated with poor oral hygiene and altered oral microbiota that can lead to the destruction of tooth support structures and tooth loss

Oral Inflammatory Lesions

- *Aphthous ulcers:* Painful superficial ulcers of unknown etiology that are sometimes associated with systemic diseases
- *Herpes simplex virus:* Primary infection is self-limited and presents with vesicles (cold sores, fever blisters) that rupture and heal without scarring; persists in latent form in nerve ganglia and may reactivate
- *Oral candidiasis:* Occurs when the oral microbiota is altered (e.g., after antibiotic use); may become invasive in immunocompromised individuals

Lesions of the Oral Cavity

- *Fibroma* and *pyogenic granuloma:* Common reactive stromal lesions of the oral mucosa that present as small masses
- *Leukoplakia* and *erythroplakia:* Mucosal squamous plaques that may undergo malignant transformation (erythroplakia more commonly than leukoplakia)
- A majority of oral cavity cancers are squamous cell carcinomas, which fall into two classes: (1) those caused by carcinogen exposure (tobacco and alcohol use) and (2) those caused by high-risk HPV infection.

Diseases of Salivary Glands

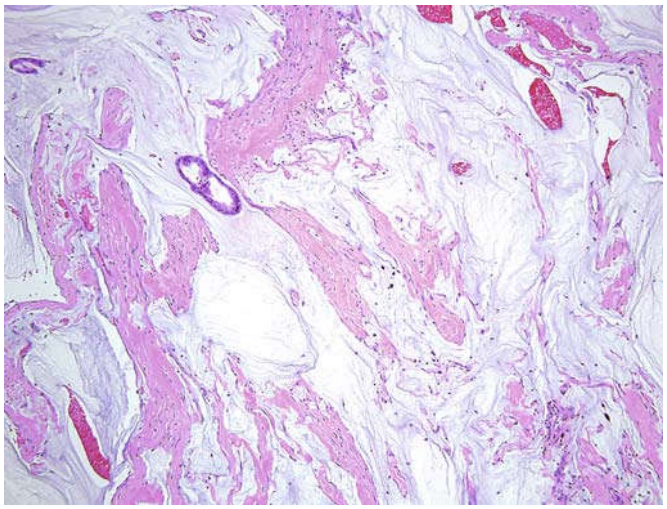
- *Sialadenitis* (inflammation of the salivary glands): Can be caused by trauma, infection (such as mumps), or an autoimmune reaction
- *Pleomorphic adenoma:* A slow-growing neoplasm composed of a heterogeneous mixture of epithelial and mesenchymal cells that is benign but may recur; sometimes undergoes malignant transformation. Most common in parotid gland.
- *Mucoepidermoid carcinoma:* A malignant neoplasm of variable biologic aggressiveness that is composed of a mixture of squamous and mucous cells. Occurs both in parotid and minor salivary glands.

Odontogenic Cysts and Tumors

- The jaws are a common site of epithelium-lined cysts derived from odontogenic remnants.
- *Keratocystic odontogenic tumor:* A locally aggressive neoplasm with a high recurrence rate
- *Periapical cyst:* A reactive, inflammatory lesion associated with caries or dental trauma
- The most common odontogenic tumors are *ameloblastoma* and *odontoma.*

Diseases of the Esophagus

- *Esophageal obstruction* may occur due to mechanical or functional anomalies; mechanical causes include developmental defects, fibrotic strictures, and tumors.
- *Esophageal varices* are caused by dilation of collateral vessels in the context of portal hypertension and are prone to massive bleeding, which may be fatal.
- Vomiting if severe and prolonged may give rise to esophageal tears, which may be superficial and cross the GE junction *(Mallory-Weiss tear)* or transmural and prone to cause mediastinitis *(Boerhave syndrome).*
- *Achalasia,* characterized by incomplete lower esophageal sphincter relaxation, increased sphincter tone, and esophageal aperistalsis; is a common form of functional esophageal obstruction.



eFIG. 13.8 *Pseudomyxoma peritonei.* Abundant mucin dissects through the wall of the appendix. Only a few nests of glandular tumor cells are seen. (From Fletcher CD: *Diagnostic Histopathology of Tumors,* ed 5, Fig. 9.33, Philadelphia, 2021, Elsevier.)

- *Esophagitis* can result from chemical or infectious mucosal injury; infections are most frequent in immunocompromised individuals.
- The most common cause of esophagitis is *gastroesophageal reflux disease (GERD),* which must be differentiated from *eosinophilic esophagitis,* an immunologic disorder occurring mainly in atopic individuals.
- *Barrett esophagus,* which may develop in patients with chronic GERD, is defined by the presence of intestinal metaplasia and carries an increased risk for esophageal adenocarcinoma.
- *Esophageal squamous cell carcinoma* is associated with alcohol and tobacco use, caustic esophageal injury, achalasia, and Plummer-Vinson syndrome.

Acute and Chronic Gastritis

- *Gastritis* is mucosal inflammation; when the architecture of gastric glands is perturbed but inflammatory cells are absent or rare, the term *gastropathy* is applied.
- Causative factors in acute gastritis include any agent or disease that interferes with gastric mucosal protection.
- The most common cause of chronic gastritis is *H. pylori* infection; most remaining cases are caused by NSAIDs, alcohol use, or autoimmune gastritis.
- *H. pylori* gastritis typically affects the antrum and is associated with increased gastric acid production and chronic inflammation.
- *Autoimmune gastritis* causes atrophy of the gastric body oxyntic glands, which results in decreased gastric acid production, antral G-cell hyperplasia, achlorhydria, and vitamin B_{12} deficiency. Parietal cell and intrinsic factor antibodies are typically present.
- *Intestinal metaplasia* may develop in chronic gastritis and is a risk factor for gastric dysplasia and adenocarcinoma.
- *Peptic ulcer disease* may be caused by *H. pylori*—associated chronic gastritis and the resultant hyperchlorhydria, or NSAID use. Ulcers in the stomach or duodenum usually heal after suppression of gastric acid production, discontinuation of NSAID use or *H. pylori* eradication.

Gastric Polyps and Tumors

- *Inflammatory* and *hyperplastic gastric polyps* are reactive lesions associated with chronic gastritis. Risk for development of neoplasia increases with polyp size. Fundic gland polyps occur in association of proton pump inhibitor use.
- *Gastric adenomas* develop in a background of chronic gastritis with intestinal metaplasia and mucosal (glandular) atrophy and are precursor lesions of adenocarcinoma.
- *Gastric adenocarcinomas* are classified according to location and gross and microscopic appearance, which may be either intestinal type or diffuse (signet ring) type. Acquired or inherited E-cadherin loss is important in the pathogenesis of the diffuse type.
- The most important risk factor for gastric adenocarcinoma is chronic gastritis with intestinal metaplasia, which is often associated with *H. pylori* infection.
- *Primary gastric lymphomas* are most often derived from mucosa-associated lymphoid tissue, the development of which is induced by *H. pylori*—associated chronic gastritis.
- *Neuroendocrine tumors (carcinoids)* arise from the diffuse components of the endocrine system and are most common in the gastrointestinal tract, particularly the small intestine. Tumors of the small intestine tend to be most aggressive, while those of the appendix are almost always benign.
- *Gastrointestinal stromal tumor (GIST),* the most common mesenchymal tumor of the abdomen, occurs most often in the stomach. GISTs arise from the interstitial cells of Cajal and usually have activating mutations in either the KIT or PDGFRA receptor tyrosine kinases.

Intestinal Obstruction

- *Intussusception* is the most common cause of intestinal obstruction in children younger than 2 years of age and can usually be treated by barium enema or air enema.
- *Hirschsprung disease* is the result of defective neural crest cell migration during development from the cecum to the rectum. It gives rise to functional obstruction due to an absence of ganglion cells.
- *Abdominal herniation* may occur through any weakness or defect in the wall of the peritoneal cavity, including inguinal and femoral canals, umbilicus, and sites of surgical scarring.

Vascular Disorders of Bowel

- *Intestinal ischemia* may occur due to arterial or venous obstruction.
- Ischemic bowel disease resulting from hypoperfusion is most common at the splenic flexure, sigmoid colon, and rectum; these are watershed zones where two arterial circulations terminate.
- Systemic vasculitides and infectious diseases (e.g., CMV infection) may cause vascular disease that can result in chronic ischemia of the gastrointestinal tract.
- *Angiodysplasia* is a common cause of lower gastrointestinal bleeding in older adults.
- *Hemorrhoids* are collateral vessels that form in response to venous hypertension.

Malabsorptive Diarrhea

- *Diarrhea* can be characterized as secretory, osmotic, malabsorptive, or exudative.
- *Cystic fibrosis* leads to malabsorption by causing pancreatic insufficiency and deficient luminal breakdown of nutrients.
- *Celiac disease* is an immune-mediated enteropathy triggered by the ingestion of gluten-containing grains. The malabsorptive diarrhea in celiac disease is due to loss of brush border surface area and, possibly, abnormal enterocyte maturation.
- *Lactase deficiency* causes an osmotic diarrhea owing to the inability to break down or absorb lactose.
- *Abetalipoproteinemia* is characterized by an inability to secrete triglyceride-rich lipoproteins due to an inherited transepithelial transport defect.
- *Microscopic colitis* takes two forms, *collagenous colitis* and *lymphocytic colitis,* that both cause chronic watery diarrhea. The intestines are grossly normal, and the diseases are identified by their histologic features.

Infectious Enterocolitis

- *Vibrio cholera* releases a preformed toxin that causes massive chloride secretion, leading to secretory diarrhea.
- *Campylobacter jejuni* is the most common bacterial enteric pathogen in higher-resource countries and is also a frequent cause of traveler's diarrhea. Most isolates are noninvasive.

- *Salmonella* and *Shigella* spp. are invasive and associated with exudative bloody diarrhea (dysentery). *Salmonella* infection is a common cause of food poisoning. *S. typhi* can cause systemic disease (typhoid fever).
- *Pseudomembranous colitis* is often triggered by antibiotic therapy that disrupts the normal microbiota and allows *C. difficile* to colonize and grow. The organism produces toxins that disrupt epithelial function and cause necrosis.
- *Rotavirus* is a common cause of severe childhood diarrhea worldwide. The diarrhea is secondary to loss of mature enterocytes, resulting in malabsorption and increased fluid secretion. Rotavirus vaccine is protective.
- *Parasitic* and *protozoal* infections affect over half of the world's population on a chronic or recurrent basis. Among the important agents causing human disease are roundworms (*Ascaris* and *Strongyloides*), hookworms (*Necator* and *Ancylostoma*), and protozoa (*Giardia* and *Entamoeba*).

Inflammatory Bowel Disease (IBD)

- IBD is an umbrella term for Crohn disease and ulcerative colitis.
- IBD is thought to arise from a combination of host interactions with intestinal microbiota, intestinal epithelial dysfunction, and aberrant mucosal immune responses.
- *Crohn disease* most commonly affects the terminal ileum and cecum, but any site within the gastrointestinal tract can be involved; inflammation is transmural; skip lesions and noncaseating granulomas are common.
- *Ulcerative colitis* is limited to the colon, always involves the rectum, and ranges in extent from rectum-only disease to pancolitis; inflammation is confined to the mucosa; neither skip lesions nor granulomas are present.
- Both Crohn disease and ulcerative colitis may have extraintestinal manifestations.
- The risk for development of colonic epithelial dysplasia and adenocarcinoma is increased in patients who have had colonic IBD for more than 8 to 10 years.

Colonic Polyps, Adenomas, and Adenocarcinomas

- *Intestinal polyps* can be classified as nonneoplastic or neoplastic; nonneoplastic polyps can be further categorized as inflammatory, hamartomatous, or hyperplastic.
- *Inflammatory polyps* form as a result of chronic cycles of injury and healing.
- *Hamartomatous polyps* occur sporadically or as a part of genetic diseases (e.g., Peutz-Jeghers syndrome); in the latter case, they are often associated with increased risk for malignancy.
- *Hyperplastic polyps* are benign epithelial proliferations that are most common in the left colon and rectum and have no malignant potential; they must be distinguished from sessile serrated adenomas, which are precursors of colon cancer.
- *Neoplastic epithelial polyps* of the colon are termed adenomas. The hallmark feature of most of these lesions, which are the precursors of colonic adenocarcinomas, is dysplasia.
- *Sessile serrated adenomas* differ in lacking dysplasia and share some morphologic features with hyperplastic polyps.
- Familial adenomatous polyposis (FAP) and hereditary nonpolyposis colorectal cancer (HNPCC) are the most common forms of familial colon cancer.
  - FAP is caused by *APC* mutations; patients typically have hundreds of adenomas and develop colon cancer before 30 years of age.
  - HNPCC is caused by mutations in DNA mismatch repair genes that result in microsatellite instability. Patients with HNPCC have far fewer polyps and develop cancer at an older age than that typical for patients with FAP, but at a younger age than in patients with sporadic colon cancer.
- Most colonic cancers are adenocarcinomas that arise due to dysregulation of the APC-β-catenin pathway or due to mutations caused by microsatellite instability. The two most important prognostic factors are depth of invasion and the presence or absence of metastases to lymph nodes or distant organs.

Appendix

- Acute appendicitis is most common in children and adolescents. It is thought to be initiated by increased intraluminal pressure consequent to obstruction of the appendiceal lumen, which compromises venous outflow.
- The most common tumor of the appendix, the *carcinoid,* or *well-differentiated neuroendocrine tumor,* is most often discovered incidentally and is almost always benign.
- Mucinous tumors of the appendix may spread to the peritoneal cavity to produce widespread disease, often referred to as *pseudomyxoma peritonei.*

Laboratory Tests[a]

Test	Reference Values	Pathophysiology/Clinical Relevance
Deamidated gliadin IgG and IgA antibody, serum	Negative: <20.0 U Weak positive: 20.0–30.0 U Positive: >30.0 U	Deamidated gliadin antibody is a serum test for celiac disease in patients with IgA deficiency (approximately 2% of patients with celiac disease), who therefore lack IgA antibodies against tissue transglutaminase (tTG), which is the first-line serologic test for celiac disease. The sensitivity of this test is reduced if patients are on a gluten-free diet prior to testing.
Endomysial antibody, IgA, serum	Negative	IgA autoantibodies to the endomysium (connective tissue surrounding muscle cells) are elevated in 70% to 80% of patients with celiac disease or dermatitis herpetiformis. By comparison, anti-tTG autoantibodies have a sensitivity and specificity of 90% to 98% and 95% to 97%, respectively; therefore, this is often the first-line test in celiac disease assessment. This test is not useful in patients with IgA deficiency. Titer generally correlates with disease severity and declines with strict adherence to a gluten-free diet.
Helicobacter pylori breath test	Negative	*H. pylori* causes chronic gastritis and predisposes to peptic ulcer disease, gastric adenocarcinoma, and lymphoma. The organism produces urease, which neutralizes gastric acid and provides ammonia for bacterial protein synthesis. Patients with suspected *H. pylori* infection ingest a small amount of urea labeled with an isotope (e.g., nonradioactive carbon-13); if *H. pylori*–derived urease is present, urea is metabolized to isotope-labeled carbon dioxide, which is detected in the patient's breath. This high-sensitivity, high-specificity test may be used for diagnosis of *H. pylori* infection and for confirming *H. pylori* eradication. Antibiotics and drugs that suppress gastric acid production may cause false-negative results, while false-positive results may be seen in the setting of achlorhydria and infection with other urease-positive organisms.
Helicobacter pylori stool test	Negative	Two types of tests evaluate the presence of *H. pylori* shedding in the stool: (1) enzyme immunoassay or immunochromatography for bacterial antigens and (2) PCR for *H. pylori* bacterial sequences. These high-sensitivity, high-specificity tests may be used for diagnosis of *H. pylori* infection and for confirming *H. pylori* eradication. Antibiotics and drugs that suppress gastric acid production may cause false-negative results by suppressing *H. pylori* growth.
Intrinsic factor (IF) antibodies, serum	Negative	Intrinsic factor is secreted by gastric parietal cells and binds to vitamin B_{12}, facilitating its absorption in the terminal ileum. In pernicious anemia, autoantibodies to IF prevent vitamin B_{12} binding, leading to vitamin B_{12} deficiency. Vitamin B_{12} deficiency can manifest with megaloblastic anemia and neurologic symptoms. Although anti-IF antibodies are very specific, they are positive in only about 50% of patients with pernicious anemia.
Parietal cell antibodies, serum	Negative: <20.0 U Equivocal: 20.1–24.9 U Positive: >25.0 U	These are IgG antibodies that bind to the H^+/K^+ ATPase pump on gastric parietal cells. They are seen in autoimmune gastritis, an inflammatory condition that leads to loss of parietal cells, atrophy and metaplasia of the oxyntic mucosa, and, if chronic, pernicious anemia, caused by loss of intrinsic factor and decreased vitamin B_{12} absorption. Parietal cell antibodies are found in more than 90% of patients with pernicious anemia but are less specific than intrinsic factor antibodies.
Tissue transglutaminase (TTG) IgA antibody, serum	<4.0 U/mL (negative) 4.0–10.0 U/mL (weak positive) >10.0 U/mL (positive)	TTG deamidates gliadin, which binds with increased affinity to HLA-DQ2 and DQ8 molecules on antigen-presenting cells, leading to a CD4+ T-cell response. TTG autoantibodies are elevated in patients in celiac disease; this test is a first-line screening test in conjunction with biopsy to confirm diagnosis. Since it assesses the presence of IgA antibodies, it is negative in patients with IgA deficiency (about 2% of patients with celiac disease). The test may be negative if patients are on a gluten-free diet and is useful in monitoring adherence to a gluten-free diet.
Molecular Tests of Relevance in Gastrointestinal Cancer		
Analyte	**Method**	**Pathophysiology/Clinical Relevance**
KIT mutation	Most commonly recognized in solid tumors by immunohistochemical staining for KIT; assessed in hematologic malignancies by DNA sequencing	Approximately 80% of gastrointestinal stromal tumors (GIST) have activating mutations in *KIT*, which encodes a receptor tyrosine kinase. *KIT* mutation correlates with strong immunohistochemical staining for KIT in GIST and is predictive of response to KIT inhibitors.

Microsatellite instability (MSI)/mismatch repair defect (MMS)	Usually assessed indirectly by immunohistochemical staining for MMR proteins (MLH1, MSH2, MSH6, PMS2); may be assessed directly by PCR amplification of microsatellites or DNA sequencing	About 2% to 3% of colorectal carcinomas arise in patients with germline defects in one of the DNA MMR genes (Lynch syndrome). Approximately 15% of sporadic colorectal carcinomas have MMR defects due to somatic hypermethylation of the *MLH1* promoter or acquired mutations in the same genes. MSI-high (MSI-H) colorectal carcinomas are often heavily infiltrated by T cells and are likely to respond to immune checkpoint inhibitors.

[a]The review of this table by Dr. Sonia S. Kupfer, Department of Medicine, University of Chicago, is greatly appreciated.

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Adapted from Deyrup AT, D'Ambrosio D, Muir J, et al. Essential Laboratory Tests for Medical Education. *Acad Pathol*. 2022;9. doi: 10.1016/j.acpath.2022.100046.

15

Pancreas

OUTLINE

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The pancreas is a transversely oriented retroperitoneal organ extending from the so-called "C loop" of the duodenum to the hilum of the spleen. Although the pancreas does not have well-defined anatomic subdivisions, adjacent vessels and ligaments serve to demarcate the organ into a head, body, and tail.

The pancreas is really two organs packaged into one. The first, the islets of Langerhans, which make up 1% to 2% of the pancreas and are scattered throughout, serve critical endocrine functions. The second, the exocrine portion, makes up the bulk of the organ and is a major source of enzymes that are essential for digestion. Diseases affecting the pancreas can be the source of significant morbidity and mortality. Unfortunately, the retroperitoneal location of the pancreas and the generally nonspecific signs and symptoms associated with disorders of the exocrine portion allow many pancreatic diseases to remain undiagnosed for extended periods of time; thus, recognition of pancreatic disorders requires a high degree of suspicion.

The *exocrine pancreas* is composed of *acinar cells* and the ductules and ducts that convey their secretions to the duodenum. The acinar cells are responsible for the synthesis of digestive enzymes, which are synthesized as inactive proenzymes that are stored in *zymogen granules*. When acinar cells are stimulated to secrete the enzymes, the granules fuse with the apical plasma membrane and release their contents into the central acinar lumen. These secretions are transported to the duodenum through a series of anastomosing ducts.

The epithelial cells lining the ducts are also active participants in pancreatic secretion. The cuboidal cells that line the smaller ductules secrete bicarbonate-rich fluid, while the columnar cells lining the larger ducts produce mucin. Ductal epithelial cells also express the *cystic fibrosis transmembrane conductance regulator (CFTR)*; aberrant function of this membrane protein affects the biochemical, in particular bicarbonate, content and viscosity of pancreatic secretions. CFTR dysfunction has a fundamental role in the pathophysiology of pancreatic disease in individuals with cystic fibrosis (Chapter 4).

As discussed later, autodigestion of the pancreas (e.g., in pancreatitis) can be a catastrophic event. A number of "fail-safe" mechanisms have evolved to minimize the risk for occurrence of this phenomenon:

- A majority of pancreatic enzymes are synthesized as inactive proenzymes and sequestered in membrane-bound zymogen granules, as mentioned earlier.
- *Proenzymes are typically activated by trypsin*, which itself is activated (from trypsinogen) by duodenal enteropeptidase (enterokinase) in the small bowel. Hence, the pancreatic enzymes are activated in the duodenum.
- Trypsin inhibitors (e.g., SPINK1, also known as *pancreatic secretory trypsin inhibitor*) are also secreted by acinar and ductal cells.
- In addition, trypsin cleaves and inactivates itself, a negative feedback mechanism that normally puts a limit on local levels of activated trypsin. Hence, trypsin is not activated within the pancreas itself, and neither are the pancreatic enzymes.
- Acinar cells are remarkably resistant to the action of activated enzymes such as trypsin, chymotrypsin, and phospholipase A_2.

Diseases of the exocrine pancreas include cystic fibrosis, congenital anomalies, acute and chronic pancreatitis, and neoplasms. Cystic fibrosis is discussed in detail in Chapter 4; the other pathologic processes are discussed in this chapter.

CONGENITAL ANOMALIES

Congenital anomalies of the pancreas are uncommon. The most significant are briefly described below:

- *Pancreas divisum* (Fig. 15.1) is the most common congenital anomaly of the pancreas, with an incidence of 3% to 10%. In most individuals, the main pancreatic duct (the duct of Wirsung) joins the common bile duct just proximal to the papilla of Vater, and the accessory pancreatic duct (the duct of Santorini) drains into the duodenum through a separate minor papilla. Pancreas divisum is caused by a failure of fusion of the fetal duct systems of the dorsal and ventral pancreatic primordia. As a result, the bulk of the pancreas (formed by the dorsal pancreatic primordium) drains into the duodenum through the small-caliber minor papilla. The duct of Wirsung in individuals with divisum drains only a small portion of the head of the gland through the papilla of Vater. More than 95% of individuals are asymptomatic. The remaining

The contributions to this chapter by Dr. Anirban Maitra, University of Texas, MD Anderson Cancer Center, Houston, Texas, in previous editions of this book are gratefully acknowledged.

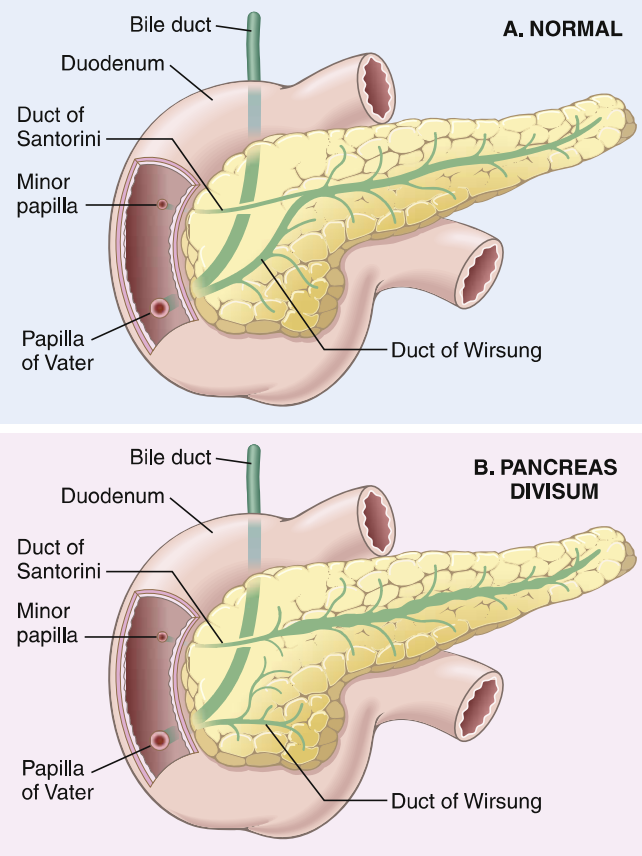


FIG. 15.1 Pancreatic ductal anatomy in (A) normal pancreas and (B) pancreatic divisum.

5% develop acute or chronic pancreatitis, possibly related to inadequate drainage of pancreatic secretions through the minor papilla.

- *Annular pancreas* is a relatively uncommon variant of pancreatic fusion in which a ring of pancreatic tissue completely encircles the duodenum. It can manifest with signs and symptoms of duodenal obstruction such as gastric distention and vomiting.
- *Ectopic pancreas* is aberrantly situated pancreatic tissue, which occurs in about 2% of the population. Common sites are the stomach and duodenum, followed by the jejunum, Meckel diverticulum, and ileum. These embryologic rests are typically small (ranging from millimeters to centimeters in diameter) and are located in the submucosa; they are composed of normal pancreatic acini with occasional islets. Though usually incidental and asymptomatic, ectopic pancreas may become inflamed, leading to pain, or—rarely—can cause mucosal bleeding. Approximately 2% of pancreatic neuroendocrine tumors (Chapter 18) arise in ectopic pancreatic tissue.
- *Congenital cysts* result from anomalous development of the pancreatic ducts. In *polycystic disease,* the kidneys, liver, and pancreas may all contain cysts (Chapter 12). Congenital cysts are generally unilocular and range from microscopic to 5 cm in diameter. They are lined by uniform cuboidal or flattened epithelium and are enclosed in a thin, fibrous capsule. These benign cysts contain clear *serous* fluid—an important point of distinction from pancreatic cystic neoplasms, which are often *mucinous* (discussed later).

PANCREATITIS

Inflammatory disorders of the pancreas are divided into acute and chronic forms. In *acute pancreatitis,* function can return to normal if the underlying cause of inflammation is removed. By contrast, *chronic pancreatitis* causes irreversible destruction of exocrine pancreas.

Acute Pancreatitis

Acute pancreatitis is a reversible inflammatory disorder that varies in severity, from focal edema and fat necrosis to widespread hemorrhagic necrosis. It is a relatively common and serious condition, with an annual incidence of 33 to 74 per 100,000 globally. In the Western world the incidence is 5 to 35 per 100,000. Overall mortality is about 5%. The incidence of acute pancreatitis is increasing due to the obesity epidemic and related increase in gallstone disease.

Etiology and Pathogenesis. The most common cause of acute pancreatitis in the United States is the impaction of gallstones within the common bile duct, impeding the flow of pancreatic enzymes through the ampulla of Vater ("gallstone pancreatitis"); this is closely followed by pancreatitis secondary to excessive alcohol intake. **Overall, gallstones and chronic excessive alcohol use account for close to 80% of acute pancreatitis cases,** with the remainder caused by a multitude of factors (Table 15.1). These include the following:

- *Nongallstone-related obstruction* of the pancreatic ducts (e.g., due to pancreatic cancer or other periampullary neoplasms, pancreas divisum, particulate solids that have precipitated from bile ("biliary sludge"), or parasites, particularly *Ascaris lumbricoides* and *Clonorchis sinensis*
- *Metabolic disorders,* in particular hypertriglyceridemia, hyperparathyroidism, and other hypercalcemic states. Hypertriglyceridemia (above 1000 mg/dL) has been reported to cause 5% to 10% of cases of acute pancreatitis.
- *Medications,* including anticonvulsants, cancer chemotherapeutic agents, thiazide diuretics, estrogens, and many others
- *Infections* with mumps virus or coxsackievirus, which can directly infect pancreatic exocrine cells

Of note, 10% to 20% of cases of acute pancreatitis have no identifiable cause *(idiopathic pancreatitis),* although a growing body of

Table 15.1 Etiologic Factors in Acute Pancreatitis

Metabolic
Alcohol use disorder[a]
Hypertriglyceridemia
Hypercalcemia
Drugs (e.g., azathioprine)
Genetic
Mutations in the cationic trypsinogen *(PRSS1)* and trypsin inhibitor *(SPINK1)* genes
Mechanical
Gallstones[a]
Trauma
Iatrogenic injury
Perioperative injury
Endoscopic procedures with dye injection
Vascular
Shock
Atheroembolism
Polyarteritis nodosa
Infectious
Mumps
Coxsackievirus

[a]Most common causes in the United States.

evidence suggests that many have an underlying genetic basis. For example, a subset of these patients with so-called *idiopathic pancreatitis* has underlying germline mutations affecting various genes: the autosomal dominant form is caused most often by mutations in the gene encoding trypsin, and most autosomal recessive disease is caused by mutations in CFTR. In those caused by mutations in the *CFTR* gene the symptoms are restricted to the pancreas (Chapter 4).

Acute pancreatitis is caused by autodigestion of the pancreas by intraacinar activation of pancreatic enzymes. Study of hereditary forms of acute pancreatitis has revealed the key role of premature activation of trypsin in this process. The feature shared by most forms of hereditary pancreatitis is a defect that increases or sustains the activity of trypsin such as mutations in the *PRSS1* gene, which encodes trypsinogen, the proenzyme of pancreatic trypsin. The pathogenic mutations alter the site through which trypsin cleaves and inactivates itself, abrogating an important negative feedback mechanism. This modification leads not only to the hyperactivation of trypsin but also to the activation of many other digestive enzymes that require trypsin cleavage for their activation. The released enzymes can inflict damage on blood vessels, causing hemorrhage within the pancreatic substance. Trypsin also converts prekallikrein to its activated form, thus initiating the kinin system, and, by activation of factor XII (Hageman factor), also sets in motion the clotting and complement systems (Chapter 3).

Three pathways can incite the initial enzyme activation that may lead to acute pancreatitis (Fig. 15.2):

- *Pancreatic duct obstruction.* Impaction of a gallstone or biliary sludge or extrinsic compression of the ductal system by a mass blocks ductal flow, increases intraductal pressure, and allows accumulation of an enzyme-rich interstitial fluid. Since lipase is secreted in an active form, local fat necrosis may result. Injured tissues, periacinar myofibroblasts, and leukocytes then release proinflammatory cytokines that promote local inflammation and interstitial edema through a leaky microvasculature. Edema further compromises local blood flow, causing vascular insufficiency and ischemic injury to acinar cells.
- *Primary acinar cell injury.* This pathogenic mechanism comes into play in acute pancreatitis caused by hypertriglyceridemia, alcohol use (discussed later), ischemia, viral infections (e.g., mumps), drugs, and direct trauma to the pancreas. The toxicity of triglycerides is not fully understood. According to one view large triglyceride-rich chylomicrons retard capillary circulation, leading to ischemic injury to pancreatic acinar cells. Injured cells release lipase into the interstitium, causing hydrolysis of triglycerides and local release of toxic free fatty acids release.
- *Defective intracellular transport of proenzymes within acinar cells.* In healthy acinar cells, digestive enzymes intended for zymogen

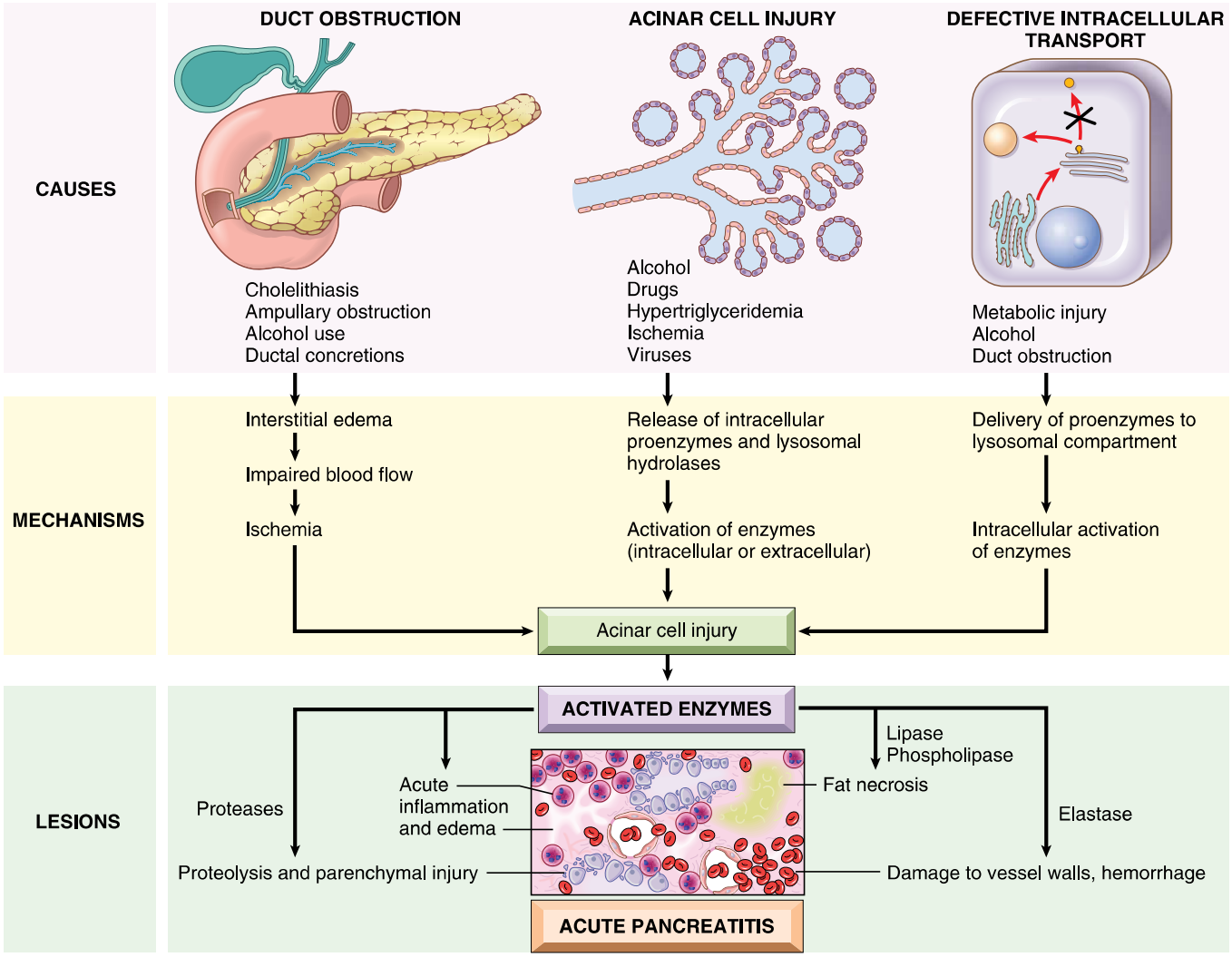


FIG. 15.2 Proposed pathogenesis of acute pancreatitis.

granules (and eventually extracellular release), and hydrolytic enzymes destined for lysosomes are transported in discrete pathways after synthesis in the endoplasmic reticulum. However, in at least some animal models of metabolic injury, pancreatic proenzymes and lysosomal hydrolases become packaged together. This results in proenzyme activation, lysosomal rupture (action of phospholipases), and local release of activated enzymes. A similar series of events is very likely in human acute pancreatitis, although definitive proof is lacking.

Alcohol consumption may cause pancreatitis by several mechanisms (see Fig. 15.2). Alcohol transiently increases pancreatic exocrine secretion and contraction of the sphincter of Oddi (the muscle regulating the flow of pancreatic secretions through the papilla of Vater). Alcohol also has direct toxic effects, including induction of oxidative stress in acinar cells, which leads to membrane damage; alcohol may also lead to delivery of proenzymes to lysosomal compartment with subsequent intracellular activation of trypsin and other digestive enzymes (discussed earlier). Finally, chronic alcohol ingestion results in the secretion of protein-rich pancreatic fluid, which leads to the deposition of inspissated protein plugs and obstruction of small pancreatic ducts.

MORPHOLOGY

The basic alterations in acute pancreatitis are **(1) microvascular leakage causing edema; (2) necrosis of fat by lipases; (3) an acute inflammatory reaction; (4) proteolytic destruction of pancreatic parenchyma and blood vessels leading to interstitial hemorrhage.**

In mild forms, there is interstitial edema and focal areas of fat necrosis in the pancreas and peripancreatic fat (Fig. 15.3A). Fat necrosis results from enzymatic destruction of fat cells; the released fatty acids combine with calcium to form insoluble salts that precipitate in situ.

In more severe forms, such as **acute necrotizing pancreatitis,** the damage also involves acinar and ductal cells, the islets of Langerhans, and blood vessels (eFig. 15.1). Macroscopically, the pancreas exhibits red-black hemorrhagic areas interspersed with foci of yellow-white, chalky **fat necrosis** (Fig. 15.3B). Fat necrosis can also occur in extrapancreatic fat, including the omentum and bowel mesentery, and even outside the abdominal cavity (e.g., in subcutaneous fat). In most cases, the peritoneum contains a serous, slightly turbid, brown-tinged fluid with globules of fat (derived from enzymatically digested adipose tissue). In the most severe form, **hemorrhagic pancreatitis,** extensive parenchymal necrosis is accompanied by diffuse hemorrhage within the substance of the gland (eFig. 15.2).

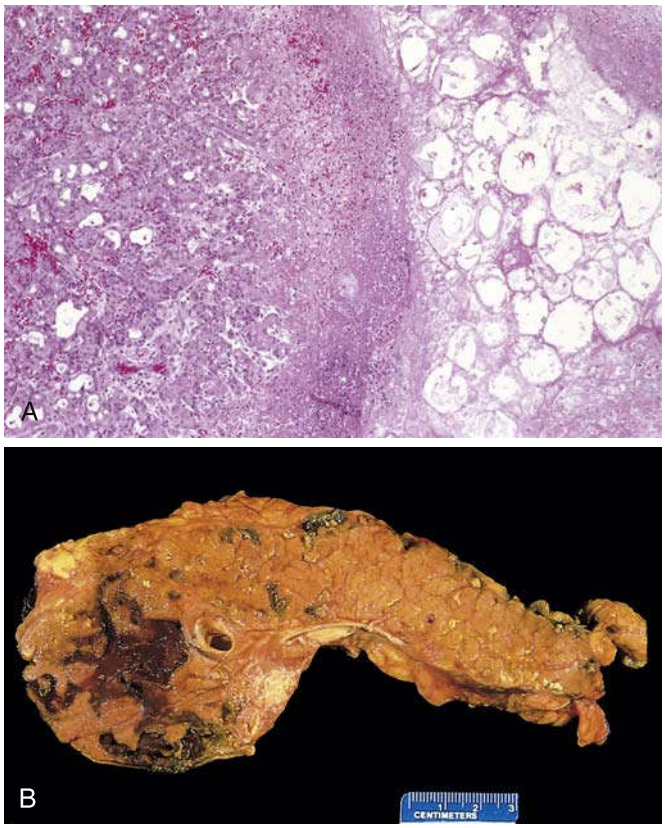


FIG. 15.3 Acute pancreatitis. (A) The microscopic field shows a region of fat necrosis *(right)* and focal pancreatic parenchymal necrosis *(center)*. (B) The pancreas has been sectioned longitudinally to reveal dark areas of hemorrhage in the pancreatic substance and a focal area of pale fat necrosis in the peripancreatic fat *(upper left)*.

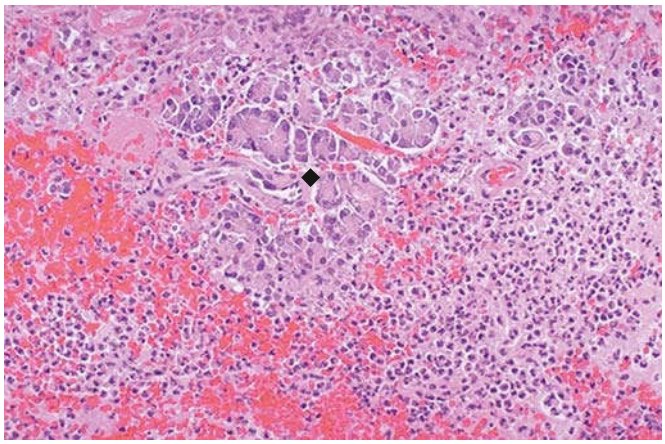
Clinical Features. **Abdominal pain is the cardinal manifestation of acute pancreatitis.** Its severity varies from mild and uncomfortable to severe and incapacitating. Acute pancreatitis is diagnosed primarily by the presence of elevated plasma levels of lipase and amylase and the exclusion of other causes of abdominal pain. In 80% of cases, acute pancreatitis is mild and self-limiting; the remaining 20% develop severe disease.

Full-blown acute pancreatitis is a medical emergency of the first order. Affected individuals usually experience the sudden calamitous onset of an "acute abdomen" with pain, guarding, and the ominous absence of bowel sounds. Characteristically, the pain is constant, intense, and referred to the upper back; it must be differentiated from similar pain due to perforated peptic ulcer, biliary colic, acute cholecystitis with rupture, and occlusion of mesenteric vessels with infarction of the bowel.

The manifestations of severe acute pancreatitis are attributable to systemic release of digestive enzymes and explosive activation of the inflammatory response. The initial clinical evaluation may reveal leukocytosis, disseminated intravascular coagulation (Chapter 10), acute respiratory distress syndrome secondary to diffuse alveolar damage (Chapter 11), and diffuse fat necrosis. Peripheral vascular collapse (shock) can rapidly ensue as a result of increased microvascular permeability and consequent hypovolemia, compounded by endotoxemia (from breakdown of the barriers between gastrointestinal flora and the bloodstream) and renal failure due to acute tubular injury (Chapter 12).

Laboratory findings include elevation of serum lipase and amylase, both of which increase 4 to 12 hours after onset of pain. **Serum lipase is the most specific and sensitive marker of acute pancreatitis, as serum amylase has a short half-life and may return to normal in 3 to 5 days, whereas lipase levels remain elevated for 8 to 14 days.** Hypocalcemia can result from precipitation of calcium in areas of fat necrosis; if persistent, it is a poor prognostic sign. The enlarged inflamed pancreas can be visualized by computed tomography (CT) or magnetic resonance imaging (MRI).

The crux of the management is supportive therapy (e.g., maintaining blood pressure and alleviating pain) and "resting" the pancreas by total restriction of oral food and fluids. In 40% to 60% of cases of acute necrotizing pancreatitis, the necrotic debris becomes infected,



eFIG. 15.1 Acute pancreatitis. Inflammation with necrosis and hemorrhage is seen here along with residual pancreatic acini *(diamond)*. The damage involves primarily the acinar cells, but the vasculature is also affected, and if severe and extensive, even the islets of Langerhans may be destroyed. (From Klatt EC: *Robbins and Cotran Atlas of Pathology,* ed 4, Fig. 9.6, Philadelphia, 2021, Elsevier.)

eFIG. 15.2 Acute hemorrhagic pancreatitis. At autopsy, the stomach *(square)* is reflected superiorly, and the spleen *(diamond)* can be seen at the far upper right. The pancreas is swollen and does not show the typical tan, lobulated architecture. Instead, hemorrhagic necrosis appears as blotchy black to red areas. (From Klatt EC: *Robbins and Cotran Atlas of Pathology,* ed 4, Fig. 9.4, Philadelphia, 2021, Elsevier.)

usually by gram-negative organisms from the alimentary tract, further complicating the clinical course. Although most individuals with acute pancreatitis eventually recover, in a minority, the systemic release of digestive enzymes may lead to serious consequences, such as the systemic inflammatory response syndrome with shock and disseminated intravascular coagulation (Chapter 3), acute respiratory distress syndrome (Chapter 11), and systemic fat necrosis. In such cases, acute pancreatitis is a dire medical emergency. In patients who survive, sequelae include sterile or infected *pancreatic "abscesses"* or *pancreatic pseudocysts.*

Pancreatic Pseudocysts

A pseudocyst is an encapsulated collection of fluid that may occur in the pancreas, but more commonly is outside the pancreas. It arises weeks after a bout of acute pancreatitis, when liquefied areas of necrotic tissue become walled off by inflammatory and fibrous tissue that lacks an epithelial lining (hence a "pseudo," or false, cyst) (Fig. 15.4). The cyst contents are rich in pancreatic enzymes and a laboratory assessment of the cyst aspirate can be diagnostic. Pseudocysts are usually solitary; they are commonly attached to the surface of the gland and involve peripancreatic tissues such as the lesser omental sac or the retroperitoneum between the stomach and transverse colon or liver. They can range in diameter from 2 cm to 30 cm. Pseudocysts account for approximately 75% of all pancreatic cysts. Most resolve spontaneously, but some persist and may lead to complications such as infection, compression, obstruction of adjacent structures, rupture, and hemorrhage.

Chronic Pancreatitis

Chronic pancreatitis is characterized by long-standing inflammation that leads to irreversible destruction of the exocrine pancreas, followed eventually by loss of the islets of Langerhans. Of note, recurrent bouts of acute pancreatitis regardless of etiology can evolve over time into chronic pancreatitis. The prevalence of chronic pancreatitis is difficult to determine but probably ranges between 0.04% and 5% of the U.S. population and 9 to 62 per 100,000 globally.

Pathogenesis. Acute and chronic pancreatitis share similar pathogenic mechanisms. By far **the most common cause of chronic pancreatitis is chronic excessive alcohol consumption,** especially among middle-aged men. How alcohol triggers chronic pancreatic injury and inflammation is not known; it may alter the activation of digestive enzymes, increase the production of oxygen-derived free radicals, or exert direct toxic effects on acinar cells. Other predisposing factors include functional or anatomic duct obstruction.

Autoimmune pancreatitis is a pathogenically distinct form of chronic pancreatitis that is associated with the presence of IgG4-secreting plasma cells in the pancreas. Autoimmune pancreatitis is one manifestation of IgG4-related disease (Chapter 5), which may involve multiple tissues. Recognizing this entity is important because it responds to anti-B cell and steroid therapy.

As many as 40% of individuals with chronic pancreatitis have no recognizable predisposing factors. It is increasingly being recognized that many "idiopathic" cases are associated with germline mutations in *CFTR*, *PRRS1*, and *SPINK1* genes, the very same genes that are often found to be mutated in familial acute pancreatitis. As discussed earlier, mutations in *PRRS1* (encoding trypsinogen) and *SPINK1* (encoding a trypsin inhibitor) both allow excessive activation of trypsin.

MORPHOLOGY

On gross evaluation, the gland is hard, sometimes with extremely dilated ducts and visible calcific concretions. Microscopically, chronic pancreatitis is characterized by **parenchymal fibrosis, reduced number and size of acini, and variable dilation of the pancreatic ducts;** initially there is a relative sparing of the islets of Langerhans (Fig. 15.5A). **Acinar loss** is a constant feature, usually with a chronic inflammatory infiltrate around remaining lobules and ducts. The ductal epithelium may be atrophied or hyperplastic or exhibit squamous metaplasia, and ductal concretions may be noted (Fig. 15.5B). The remaining islets of Langerhans become embedded in the sclerotic tissue and may fuse and appear enlarged; eventually they also disappear (eFig. 15.3).

Autoimmune pancreatitis is characterized by striking infiltration of the pancreas by lymphocytes and plasma cells, many of which are positive for IgG4, accompanied by a "swirling" fibrosis and venulitis **(lymphoplasmacytic sclerosing pancreatitis).**

Clinical Features. Abdominal pain is the most common symptom of chronic pancreatitis. It can also manifest as repeated bouts of jaundice, vague indigestion, or persistent or recurrent abdominal and back pain, or it may be entirely silent until exocrine pancreatic insufficiency and diabetes develop (the latter as a consequence of islet destruction).

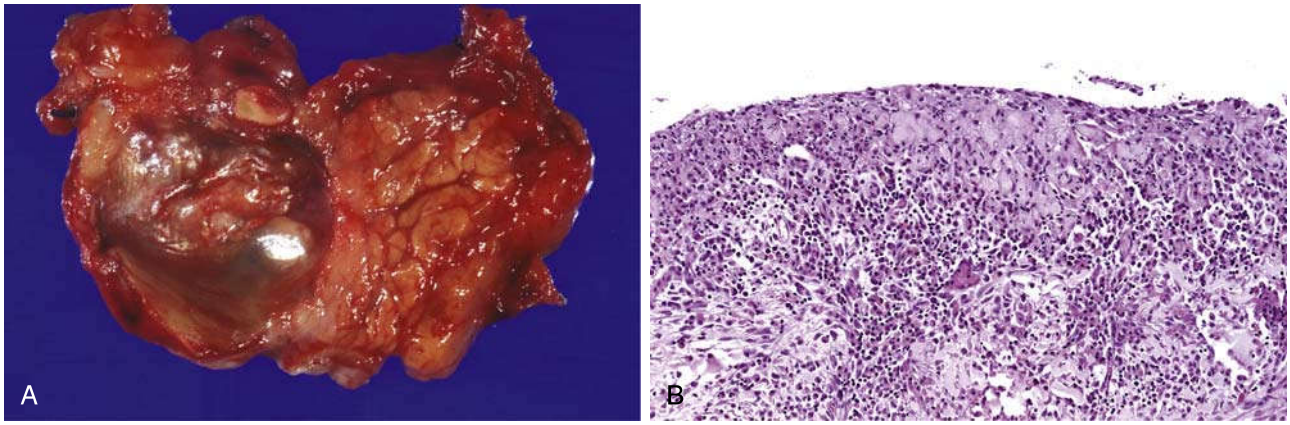
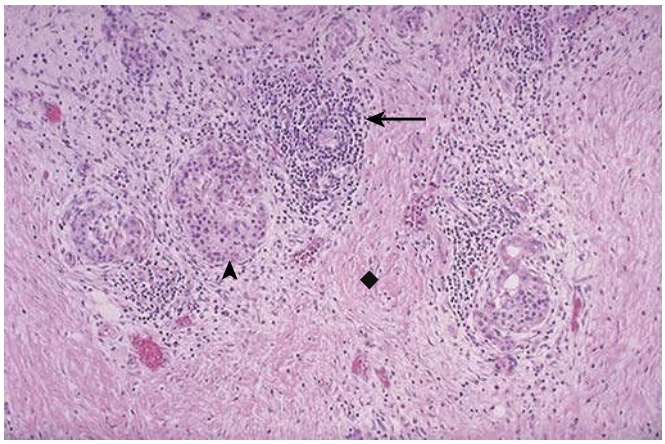


FIG. 15.4 Pancreatic pseudocyst. (A) Cross-section revealing a poorly defined cyst with a necrotic brownish wall. (B) Histologically, the cyst lacks an epithelial lining and instead is lined by fibrin and granulation tissue, with typical changes of chronic inflammation.



eFIG. 15.3 Chronic pancreatitis. Chronic inflammatory cells *(arrow)* are seen in a collagenous stroma *(diamond)* that lacks acini and retains a few islets of Langerhans *(arrowhead)*. (From Klatt EC: *Robbins and Cotran Atlas of Pathology,* ed 4, Fig. 9.11, Philadelphia, 2021, Elsevier.)

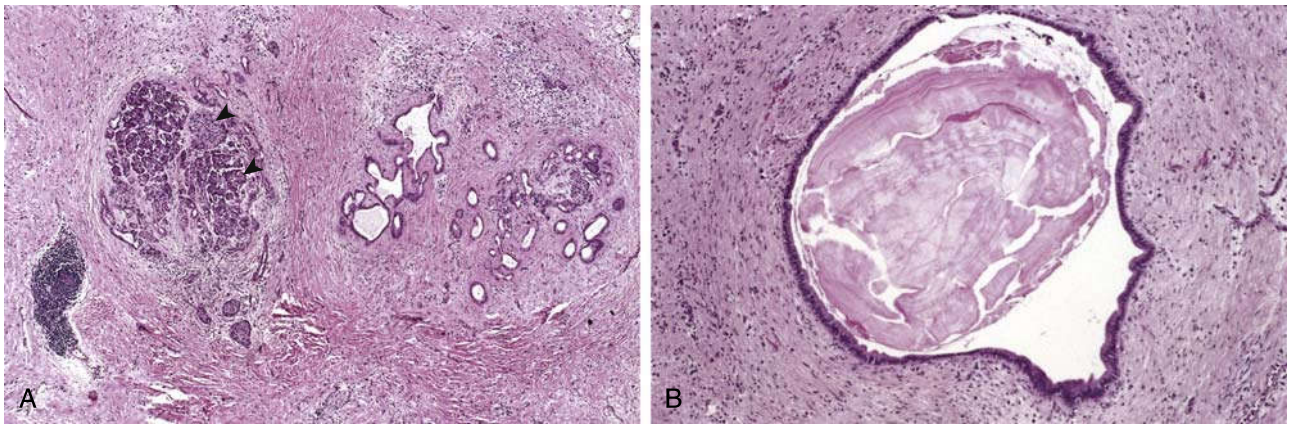


FIG. 15.5 Chronic pancreatitis. (A) Extensive fibrosis and atrophy have left only residual islets *(arrowheads)* and ducts *(right),* with a sprinkling of chronic inflammatory cells and acinar tissue. (B) A higher-power view demonstrating a dilated duct with inspissated eosinophilic concretions in a patient with alcohol-related chronic pancreatitis.

Attacks can be precipitated by excessive alcohol use, overeating (which increases demand on pancreatic secretions), or opiates or other drugs that increase the muscle tone of the sphincter of Oddi.

The diagnosis of chronic pancreatitis requires a high degree of clinical suspicion. With extensive fibrosis, acinar destruction may be so advanced that enzyme elevations are absent. Weight loss and hypoalbuminemic edema from malabsorption caused by pancreatic exocrine insufficiency can also point to the disease. Deficiency of fat-soluble vitamins, particularly vitamin D, can cause osteopenia. A helpful finding is visualization of calcifications within the pancreas by CT or ultrasonography.

Although chronic pancreatitis is usually not acutely life threatening, the long-term outlook is poor, with a 50% mortality rate over 20 to 25 years. *Pancreatic pseudocysts* (discussed earlier) develop in about 10% of patients. The most serious long-term complication of chronic pancreatitis is pancreatic cancer. Patients with hereditary pancreatitis associated with *PRSS1* mutations have a 40% lifetime risk of developing pancreatic cancer; the risk of pancreatic cancer is only modestly elevated in other forms of chronic pancreatitis.

PANCREATIC NEOPLASMS

Pancreatic exocrine neoplasms can be cystic or solid. Some tumors are benign, while others are among the most lethal of all malignancies.

Cystic Neoplasms

Cystic neoplasms are diverse tumors that range from harmless benign cysts to invasive, potentially lethal, cancers. Approximately 5% to 15% of all pancreatic cysts are neoplastic; these constitute less than 5% of all pancreatic neoplasms. There are three variants of cystic neoplasms. Some, such as serous cystic neoplasms, are almost always benign, whereas others, such as intraductal papillary mucinous neoplasms (IPMN) and mucinous cystic neoplasms, are precancerous. Most cystic neoplasms are detected incidentally when abdominal imaging is done for other reasons. IPMNs account for 38% of lesions, mucinous cystic neoplasms for 23%, and serous cystic tumors for 16%. Each of these is described next.

Serous Cystadenomas

These tumors are composed of glycogen-rich cuboidal cells surrounding small cysts containing clear, straw-colored fluid (Fig. 15.6). The cysts are small (1—3 mm) and can be solitary, multiple, or present as a honeycomb of microcystic lesions. They typically manifest in the seventh decade of life with nonspecific symptoms such as abdominal

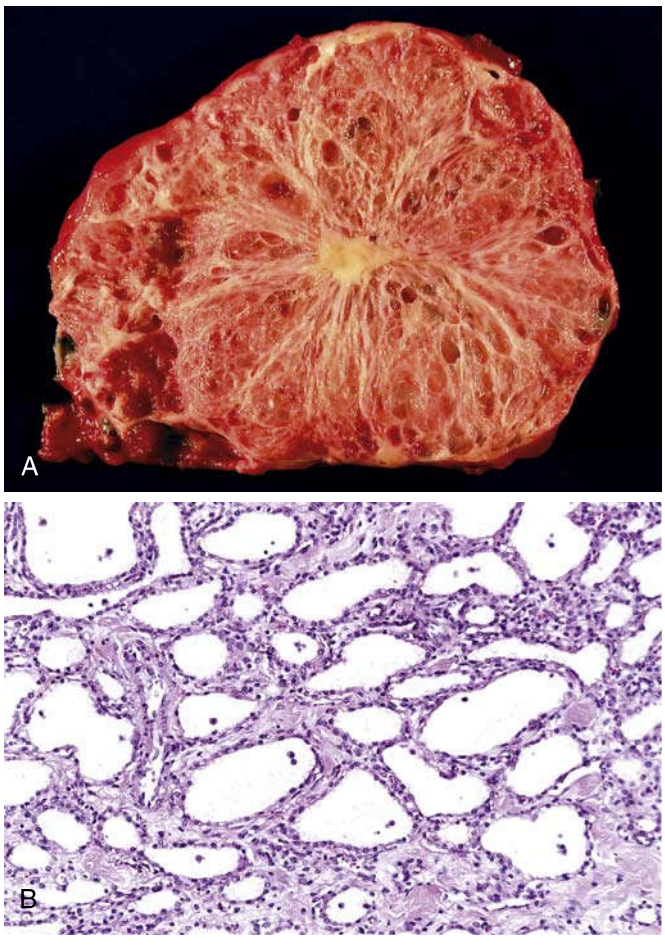


FIG. 15.6 Serous cystadenoma. (A) Cross-section through a serous cystadenoma. The lesion consists of honeycomb of microcystic lesions, only a thin rim of normal pancreatic parenchyma remains. The cysts are relatively small and contain clear, straw-colored fluid. (B) The cysts are lined by cuboidal epithelium without atypia.

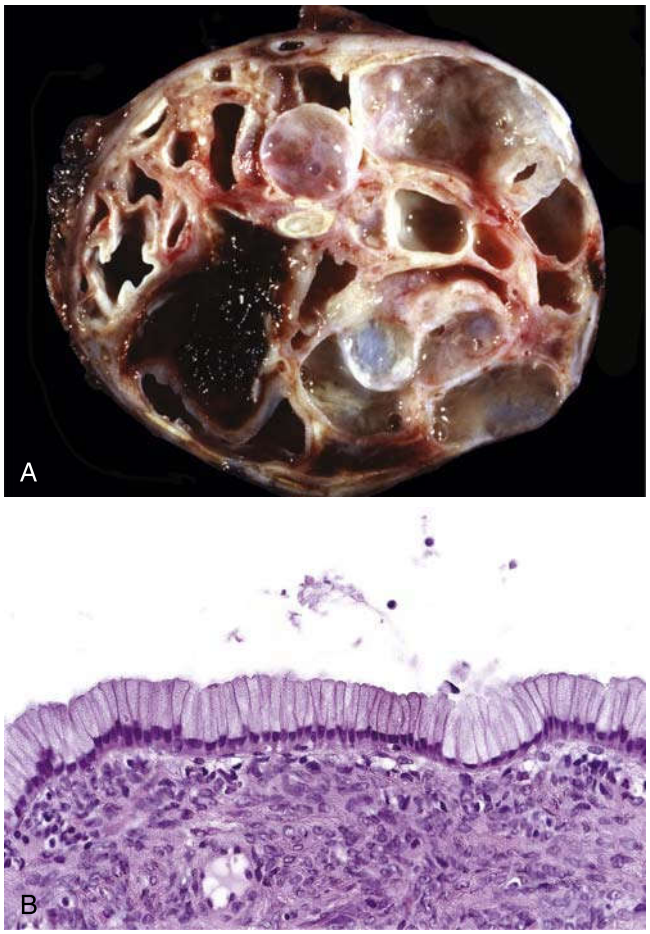


FIG. 15.7 Mucinous cystic neoplasm. (A) Cross-section through a mucinous multiloculated cyst in the tail of the pancreas. The cysts are large and filled with tenacious mucin. (B) The cysts are lined by columnar mucinous epithelium, with a densely cellular "ovarian" stroma.

pain; the female-to-male ratio is 2 :1. They are almost uniformly benign, and surgical resection is curative in the vast majority of patients. Most serous cystadenomas carry somatic loss-of-function mutations of the von Hippel-Lindau *(VHL)* tumor suppressor gene, which you will recall is a negative regulator of hypoxia-inducible factor-α (HIFα) (Chapter 12).

Mucinous Cystic Neoplasms

In contrast to serous cysts, close to 95% of mucinous cystic neoplasms arise in women, and they are precursors to invasive carcinomas. They usually arise in the tail of the pancreas and present as painless, slow-growing masses. The cystic spaces are filled with thick, tenacious mucin, and are lined by a columnar mucinous epithelium with an associated densely cellular stroma resembling that of the ovary (Fig. 15.7). Up to one-third of these cysts are associated with an invasive adenocarcinoma, another important difference from serous tumors. Mucinous cystic neoplasms harbor oncogenic *KRAS* mutations in approximately half of cases.

Intraductal Papillary Mucinous Neoplasms

In contrast with mucinous cystic neoplasms, IPMNs occur more frequently in men than in women and more commonly involve the head of the pancreas. IPMNs arise in the main pancreatic ducts, or one of its major branch ducts, and lack the cellular stroma seen in mucinous cystic neoplasms (Fig. 15.8). As with mucinous cystic neoplasms, IPMNs can progress to an invasive cancer: 70% of those affecting the main duct incur the risk of developing an invasive cancer, whereas the risk is smaller in branch duct lesions. In particular, "colloid" carcinomas of the pancreas, which are adenocarcinomas associated with abundant extracellular mucin production, nearly always arise through malignant transformation of an IPMN. Up to 80% of IPMNs harbor *KRAS* mutations, and two-thirds have oncogenic mutations of *GNAS,* which encodes the alpha subunit of a stimulatory G protein, G_s (Chapter 18). Constitutive activation of this G protein elevates levels of the second messenger cyclic AMP, activating certain kinases that promote cell proliferation.

Pancreatic Carcinoma

Infiltrating ductal adenocarcinoma of the pancreas (more commonly referred to as *pancreatic cancer*) is the third leading cause of cancer deaths in the United States, exceeded only by lung and colon cancers. Although it is substantially less common than the other two malignancies, pancreatic carcinoma is near the top of the list of killers because it carries one of the highest mortality rates. Close to 60,000 Americans were diagnosed with pancreatic cancer in 2021, and virtually all will die in a short period after diagnosis; the 5-year survival rate is a dismal 8%. The global rate is 8 to 14 per 100,000 with 6.92 per 100,000 dying.

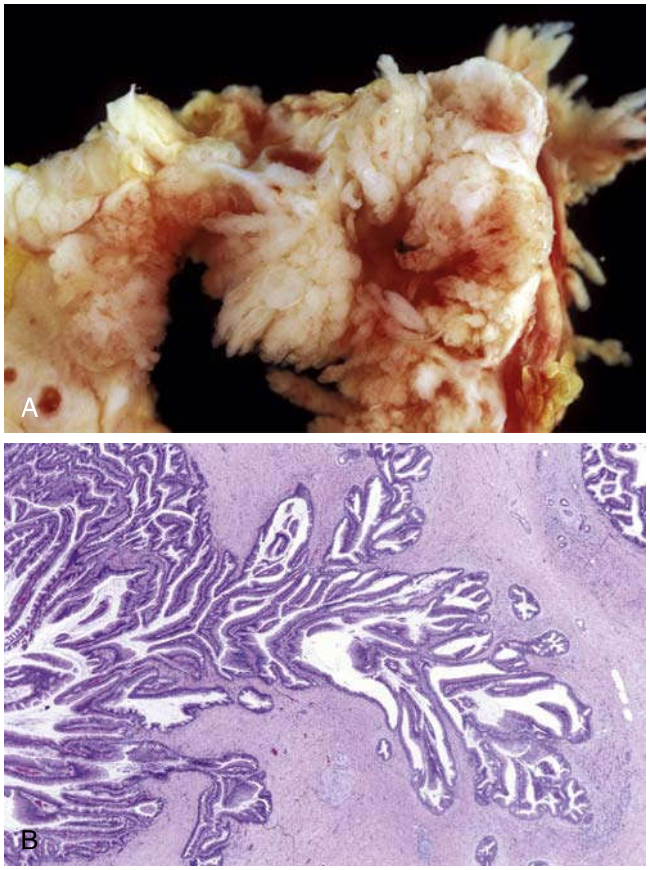


FIG. 15.8 Intraductal papillary mucinous neoplasm. (A) Cross-section through the head of the pancreas showing a prominent papillary neoplasm distending the main pancreatic duct. (B) The papillary mucinous neoplasm involves the main pancreatic duct *(left)* and is extending down into the smaller ducts and ductules *(right).*

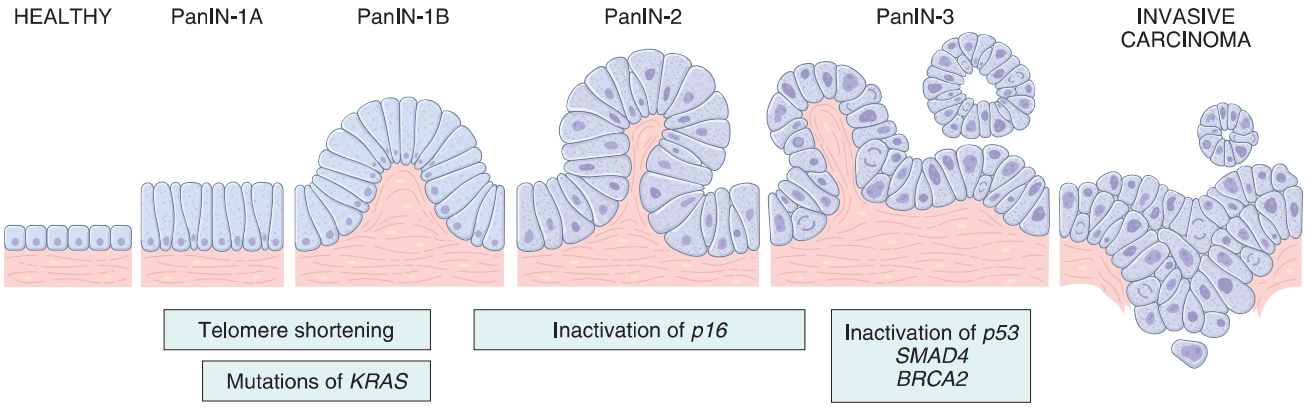


FIG. 15.9* Progression model for the development of pancreatic cancer. It is postulated that telomere shortening and mutations of the *KRAS* oncogene occur at early stages of carcinogenesis, inactivation of the *p16* tumor suppressor gene occurs at intermediate stages, and inactivation of the *TP53, SMAD4,* and *BRCA2* tumor suppressor genes occurs at late stages. Note that while there is a general temporal sequence of changes, the accumulation of multiple mutations is more important than their occurrence in a specific order. *PanIN,* Pancreatic intraepithelial neoplasm. The numbers following the labels on the top refer to stages in the development of PanINs. *Stages refers to stages of development of the cancer, not clinical staging. (Modified from Maitra A, Hruban RH: Pancreatic cancer, *Annu Rev Pathol Mech Dis* 3:157, 2008.)

Pathogenesis. Like all cancers, pancreatic cancer arises as a consequence of inherited and acquired mutations in cancer-associated genes. In a pattern analogous to that seen in the multistep progression of colon cancer (Chapters 6 and 13), there is a progressive accumulation of genetic changes in pancreatic epithelium as it proceeds from nonneoplastic, to noninvasive precursor lesions, to invasive carcinoma (Fig. 15.9). While both intraductal papillary mucinous neoplasms (IPMNs) and mucinous cystic neoplasms can progress to invasive adenocarcinoma, **the most common antecedent lesions of pancreatic cancer arise in small ducts and ductules and are called pancreatic intraepithelial neoplasias (PanINs).** Evidence in favor of the precursor relationship of PanINs to frank malignancy includes the observations that these microscopic lesions are often found adjacent to infiltrating carcinomas and the two share a number of genetic alterations. Moreover, the epithelial cells in PanINs show dramatic telomere shortening, potentially predisposing these lesions to pathogenic chromosomal abnormalities that may contribute to acquisition of the full spectrum of cancer hallmarks. It should be noted that while most pancreatic cancers arise from PanINs, the vast majority of PanINs do not progress to frank malignancy.

The recent sequencing of the pancreatic cancer genome has confirmed that **four genes are most commonly affected by somatic mutations in this neoplasm: *KRAS, CDKN2A/p16, SMAD4,* and *TP53:***

- *KRAS is the most frequently altered oncogene in pancreatic cancer*; it is activated by a point mutation in greater than 90% of cases. These mutations impair the intrinsic GTPase activity of the KRAS protein so that it is constitutively active. In turn, KRAS activates a number of intracellular signaling pathways that promote carcinogenesis (Chapter 6).
- *CDKN2A is inactivated in 30% of cases.* This complex locus encodes two tumor suppressor proteins (Chapter 6): p16/INK4a, a cyclin-dependent kinase inhibitor that antagonizes cell cycle progression, and ARF, a protein that augments the function of the p53 tumor suppressor protein.
- *The SMAD4 tumor suppressor gene is inactivated in 55% of pancreatic cancers* and only rarely in other tumors; it codes for a protein that plays an important role in signal transduction downstream of the transforming growth factor-β receptor.
- *Inactivation of the TP53 tumor suppressor gene occurs in 70% to 75% of pancreatic cancers.* Its gene product, p53, acts both to enforce cell-cycle checkpoints and as an inducer of apoptosis or senescence (Chapter 6). *BRCA2* is also mutated late in a subset of pancreatic cancers.

Pancreatic cancer is primarily a disease of older adults, with 80% of cases occurring between 60 and 80 years of age. The strongest environmental influence is smoking, which doubles the risk. Long-standing chronic pancreatitis and diabetes are also associated with a modestly increased risk for pancreatic cancer. In addition to being risk factors for pancreatic cancer, chronic pancreatitis and diabetes may be manifestations of pancreatic cancer. Thus, for example, tumors arising in the head of the pancreas often cause chronic pancreatitis in the distal parenchyma, while diabetes caused by duct obstruction and subsequent pancreatitis may be the manifestation of an underlying neoplasm. In fact, approximately 1% of the older adult population with new-onset diabetes harbors an unsuspected pancreatic cancer. Familial clustering of pancreatic cancer has been reported, and a growing number of inherited genetic defects are now recognized that increase pancreatic cancer risk. Included in these are germline mutations of the familial breast/ovarian cancer gene *BRCA2* and mismatch repair genes, both seen in approximately 10% of cases.

MORPHOLOGY

Approximately 60% of pancreatic cancers arise in the head of the gland, 15% in the body, and 5% in the tail; in the remaining 20%, the neoplasm diffusely involves the entire organ. Carcinomas of the pancreas are usually hard, gray-white, stellate, poorly defined masses (Fig. 15.10A).

Two features are characteristic of pancreatic cancer: It is highly invasive (even "early" invasive pancreatic cancers invade peripancreatic tissues extensively), and it elicits an intense host reaction in the form of dense fibrosis **(desmoplastic response).**

Most carcinomas of the head of the pancreas obstruct the distal common bile duct as it courses through the head of the pancreas. In 50% of such cases, there is marked distention of the biliary tree, and patients typically exhibit jaundice. By contrast, carcinomas of the body and tail of the pancreas do not impinge on the biliary tract. Pancreatic cancers often extend through the retroperitoneal space, entrapping adjacent nerves (thus, accounting for the pain), and occasionally invade the spleen, adrenal glands, vertebral column, transverse colon, and stomach. Peripancreatic, gastric, mesenteric, omental, and portahepatic lymph nodes are frequently involved, and the liver is often enlarged with metastatic deposits. Distant metastases may occur, principally to the lungs and bones.

On microscopic examination, pancreatic carcinoma is usually a **moderately to poorly differentiated adenocarcinoma,** forming abortive glands with mucin secretion or cell clusters and exhibiting an aggressive, deeply infiltrative growth pattern (Fig. 15.10B). Dense stromal fibrosis accompanies tumor invasion, and there is a tendency for perineural invasion within and beyond the organ. Lymphatic invasion is also commonly seen.

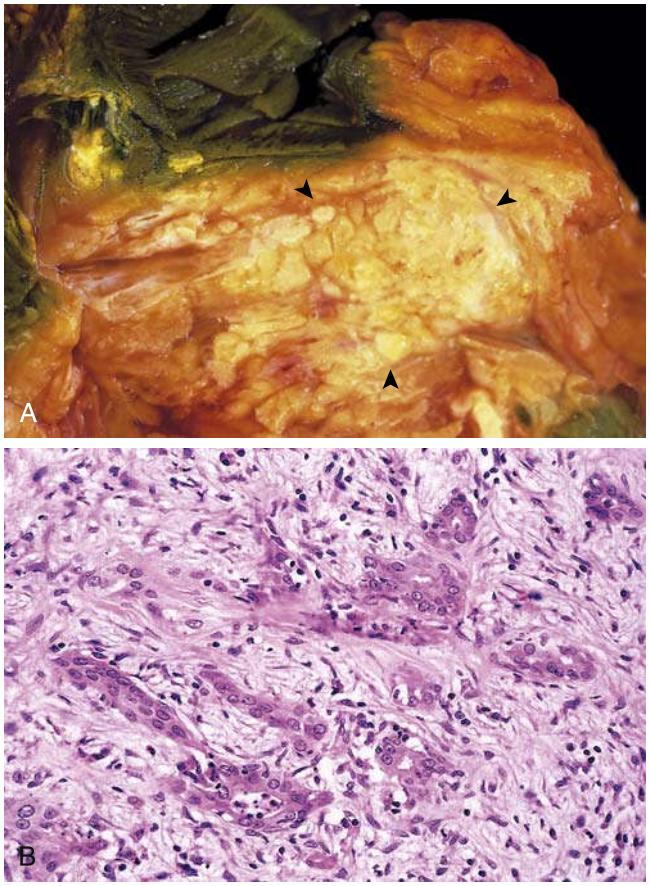


FIG. 15.10 Carcinoma of the pancreas. (A) Cross-section through the head of the pancreas and adjacent common bile duct showing both an ill-defined mass in the pancreatic substance *(arrowheads)* and the green discoloration of the duct resulting from total obstruction of bile flow. (B) Poorly formed glands are present in a densely fibrotic (desmoplastic) stroma within the pancreatic substance.

Clinical Features. **Carcinomas of the pancreas typically remain silent until their extension impinges on some other structure.** Pain is usually the first symptom, but by that point these cancers are often beyond cure. *Obstructive jaundice* can be associated with carcinoma in the head of the pancreas, but it rarely draws attention to the cancer sufficiently early for timely intervention. Weight loss, anorexia, and generalized malaise and weakness are manifestations of advanced disease. *Migratory thrombophlebitis (Trousseau syndrome)* occurs in about 10% of patients and is attributable to the elaboration of platelet-aggregating factors and procoagulants from the tumor or its necrotic products (Chapter 6). As previously stated, new-onset diabetes is the first manifestation of pancreatic cancer in some patients.

The clinical course of pancreatic carcinoma is rapidly progressive and often distressingly brief. Less than 20% of pancreatic cancers are resectable at the time of diagnosis. It has long been recognized that there is a profound need for biomarkers capable of detecting early, potentially curable, pancreatic cancers. Although serum levels of many enzymes and antigens (e.g., carcinoembryonic and CA19-9 antigens) are elevated, these markers are neither specific nor sensitive enough to be useful for screening. Several imaging techniques, such as endoscopic ultrasonography and high-resolution CT scans, are helpful for investigation in cases of suspected cancer but are not practical screening tests.

RAPID REVIEW

Pancreatitis

- *Acute pancreatitis* is characterized by inflammation and reversible parenchymal damage that ranges from focal edema and fat necrosis to widespread parenchymal necrosis and hemorrhage; the clinical presentation varies widely, from mild abdominal pain to rapidly fatal vascular collapse.
- *Chronic pancreatitis* is characterized by irreversible parenchymal damage and scar formation; clinical presentations include chronic malabsorption (due to pancreatic exocrine insufficiency) and diabetes (due to islet cell loss).
- Both entities share similar pathogenic mechanisms, and indeed recurrent acute pancreatitis can result in chronic pancreatitis. *Ductal obstruction by gallstones* and *chronic alcohol excess* are the most common causes in both forms. Inappropriate activation of pancreatic digestive enzymes (due to mutations in genes encoding trypsinogen or trypsin inhibitors) and primary acinar injury (due to toxins, infections, ischemia, or trauma) also cause pancreatitis. In some cases mutation in *CFTR* underlies pancreatitis.

Pancreatic Neoplasms

- Virtually all serous cystic neoplasms are benign; mucinous cystic neoplasms and intraductal papillary mucinous neoplasms are curable but incur a higher risk of developing cancer.
- Pancreatic cancer probably arises from noninvasive precursor lesions (most commonly, PanINs), developing by progressive accumulation of mutations of oncogenes (e.g., *KRAS*) and tumor suppressor genes (e.g., *CDKN2A/p16, TP53,* and *SMAD4*).
- Typically, these neoplasms are ductal adenocarcinomas that produce an intense desmoplastic response.
- Most pancreatic cancers are diagnosed at an advanced stage, accounting for the high mortality rate.
- Obstructive jaundice is a feature of carcinoma of the head of the pancreas; many patients also experience debilitating pain.
- Carcinomas of the tail of the pancreas are often not detected until late in their course.

Laboratory Tests

Test	Normal Value	Pathophysiology/Clinical Relevance
Amylase, serum	28–100 U/L	Amylase hydrolyzes complex carbohydrates and is primarily secreted by the salivary glands and the pancreas. Its level increases with gland inflammation or duct obstruction. Amylase level is increased in acute pancreatitis, pancreatic pseudocyst, and pancreatic duct obstruction (e.g., choledocholithiasis, pancreatic cancer). In acute pancreatitis, amylase levels increase rapidly (4–12 hours of symptom onset) but decline to normal in 3–5 days. Serum lipase level is currently the preferred test for the diagnosis of acute pancreatitis (see below). Amylase levels are decreased in pancreatic insufficiency and chronic pancreatitis.
Lipase, serum	13–60 U/L	Lipase is a digestive enzyme produced in pancreatic acinar cells and secreted into the duodenum to digest lipids. With acinar cell injury (e.g., acute pancreatitis), lipase is released into the pancreas, where it contributes to local tissue damage, including acute inflammation, autodigestion of pancreatic parenchyma, fat necrosis, and vascular damage. In acute pancreatitis, serum lipase is elevated, typically >3 times the upper limit of normal. The rise occurs within 4–8 hours and levels may remain elevated for up to 14 days. It is a more sensitive and specific test for acute pancreatitis. The degree of lipase elevation does not correlate with the severity of pancreatitis.

References values from https://www.mayocliniclabs.com/ by permission of Mayo Foundation for Medical Education and Research. All rights reserved.

Adapted from Deyrup AT, D'Ambrosio D, Muir J, et al. Essential Laboratory Tests for Medical Education. *Acad Pathol*. 2022;9. doi: 10.1016/j.acpath.2022.100046.